

Monopoly, Quality, and Costly Information *

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Abstract

A seller offers a buyer a schedule of transfers and associated product qualities. After observing this schedule, the buyer chooses a flexible costly signal about his type. We show it is without loss to focus on a class of allocations that compensate the buyer for his marginal learning costs. In the optimal menu, all types typically receive lower-than-efficient quality (including “at the top”). Moreover, profits are non-monotonic in the level of information costs, and the consumer may be better off when such costs are low than when information is free. When learning costs are steep, the optimal menu contains at most two purchasing options.

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1. Introduction

Consumers are often unsure about which product is right for them. Frequently, additional information that could aid the buyer’s decision is available, but acquiring and processing it requires effort. In health insurance markets, for instance, potential buyers struggle to select the most suitable plan, incurring large losses for choosing the wrong product (Abaluck and Gruber, 2011; Handel and Kolstad, 2015; Brown and Jeon, 2024). Consumers may consider their past experience and investigate their family health history to forecast coverage needs, or they may compare how different plans cover various medical conditions. Whether they are assessing their personal needs or evaluating how different products align with their preferences, acquiring information entails costs. We study how this costly information acquisition in demand affects equilibrium outcomes when a firm screens consumers.

Understanding equilibrium responses to information frictions is important for several reasons. First, extensive evidence shows that such frictions significantly affect consumer behavior (Abaluck and Gruber, 2011; Taubinsky and Rees-Jones, 2018; Brown and Jeon, 2024). Second, supply responses shed light on producers’ incentives to aid or hinder consumer learning. Indeed, insurance company websites offer tools that help consumers compare plans, suggesting that sellers act to influence information acquisition. Finally, information frictions justify public interventions—in health care, governments may provide instruments for price transparency (Brown, 2019), or standardize and simplify information on insurance plans (Ericson and Starc, 2016). The welfare consequences of these interventions cannot be ascertained without considering equilibrium effects. In fact, we show that helping buyers choose may reduce consumer welfare.

The Model We study a model in which a seller decides what menu of vertically differentiated products to offer to a potential buyer. Unlike the classical model of Mussa and Rosen (1978) and Maskin and Riley (1984), we do not assume the buyer possesses private information when he first sees the monopolist’s menu. Instead, the buyer sees this menu, and then *chooses* what to learn about his type. The buyer’s information choice is flexible and costly; that is, the consumer is rationally inattentive about his type. Because learning takes place after the seller makes her offer, the menu of products affects information choices: the monopolist can influence which information the buyer acquires by choosing which products to provide at which prices. Our main interest is in the structure of this menu and the efficiency of the resulting allocation with respect to the buyer’s chosen information.

Our key modeling assumption is that information costs are *posterior-mean separable* (Mensch and Malik, 2023). That is, we assume that for the buyer to learn a certain signal about his type, he faces a cost that is linear in the distribution of posterior means generated by that signal. With the standard payoff specification in Mussa and Rosen (1978), this assumption implies that the buyer’s optimal learning program is a special case of mean-measurable information design (Gentzkow and Kamenica, 2016; Dworzak and Martini, 2019; Arieli et al., 2020; Kleiner, Moldovanu, and Strack, 2021). Unlike their more general counterpart, mean-measurable information design problems remain tractable even when the underlying state space is very large. Consequently, we can accommodate discrete and continuous prior distributions. When the prior distribution is binary, the class of cost functions we entertain includes the standard posterior-separable costs (Caplin, Dean, and Leahy, 2022), commonly used in the rational inattention literature.

Results Information acquisition brings a new dimension to the monopolist’s problem: the product line must induce the buyer to learn what the seller wants him to. Optimal learning for the buyer then imposes joint constraints on the amount of product differentiation and the resulting information asymmetry. Using Dworzak and Martini (2019) duality-based tools, we can show it is without loss to focus on the class of *cost-compensating* allocations. These allocations decompose the buyer’s information rents in two parts: one part that compensates the buyer’s cost of learning, and a residual that comes from the shadow value of information. The reduction to cost-compensating allocations enables us to reason about the monopolist optimal outcome using two familiar-looking programs. The first program involves maximizing the monopolist’s profits across all cost canceling allocations that induce the buyer to learn a certain signal. This program resembles the standard mechanism design problem. The second program searches for the most profitable signal for the monopolist among those signals that the buyer can be induced to learn given a fixed allocation. When this allocation is cost-compensating, the second program turns out to be a “constrained” mean-measurable information design problem.

Using these two programs, we prove three main results about the monopolist-optimal outcome. First, we show that information acquisition exacerbates the inefficiencies generated by information asymmetry alone. As in standard screening, the allocation of quality to consumers is always weakly below the efficient one. However, when information costs are steep, quality distortions are more widespread than in standard screening: agents with all valuations receive below-efficient quality levels—in contrast to the standard “no distortion

at the top” result, where the highest-valuation buyer’s quality is efficient. The reason is that the consumer can deviate from the principal-preferred information, learning what the seller does not want him to. Thus, the buyer receives rents not only due to the private information he actually acquires, but also because of the information he could have obtained. To reduce these rents, the seller distorts quality provision.

For an intuition, consider the simple case in which information is symmetric at the optimum—i.e., the buyer chooses to remain uninformed.¹ In standard screening, if information is symmetric, the seller offers a single product to the buyer, with efficient quality tailored to his valuation; prices are such that the consumer is indifferent between buying or not. Such indifference cannot occur in our model. If the consumer is ex-ante indifferent between buying and not buying, then he can strictly benefit from learning whether his true value is just slightly lower than his expected valuation: If he finds out his type is low, he does not purchase the product. Hence, the buyer avoids learning only if the seller offers a price discount and, to save on costs, degrades the product’s quality. Thus, the contract is inefficient even when information is symmetric. This argument highlights the distinction between this model and standard screening: the key distortion here is the *threat* of obtaining information, rather than the presence of private information.

The second main result is that profits and consumer surplus can be non-monotonic in the level of consumer inattention. Profits increase and consumer surplus decreases when the scale of information costs is high, and they move in the opposite direction when such scale is low, under suitable conditions. Heuristically, this is because the value of the agent’s preferred deviations varies as costs change. For an intuition, when information costs are high, the seller induces the buyer to acquire very little information. In that case, the buyer extracts rents by threatening to learn more than the seller wants. But that threat becomes less pressing when information costs grow, benefiting the seller at the buyer’s expense. The symmetric logic delivers a similar result when information costs are low: in that region, profits fall with information costs, but consumers receive further rents. In that case, the increased consumer rents are offset by the growth of information costs, and we characterize when the net consumer surplus grows or falls as a function of the shape of learning costs.

Finally, we show that simple menus are optimal when information costs are steep. In other words, the possibility of information acquisition by the buyer can make it optimal for the seller to offer just a few product options, even when the set of possible buyer types is

¹We prove that information is indeed symmetric in equilibrium when information costs are sufficiently high.

very large. Roughly speaking, for steep information costs the optimal signal for the buyer turns out to be binary even if the support of the prior distribution is a whole interval. In this case, a revelation-principle-style argument immediately delivers that the monopolist can maximize profits by offering the buyer at most two different products. We further identify conditions in which the buyer’s signal is binary, but the seller can attain the optimum offering exactly one purchasing option. This explanation for menu simplicity stands in contrast to the full-information case, where small menus typically arise only in restrictive cases, such as when the set of possible buyer types is small, or when the marginal production costs are constant in quality.

Related Literature. Our paper lies in the literature studying the interaction between flexible information acquisition and trade. Very closely related is Mensch (2022), who studies the optimal way to auction an indivisible good to buyers who flexibly acquire information about their value after observing the menu. His single-buyer case corresponds to a monopolistic screening model with endogenous information that differs from ours in that information costs are posterior-separable and “quality” (which maps to the probability of receiving the good in his model) costs zero to produce. By contrast, we assume that learning costs take a *posterior-mean* separable form, and that quality has strictly increasing marginal costs. These assumptions yield new insights that cannot be obtained in his model. For example, his results yield efficiency (i.e. sale with probability 1) at the highest signal, whereas we find inefficiently low quality provision at all signals, including the highest, whenever marginal costs are strictly increasing. We also demonstrate comparative statics and results regarding optimality of simple menus with multiple states, which do not have corresponding analogs in Mensch (2022).

In addition to Mensch (2022), several other papers study the interaction between information design (Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011; Bergemann and Morris, 2013) and trade. The closest papers to ours are Condorelli and Szentes (2020) and Ravid, Roesler, and Szentes (2022), both of which study models of bilateral trade with a single indivisible good, but differ in the timing of information acquisition. Several other papers study the set of possible outcomes in bilateral trade settings with indivisible goods as one varies each party’s information; see Bergemann, Brooks, and Morris (2015); Roesler and Szentes (2017); Kartik and Zhong (2019); Haghpahan and Siegel (2022a); and

Haghpanah and Siegel (2022b).²

More broadly, this paper contributes to the burgeoning literature on rational inattention, started by the seminal papers of Sims (1998; 2003), and developed into models of flexible information acquisition by Caplin and Dean (2013; 2015), Matějka and McKay (2015), and Caplin, Dean, and Leahy (2022) using a posterior-separable approach to modeling information costs. Since then, there have been a number of applications of rational inattention to various economic problems, such as global games (Yang, 2015; Morris and Yang, 2022; Denti, 2022), bargaining (Ravid, 2020), and attention management (Lipnowski, Mathevet, and Wei 2020). The most relevant paper is Yang (2020), who studies a security-design problem related to our model.

Several papers use more structured learning models to explore how the buyer’s incentives to acquire information depends on the selling mechanism. For example, Crémer and Khalil (1992) consider a buyer who contracts with a seller who decides whether or not to pay in order to observe their cost of production, and show the seller remains uninformed at the optimum. Persico (2000) shows buyers acquire less information in a second-price auction than in a first-price one, provided that their signals are affiliated. Bergemann and Välimäki (2002) shows that with information acquisition, the classic Vickrey-Clark-Groves mechanism still implements the efficient allocation when values are private, but that efficiency may fail when values are common. Compte and Jehiel (2007) show simultaneous auctions generate lower revenue than dynamic ones when buyers have an opportunity to learn. Shi (2012) characterizes the revenue-maximizing auction in private-value settings. In addition to their focus on auctions, these models differ from ours in that they require the buyer to choose among a set of signal structures that can be linearly ordered in their informativeness.³

²Armstrong and Zhou (2022) studies the effect of information on profits and consumer surplus in oligopolistic competition. In addition, several papers use information design tools to study information provision in markets. For example, see Hwang, Kim, and Boleslavsky (2019), Smolin (2020), and Yang (forthcoming).

³Another strand of the literature studies the seller’s benefits from revealing information about the buyers’ valuations prior to participating in an auction; see, for example, Milgrom and Weber (1982), Ganuza (2004), Bergemann and Pesendorfer (2007), Ganuza and Penalva (2010), and Li and Shi (2017).

2. Model

There is a monopolist (she) and a buyer (he). The game begins with the monopolist offering the buyer a menu, which is a compact set of pairs, $M \subseteq [0, \bar{q}] \times \mathbb{R}$.⁴ Each menu item $(q, t) \in M$ corresponds to a transfer of t to be paid to the monopolist by the buyer, and the quality q of the product the buyer gets in exchange. The buyer's utility from (q, t) depends on his quality valuation, θ , a random variable distributed over $\Theta = [\underline{\theta}, \bar{\theta}] \subseteq \mathbb{R}_+$ according to a CDF F_0 , with mean θ_0 . Without loss of generality, F_0 includes $\underline{\theta}$ and $\bar{\theta}$ in its support. Given θ , the buyer's utility from (q, t) is

$$U(\theta, q, t) = \theta q - t.$$

The monopolist's payoff from the buyer's chosen menu item (q, t) is

$$\Pi(q, t) = t - \kappa(q),$$

where $\kappa : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is an increasing, continuously differentiable, and strongly convex function satisfying $\kappa(0) = 0$.⁵ We also assume it is efficient to serve all types, meaning $\kappa'(0) < \underline{\theta}$, though this assumption can often be relaxed—see Section 7. The monopolist's menu must give the buyer the option of not buying anything, meaning M must include the option $(0, 0)$. Both the monopolist and the buyer are risk-neutral expected utility maximizers.

Neither the monopolist nor the buyer knows θ , but the buyer can choose to learn about it after observing the monopolist's menu. The buyer's information acquisition is flexible, meaning he can use any signal s to learn about θ . The flexibility assumption expresses two ideas. First, the buyer has access to detailed data that helps him determine his exact value for the item. Second, the buyer can collect only the information he finds useful, avoiding any unnecessary effort to gather or process data he wishes to disregard.

The functional form of the buyer's utility means that his expected payoff from any menu item depends on his posterior mean, $\mathbb{E}[\theta|s]$. Therefore, the marginal distribution of $\mathbb{E}[\theta|s]$ pins down the buyer's expected trade surplus from any menu. This distribution also determines the probability the buyer purchases any menu item, which, in turn, is sufficient for calculating the monopolist's profits and optimal menu. In other words, trade outcomes

⁴We require the menu to be compact to ensure existence of an optimal choice for the buyer.

⁵A twice continuously differentiable function f is strongly convex if $\min_x f''(x) > 0$.

depend only on the marginal distribution of the buyer's posterior mean, and so we identify each signal with the CDF of this marginal.⁶ More precisely, letting $\mathcal{F}$ be the set of all CDFs over Θ , we let the buyer choose any element of $\mathcal{F}$ that can arise as the marginal CDF of $\mathbb{E}[\theta|s]$ for some s . We denote this set by $\mathcal{I}$ and describe it formally below.

As observed by Gentzkow and Kamenica (2016), F is the CDF of the marginal distribution of the buyer's posterior mean for some signal if and only if it is a mean-preserving contraction of the prior, F_0 . Recall that $F \in \mathcal{F}$ is a **mean-preserving spread** of $G \in \mathcal{F}$ (denoted by $F \succeq G$) if and only if

$$\int_{\tilde{\theta} \leq \theta} (F - G)(\tilde{\theta}) d\tilde{\theta} \geq 0 \text{ for all } \theta \in \Theta, \text{ with equality at } \theta = \bar{\theta}.$$

Therefore, one can describe the set of feasible posterior mean distributions via

$$\mathcal{I} = \{F \in \mathcal{F} : F_0 \succeq F\}.$$

We refer to CDFs in $\mathcal{I}$ as **signals**. Given $F \in \mathcal{I}$, we denote the lowest and highest realizations F can generate by $\underline{\theta}_F := \min[\text{supp } F]$ and $\bar{\theta}_F := \max[\text{supp } F]$, respectively.

It turns out to be convenient to describe the set $\mathcal{I}$ via a continuum of inequality constraints. For every CDF F , define the function $I_F : \Theta \rightarrow \mathbb{R}$ as

$$I_F(\theta) := \int_{\tilde{\theta} \in [\underline{\theta}, \theta]} (F_0 - F)(\tilde{\theta}) d\tilde{\theta}.$$

In other words, $I_F(\theta)$ gives the difference between the integral of F_0 and the integral of F over the range $[\underline{\theta}, \theta]$. A CDF F is a signal if $I_F(\theta) \geq 0$ holds for all θ , and $I_F(\bar{\theta}) = 0$. Intuitively, $I_F(\theta)$ measures the degree to which the signal that generates F pools states above and below θ . Indeed, $I_F(\theta) = 0$ if and only if every signal structure that generates F does not pool types strictly below θ with types strictly above θ with positive probability; that is, the signal must separate types above and below θ . We follow Ravid, Roesler, and Szentes (2022) and refer to any θ with $I_F(\theta) = 0$ as **F -separating**, and refer to any θ that is not F -separating as **F -pooling**.

Information acquisition comes at a cost. In general, different information structures generating the same distribution of posterior expectations might come at different costs.

⁶This method of modeling flexible information is common in the information-design literature; see, for example, Gentzkow and Kamenica (2016), Roesler and Szentes (2017), Kolotilin (2018), and Dworczak and Martini (2019).

However, because the buyer’s expected payoff from trade depends only on the distribution of this posterior expectation, F , she would always use the least expensive signal structure that leads to F . In fact, the buyer may even randomize to get F . Thus, we can evaluate the cost of F by the expected cost of the cheapest randomization that generates it, resulting in an indirect cost function, $C : \mathcal{I} \rightarrow \mathbb{R}_+$. We follow Ravid, Roesler, and Szentes (2020) and state our assumptions directly in terms of this C . We assume C is continuous, affine,⁷ and strictly increasing in informativeness; that is, $C(F) > C(F')$ whenever F is a strict mean-preserving spread of F' . In the online appendix, we prove these properties imply the existence of some continuous, strictly convex function $c : \Theta \rightarrow \mathbb{R}_+$ such that

$$C(F) = \int c(\theta) F(d\theta).$$

Moreover, we show it is without loss for c to attain its minimum at θ_0 . In addition, we require c to be thrice differentiable function with a strictly positive and bounded second derivative. In Section 7, we discuss which results extend to the case in which the derivatives of c are unbounded.

After choosing F , the buyer gets to see its realization, $\theta \in \Theta$, and decides whether to purchase, and if so, what item to select from the menu to maximize his expected utility.

To summarize, the game begins with the monopolist choosing a menu. Next, the buyer observes the menu, and chooses what signal $F \in \mathcal{I}$ to acquire. The buyer then sees his signal realization $\theta \in \Theta$, and chooses an item from the monopolist’s menu. We are interested in the menu and signal that maximize the monopolist’s expected profits, subject to the buyer behaving optimally, which exists by the following lemma.

Lemma 1. *A monopolist-optimal menu exists.*

Our timing assumptions mean the buyer’s interim expected payoff is fully determined by his posterior mean. Hence, this mean completely determines the buyer’s decision from the monopolist’s menu. As such, by the revelation principle, it is sufficient to focus on direct revelation mechanisms in which the buyer reports his posterior mean. Such menus

⁷Some readers may be interested in deriving these properties from a more primitive object, such as cost function defined over the distribution of the agent’s posterior belief. To do so, one can assume the cost of a distribution over posteriors is posterior separable (Caplin, Dean, and Leahy, 2022), with a divergence function that depends only on posterior’s mean. Mensch and Malik (2023) provides a revealed preference characterization of such cost functions.

can be described with two maps,

$$Q : \Theta \rightarrow [0, \bar{q}], \quad T : \Theta \rightarrow \mathbb{R}_+,$$

where $Q(\theta)$ and $T(\theta)$ correspond to the quality and transfer pair chosen by a buyer with posterior mean θ . These mappings must satisfy the standard incentive compatibility and individual rationality constraints,⁸

$$\theta Q(\theta) - T(\theta) \geq \theta Q(\theta') - T(\theta') \quad \forall \theta, \theta' \in \Theta, \quad (\text{IC})$$

$$\theta Q(\theta) - T(\theta) \geq 0 \quad \forall \theta \in \Theta. \quad (\text{IR})$$

Usual envelope-style reasoning (Myerson, 1981) delivers that a Q and T satisfy the above two conditions if and only if Q is increasing and

$$T(\theta) = \theta Q(\theta) - \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta}) d\tilde{\theta} - \underline{u}, \quad (1)$$

where $\underline{u} \geq 0$ is the utility granted to the lowest possible type,

$$\underline{u} = \underline{\theta} Q(\underline{\theta}) - T(\underline{\theta}).$$

It follows that $\underline{u}$ and Q are sufficient for pinning down every feasible IC and IR menu. Let $\mathcal{Q}$ be the set of all increasing functions from Θ to $[0, \bar{q}]$. We refer to a $Q \in \mathcal{Q}$ as an **allocation**. For essentially the usual reasons, setting $\underline{u} = 0$ is always optimal for the monopolist. Therefore, the allocation alone pins down the payoffs of both players, and we henceforth omit $\underline{u}$. Given an allocation Q , we let T_Q denote the transfer implied by (1) with $\underline{u} = 0$. This description implies that a type- θ buyer's utility from truthful reporting under Q is

$$V_Q(\theta) := \theta Q(\theta) - T_Q(\theta) = \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta}) d\tilde{\theta}.$$

Given an allocation Q , let $\underline{\theta}_Q$ be the highest θ type that Q excludes, and $\bar{\theta}_Q$ to be the lowest type to which Q gives the highest quality. If no type is excluded (gets the highest quality), set $\underline{\theta}_Q = \underline{\theta}$ ($\bar{\theta}_Q = \bar{\theta}$).⁹

⁸Note that for every menu, every type $\theta \in \Theta$ admits some optimal selection from that menu. Letting $(Q(\theta), T(\theta))_{\theta \in \Theta}$ be this selection then delivers a Q and a T that satisfy (IC) and (IR). Note these constraints are satisfied even for posterior means that do not arise under the buyer's chosen CDF.

⁹Formally, set $\bar{\theta}_Q := \sup [\{\underline{\theta}\} \cup Q^{-1}[0, \bar{q}]]$ and $\underline{\theta}_Q = \inf [\{\bar{\theta}\} \cup Q^{-1}(0, \bar{q}]]$.

We now state the monopolist's problem of choosing a profit-maximizing mechanism. Given an allocation Q , the buyer's utility from using signal $F \in \mathcal{I}$ is given by his expected utility net of information costs. We refer to an allocation-signal tuple (Q, F) as **an outcome**, and say the outcome is **incentive compatible** (IC) if F maximizes the buyer's utility given Q ,

$$F \in \operatorname{argmax}_{\tilde{F} \in \mathcal{I}} \int (V_Q(\theta) - c(\theta)) \tilde{F}(\mathrm{d}\theta). \quad (2)$$

Denote the monopolist's payoff when the buyer reports a signal realization of θ by

$$\pi_Q(\theta) := T_Q(\theta) - \kappa(Q(\theta)) = \theta Q(\theta) - V_Q(\theta) - \kappa(Q(\theta)).$$

Then, we can write the monopolist's expected profit from offering Q when the buyer uses F as $\int \pi_Q(\theta) F(\mathrm{d}\theta)$, and so the monopolist's program is given by

$$\max_{(Q, F)} \int \pi_Q(\theta) F(\mathrm{d}\theta) \text{ s.t. } (Q, F) \text{ is IC.}$$

The goal of this paper is to study the above program.

Before proceeding, we introduce some definitions. Fix an increasing function $\varphi : \Theta \rightarrow [x, y]$. We say φ is **constant around** θ whenever it is constant in some open neighborhood of θ . If φ is not constant around θ , we say that φ is **strictly increasing** at θ .

Binary types Throughout the paper we utilize the case in which F_0 is binary— $|\operatorname{supp} F_0| = 2$ —to help illustrate and provide intuition about the general problem. Formally, we assume F_0 is the CDF with mean θ_0 , and $\operatorname{supp} F_0 = \{\underline{\theta}, \bar{\theta}\}$. In this case, any CDF $F \in \mathcal{F}$ with mean θ_0 is a mean-preserving contraction of F_0 . That is:

$$\mathcal{I} = \left\{ F \in \mathcal{F} : \int \tilde{\theta} F(\mathrm{d}\tilde{\theta}) = \theta_0 \right\}.$$

Finally, the assumption of affine information costs is less restrictive when the prior is binary. With only two states, the posterior mean fully identifies the posterior distribution. Therefore, a function is affine in posterior means whenever it is affine in the posterior distributions themselves. Indeed, for binary values, the affinity assumption is satisfied by the class of posterior-separable information costs (Caplin, Dean, and Leahy, 2022), which are widely used in the applied literature.

3. Cost-Compensating Allocations

In this section, we show that it is without loss of generality to restrict attention to a convenient class of allocations, which we call cost-compensating. This observation, which builds on Dworczak and Martini (2019) duality results, allows us to relate the monopolist problem to two familiar maximization programs: one that involves maximization over allocations, and the other that optimizes over the buyer's information.

We start by characterizing the solution to the buyer's information acquisition problem.

Lemma 2. *Fix an allocation Q . Then, F solves (2) if and only if a Lipschitz continuous and convex function $P : \Theta \rightarrow \mathbb{R}$ exists such that $P \geq V_Q - c$,*

$$P(\theta) = V_Q(\theta) - c(\theta), \text{ for all } \theta \in \text{supp} F,$$

and P is affine on any interval (θ_1, θ_2) of F -pooling types.

The proof of the above lemma relies heavily on Dworczak and Martini's (2019) duality-based approach for solving mean-measurable information design problems.¹⁰ In the language of their paper, P is a price function. Intuitively, one can think of P as the shadow value of producing mean-preserving spreads and contractions that involve θ . The value of information is always convex and if it is strictly convex around θ , the buyer strictly benefits from information acquisition, that is, from producing a local mean-preserving spread. Therefore, optimality implies that whenever such spread is feasible, P must provide the buyer with no incentives to acquire information: in other words, when θ is F -pooling, P must be affine. If, for a given F , a convex and Lipschitz continuous function P is affine on any F -pooling interval, we call it an **F -price**.

Lemma 2 motivates us to restrict our attention to a subset of allocations in which the net value for the buyer, $V_Q - c$, coincides with the associated price P over the relevant range of types. To achieve such net value, the allocation offered by the seller must provide rents V_Q to the buyer that compensate them for the information costs they incur, c , on top of the utility given by the price function. We call such allocations F -cost-compensating.

Definition 1 (Cost-Compensating Allocations). *An allocation Q is **F -cost-compensating***

¹⁰To apply the result to the case where c' is unbounded, one needs to use Dizdar and Kováč's (2020) generalization of Dworczak and Martini (2019). See the Appendix for precise formulation.

(*F*-CC) if an *F*-price exists such that $V_Q \leq P + c$, and

$$V_Q(\theta) = P(\theta) + c(\theta) \quad \text{for all } \theta \in [\underline{\theta}_Q, \bar{\theta}_Q]. \quad (3)$$

Because P is convex and Lipschitz continuous, the usual envelope condition implies that every *F*-CC allocation Q can be rewritten as:

$$Q(\theta) = p(\theta) + c'(\theta), \quad \forall \theta \text{ s.t. } p(\theta) + c'(\theta) \in [0, \bar{q}], \quad (4)$$

for some function $p : \Theta \rightarrow \mathbb{R}$ such that

$$P(\theta) = P(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} p(x) dx, \quad \theta \in \Theta.$$

Note we can choose p so that it satisfies two useful properties. First, because the price function is convex and affine for *F*-pooling types, we can select p to be nondecreasing and constant on every *F*-pooling interval. Second, because the allocation Q must lie between 0 and $\bar{q}$:

$$-c'(\underline{\theta}_F) \leq p(\theta) \leq \bar{q} - c'(\bar{\theta}_F) \quad \text{for all } \theta \in [\underline{\theta}_F, \bar{\theta}_F]. \quad (5)$$

We name any p satisfying the two properties above an *F*-**marginal price**, as it is the derivative of a price function.

The equations above show that, in a cost-compensating allocation, the allocation at type θ is fully determined by the marginal price p and the marginal cost of information c' . This also implies that, given any *F*-marginal price, p , one can construct a unique *F*-CC allocation, using equation (4). We denote such allocation by Q_p .

Economically, *F*-CC allocations require the change in the accumulation rate of the buyer's information rents over $[\underline{\theta}_F, \bar{\theta}_F]$ to satisfy two bounds: the change must be at least as high as the change in his marginal learning costs, and can be strictly higher only over *F*-separating regions. These two bounds are constructed in order to satisfy the buyer's incentive to acquire *F*. The lower bound guarantees the net utility is not concave, meaning that the buyer does not benefit from conducting mean-preserving contractions. The upper-bound dissuades the buyer from conducting mean-preserving spreads by ensuring that the net utility is only strictly convex over intervals where such spreads are infeasible.

Note that to induce a signal *F*, the seller must provide a minimal degree of quality differentiation. In particular, equation (4) implies

$$Q(\bar{\theta}_F) - Q(\underline{\theta}_F) \geq c'(\bar{\theta}_F) - c'(\underline{\theta}_F). \quad (6)$$

We use this inequality repeatedly in the sequel.

Example. (Binary types and Variance reduction) In Figure 1, we illustrate the construction of an F -CC allocation from an incentive compatible outcome (Q, F) . We assume the prior F_0 is binary, and that information costs are quadratic:

$$c(\theta) = \frac{1}{2}(\theta - \theta_0)^2.$$

Under this specification, the cost of learning a signal is proportional to the reduction in the variance of F_0 obtained by observing that signal.

The left panel plots the net utility of the buyer, $V_Q - c$, for some allocation Q , which is pictured in the right panel. The optimal signal F for the buyer is a binary distribution with $\text{supp} F = \{\underline{\theta}_F, \bar{\theta}_F\}$. Following Lemma 2, the price function P in the left panel certifies optimality of F given the allocation Q . Note that P must be affine because, for a binary prior, any interval $(\theta_1, \theta_2) \subset (\underline{\theta}, \bar{\theta})$ is F -pooling as long as $F \neq F_0$. In other words, for binary values, Lemma 2 consists of standard concavification of the buyer's objective function (Kamenica and Gentzkow, 2011).

On the right panel, we construct an F -CC allocation from the derivative of the price P , which is a constant F -marginal price p . We start by plotting the marginal cost of information, c' , which is linear. Following equation (4), we obtain the F -CC allocation Q_p by summing p to c' . Because, for high types, that sum is higher than our quality upper-bound, $\bar{q}$, the allocation Q_p is capped at the upper bound. Similarly, the sum $c' + p$ is negative for low types, so we impose the lower-bound of zero.

If instead of the signal above the seller aimed to induce full information—i.e. $F' = F_0$ —, then, every interval $(\theta_1, \theta_2) \subseteq \Theta$ would be F' -separating. In that case, an allocation Q is F' -CC if and only if $Q - c'$ is increasing whenever $Q(\theta) - c'(\theta) \in (0, \bar{q})$. ■

As the example above highlights, by constructing the F -CC allocation Q_p from a price function that makes (Q, F) incentive-compatible, we obtain an F -CC allocation Q_p that coincides with Q in the support of F . Moreover, one can show this F -CC allocation gives the buyer a weakly lower utility for every signal realization—that is, the value function, V_{Q_p} lies weakly below V_Q over the support of F . It follows that F -CC allocations are without

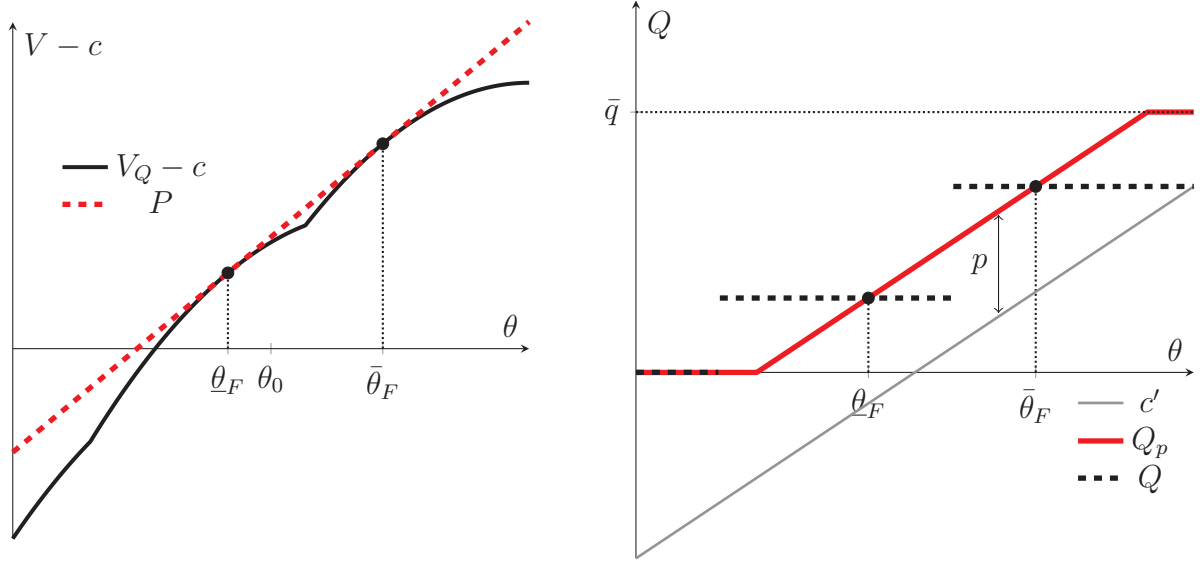


Figure 1: From an IC outcome to an F -CC allocation

loss for evaluating optimality and efficiency of the resulting allocation: the monopolist can replace any allocation with an F -CC one without foregoing any profits or changing the quality it provides to the buyer's realized types. The next Theorem, which is the main result of this section, formalizes this observation.

Theorem 1. *F solves the buyer's information acquisition problem for any F -CC allocation. Moreover, if F solves the buyer's information acquisition problem for allocation $\hat{Q}$, there exists an F -CC allocation Q with:*

- (i) $Q(\theta) = \hat{Q}(\theta)$ for all $\theta \in \text{supp} F$;
- (ii) $V_Q(\theta) \leq V_{\hat{Q}}(\theta)$ for all $\theta \in \text{supp} F$.

That every F -CC allocation Q makes F a solution to the buyer's information acquisition problem follows from Lemma 2. To conclude the converse, we use Lemma 2 to obtain an F -price function P that certifies the optimality of F , show that the associated $Q_p = Q$ over the support of F , and prove that $V_{Q_p} \leq V_{\hat{Q}}$.

The restriction to cost-compensating allocations is useful because it enables us to reason about the monopolist's problem "one dimension at a time". Specifically, we note the monopolist optimal allocation must solve a familiar-looking one-dimensional allocation design program, whereas the optimal signal must solve a particular mean-measurable information design problem. Thus, one can use familiar techniques to make inferences about the monopolist's optimal outcome.

Corollary 1 below outlines the above-mentioned programs. The starting point for both programs is a monopolist optimal outcome (Q, F) . The allocation design problem is based on the observation that, holding F fixed, the allocation Q must maximize the monopolist's profit across all allocations for which the buyer find optimal to acquire signal F . However, since each such allocation is outcome equivalent (under F) to some F -CC allocation, there is no loss in focusing solely on F -CC allocations. The allocation design program then follows from observing that F -CC allocations are indexed by their corresponding F -marginal price functions. Denote the set of all F -marginal price functions by $\mathcal{P}(F)$.

A similar logic delivers the information design program: holding Q fixed, the signal F must maximize the monopolist's profit across all signals that are optimal for the buyer given Q . In particular, if Q is F -CC, the signal F must be better for the monopolist than any other signal for which Q is cost-compensating. Given a cost-compensating mechanism Q , we let $\mathcal{I}(Q)$ be the set of signals for which Q is cost-compensating.¹¹ It is straightforward to verify that $\mathcal{I}(Q)$ is a compact and convex set.

Corollary 1. *Suppose (Q, F) is monopolist optimal. For any F -CC allocation $\tilde{Q}$, the outcome $(\tilde{Q}, F)$ is monopolist optimal if and only if $p_{\tilde{Q}}$ solves*

$$\max_{p \in \mathcal{P}(F)} \int \pi_{Q_p}(\theta) F(d\theta). \quad (7)$$

Moreover, if Q is F -CC, then $(Q, \tilde{F})$ is monopolist optimal if and only if $\tilde{F}$ solves

$$\max_{\tilde{F} \in \mathcal{I}(Q)} \int \pi_Q(\theta) \tilde{F}(d\theta). \quad (8)$$

Thus, once we know the monopolist optimal signal, finding the monopolist optimal allocation amounts to finding the F -marginal price that solves the program (7). Similarly, once we know the monopolist optimal cost-compensating allocation, finding the optimal signal amounts to maximizing the monopolist's profit across all signals for which that allo-

¹¹That is, $\mathcal{I}(Q)$ is the set of all signals $F \in \mathcal{I}$ such that $\text{supp}(F) \subseteq [\underline{\theta}_Q, \bar{\theta}_Q]$, and $I_F(\theta) > 0$ only if p_Q is constant around θ .

cation is CC.¹² Next, we use these two programs to make inferences about the monopolist optimal outcome.

4. Downward Quality Distortions

This section studies the inefficiencies in the optimal menu. We prove that the monopolist distorts quality downward for all of the buyer’s realized types. Moreover, we provide sufficient conditions for this result to hold strictly, therefore reversing the celebrated “no distortion at the top” observation from standard screening with exogenous information. Concretely, we prove that the downward distortion is strict for all realizations below $\bar{\theta}$, the highest possible under the prior. Thus, whenever the optimal outcome involves a signal whose highest realization lies strictly below $\bar{\theta}$, one gets that quality is strictly distorted downwards for all posterior means that can realize under the buyer’s chosen signal structure. We then prove that the optimal signal’s highest realization is strictly below $\bar{\theta}$ as long as information costs are sufficiently steep.

Theorem 2. *Every monopolist optimal outcome (Q^*, F^*) admits an allocation Q such that $Q = Q^*$ holds F^* -almost surely, (Q, F^*) is monopolist optimal, and every $\theta \in \text{supp } F^*$ has $\theta \geq \kappa'(Q(\theta))$, with equality holding if and only if $\theta = \bar{\theta}$.*

The above theorem states that the monopolist underprovides quality to all realized types except for $\bar{\theta}$, whose allocation is efficient. An immediate implication is that all realized types are underprovided if and only if $\bar{\theta} \notin \text{supp } F$. Theorem 2 highlights a fundamental difference between the exogenous and endogenous information cases. When information is exogenous, Mussa and Rosen’s (1978) analysis shows that quality is distorted downward at all types *except* the highest one. Our results demonstrate that, when information is endogenous, quality is still distorted downwards, but the “no distortion” at the top observation remains if and only if the highest type under the buyer’s chosen signal is the highest possible type when the buyer is fully informed. Proposition 1 below finds a sufficient condition

¹² In particular, whenever F_0 has a density, it is without loss of optimality to focus on signals that have a bipooling structure (Kleiner, Moldovanu, and Strack, 2021; Arieli et al., 2023). The reason is that the signals for which the allocation is cost-compensating is equivalent to the set of signals that maximize the expectation of $V_Q - c$. Note, however, that the set of maximizers of an affine function over a convex set is a face of that set, and the extreme points of a face are a subset of the extreme points of the original convex set. Therefore, the extreme points of the set of signals that are IC for the buyer for a given allocation Q is a subset of the extreme points of all signals, all of which take a bipooling structure by Kleiner, Moldovanu, and Strack (2021) and Arieli et al. (2023). We note that we do not rely on this fact for most of our analysis.

for the buyer to choose to not learn $\bar{\theta}$ and, consequently, for the failure of “no distortion of the top”.

The theorem’s logic relies on two economic forces. The first force is familiar from Mussa and Rosen’s (1978) model of monopolistic screening with exogenous information. In that model, the monopolist distorts the allocation of lower types downwards in order to reduce the information rents given to higher types. Using the optimal allocation program (7) from Corollary 1, we show a similar concern arises in the current setting for any type at which the monopolist’s chosen marginal price is strictly increasing.

The second force comes from the endogenous nature of the buyer’s information. By having the buyer learn different signals, the monopolist can trade-off the benefit of having different type realizations. Whether the monopolist benefits from a marginal increase to the buyer’s realized type depends on whether or not that type’s quality is above or below the efficient level. Roughly speaking, an increase in the buyer’s posterior-realization affects the monopolist’s profits in two ways: First, it changes the total surplus generated by the transaction, $\theta Q(\theta) - \kappa(Q(\theta))$, and second, it impacts the information rents the monopolist cedes to the buyer, $V_Q(\theta)$. By the envelope theorem, the change in the buyer’s information rents is second order, meaning the difference in the available social surplus dominates. Heuristically:

$$\pi'_Q(\theta) = \underbrace{q(\theta) - V'_Q(\theta)}_{0 \text{ by envelope}} + (\theta - \kappa'(Q(\theta))) Q'(\theta).$$

Thus, since the provided quality is increasing with the buyer’s type, a small increase in θ raises the seller’s profits if and only if quality is under-provided at θ . Thus, π_Q increases at θ if and only if $Q(\theta)$ is below the efficient level.¹³

To see the relevance of this observation to Theorem 2, take any $\theta \in \text{supp } F$ with $\theta < \bar{\theta}$ around which p_Q is constant. Let $\tilde{\theta}$ be the highest type below θ at which p_Q is strictly increasing. Suppose, for the sake of intuition, that both θ and $\tilde{\theta}$ occur with positive probability under F . Consider the change in profits due to a small mean-preserving contraction that replaces each realization of θ with a slightly lower realization, and each realization of $\tilde{\theta}$ with a slightly higher one. By choice of $\tilde{\theta}$, p_Q must be constant over $[\tilde{\theta}, \theta]$, meaning this contraction is incentive compatible for the buyer. Therefore, because F is optimal, the total

¹³More formally, suppose Q is an F -CC allocation such that p_Q is constant on $[\theta_1, \theta_2]$. One can show that, in this case, the right derivative $\pi'_{Q+}(\theta_1)$ and the left derivative $\pi'_{Q-}(\theta_2)$ are well defined, and that $\pi'_{Q+}(\theta)$ ($\pi'_{Q-}(\theta)$) is strictly positive (negative) if and only if $Q(\theta)$ is strictly below (above) the efficient level.

effect of this contraction must be negative. Recall, however, that because p is increasing at $\tilde{\theta}$, the quality $Q(\tilde{\theta})$ must be below its efficient level. Hence, slightly increasing $\tilde{\theta}$ must strictly increase the seller's profit, as explained above. Since this contraction cannot be profitable, we get that the slight reduction in θ must strictly decrease the monopolist's profits, and so θ 's quality must be strictly below its efficient level as well.

We next establish sufficient conditions for all of the inequalities in Theorem 2 to be strict: that is, for there to be a downward distortion even at the top of the support of F . In particular, we show that quality is strictly under-provided to the highest signal realization whenever the scale of information costs is sufficiently large.

It seems intuitive that one can also establish the converse, namely that when the scale of information costs is low, the highest type in the support is $\bar{\theta}$, and so there is no distortion at the top. Our result establishes this converse, but only under the condition below. We note this condition also plays a role for the comparative statics in Section 5.

Condition 1 (Finiteness and Regularity). *We say that the type distribution F_0 is **Finite and Regular** (FR) if it has finite support— $\text{supp} F_0 = \{\theta_1, \dots, \theta_L\}$, where $\theta_1 < \theta_2 < \dots < \theta_L$ —, and its virtual values,*

$$v_i := \theta_i - \frac{1 - F_0(\theta_i)}{F_0(\theta_i) - F_0(\theta_{i-1})}(\theta_{i+1} - \theta_i), \quad i \leq L,$$

are strictly increasing in i , and are strictly larger than $\kappa'(0)$ for all $i \geq 2$.

The assumption on virtual values is equivalent to the seller finding it optimal to fully separate the buyer's types under full learning. Moreover, one can show Condition 1 implies that, when learning is free, the seller finds it uniquely optimal to induce the buyer to acquire full information. For an intuitive explanation, note that when information is free, a buyer's optimal signal must result in an ex-post optimal selection from the monopolist's menu. Therefore, when information is costless, the seller's profits from any menu must be the profit she gets under perfect learning. Hence, the best the monopolist can do under free-information is to offer the buyer the optimal perfect-learning menu. Condition 1 ensures this menu is fully-separating, which in turn implies that full-learning is the uniquely optimal monopolist signal when information is free—this property is guaranteed by the condition's “regularity” portion. The assumption that the value distribution is finite guarantees that perfect learning remains optimal for the monopolist when the scale of information costs is sufficiently low.

Proposition 1. *Suppose information costs satisfy $C(F) = \gamma \int c(\theta)F(d\theta)$, for $\gamma > 0$. Then:*

- (i) *When γ is sufficiently large, $\bar{\theta} \notin \text{supp } F^*$.*
- (ii) *If F_0 is FR and γ is sufficiently small, $\bar{\theta} \in \text{supp } F^*$.*

Part (i) of the proposition establishes sufficient conditions for the monopolist optimal outcome to provide the highest realized type a quality that is strictly below efficient. When γ parametrizes the scale of marginal information costs, the result states that all qualities are strictly below efficient for high γ . Specifically, letting $\bar{q}_e$ be the efficient quality level for type $\bar{\theta}$ (i.e., $\bar{q}_e$ solves $\kappa'(\bar{q}_e) = \bar{\theta}$), one has that $\bar{\theta} \notin \text{supp } F^*$ whenever

$$\gamma \geq \frac{\bar{q}_e}{c'(\bar{\theta}) - c'(\theta_0)}.$$

Intuitively, whenever the above inequality holds, the monopolist can make a signal F with $\bar{\theta}_F = \bar{\theta}$ incentive compatible only if she upward distorts the quality she provides $\bar{\theta}$, which is suboptimal. For an explanation, recall that by Theorem 1, the monopolist can make F incentive compatible only if she provides a minimal amount of quality differentiation, described by equation (6). Specializing this equation to the case where costs are parameterized by γ and observing that $\bar{\theta}_F = \bar{\theta}$ only if $\underline{\theta}_F < \theta_0$ gives the following inequality for any Q that makes F incentive compatible:

$$Q(\bar{\theta}_F) \geq Q(\bar{\theta}_F) - Q(\underline{\theta}_F) \geq \gamma \left(c'(\bar{\theta}_F) - c'(\underline{\theta}_F) \right) > \gamma \left(c'(\bar{\theta}_F) - c'(\theta_0) \right).$$

When γ surpasses the above-mentioned threshold, the inequality's right-hand side becomes larger than $\bar{q}_e$, meaning that Q makes F IC only if Q gives $\bar{\theta}_F$ too high of a quality. Since quality must be under-provided to all realized types at the optimum, F cannot be part of a profit maximizing outcome.

The proposition's part (ii) provides sufficient conditions for our model to exhibit the classic "efficiency at the top" result. These conditions require the true type distribution F_0 to be finite and regular, and for learning costs to be sufficiently small. This part follows from the proof of Theorem 4. In that result, we show that when F_0 is finite and regular, the monopolist finds full-learning optimal for all sufficiently low γ . It then follows from Theorem 2 that the monopolist optimal outcome induces the buyer to obtain the efficient quality whenever he learns that his type is the highest possible—i.e., when $\theta = \bar{\theta}$.

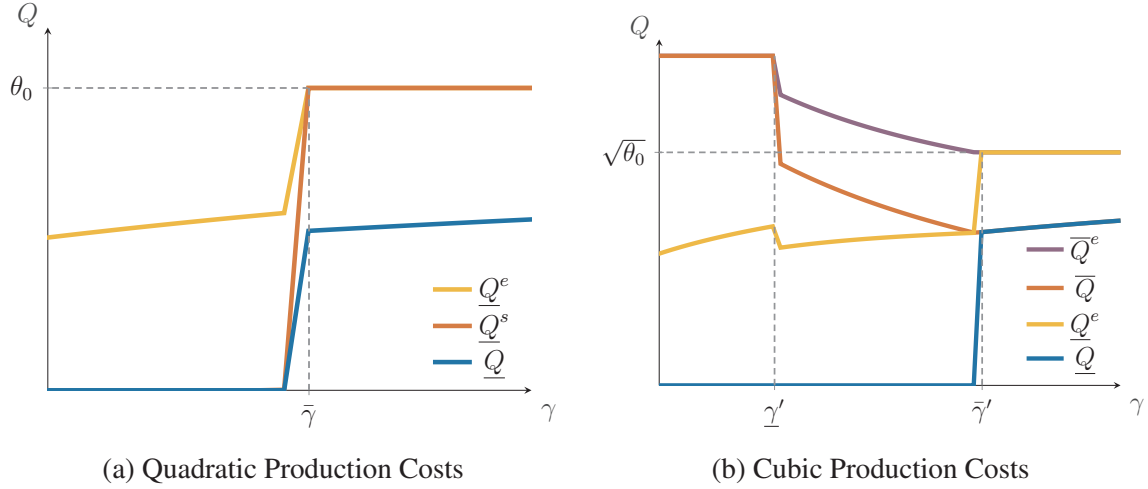


Figure 2: Optimal Qualities for Variance-Reduction Information Costs

Example 1. (Binary types and Variance reduction) We now return to our example with binary types and quadratic information costs to illustrate Theorem 1. Throughout, we assume that there is a constant γ which scales the quadratic information costs: $C(F) = \frac{\gamma}{2} \int (\theta - \theta_0)^2 F(d\theta)$. When F_0 is binary, any optimal signal is at most binary: it contains one or two points in its support. We then denote an optimal signal F by its support $\text{supp} F = \{\theta_L, \theta_H\}$, with $\theta_L \leq \theta_0 \leq \theta_H$. Moreover, we denote the associated qualities each of these types receives in the optimal allocation as $\underline{Q}$ and $\overline{Q}$, respectively. Given the binary signal F , we denote by $(\underline{Q}^e, \overline{Q}^e)$, the efficient qualities for these types, which would obtain under symmetric information, and by $(\underline{Q}^s, \overline{Q}^s)$ the qualities these types would receive in the second-best, that is under asymmetric, but exogenous information as in Mussa and Rosen's (1978).

In what follows, we illustrate these optimal qualities using two examples of production costs. We start describing the optimal signal F when production costs are quadratic: $\kappa(q) = \frac{q^2}{2}$. In this case, when the level of information costs, γ , is lower than some $\bar{\gamma}$, one of the signal realizations fully reveals the high type, $\theta_H = \bar{\theta}$. The other realization of the signal pools some realizations of the high type with realizations of the low type, leading to a posterior $\theta_L \in [\underline{\theta}, \theta_0]$. As γ grows, this low signal deteriorates monotonically, increasing the low posterior θ_L and the probability that this posterior occurs, until θ_L reaches θ_0 at $\gamma = \bar{\gamma}$. At that point, the low signal happens with probability 1 and the optimal signal becomes uninformative— $\text{supp} F = \{\theta_0\}$ —and remains so for any higher γ . In fact, this is a general property in our model: for any choice of information costs and production costs,

there is a high enough γ such that the optimal signal is uninformative for all higher levels of the cost.

The left panel in Figure 2 depicts the optimal qualities for quadratic production costs as γ varies. Because the high signal is fully revealing, the high type receives efficient quality by Theorem 2, and all the distortions in this example come from the allocation of the low type. In the Figure, we plot the allocation for that type with endogenous information, compared to the one that would hold with asymmetric but exogenous information, $\underline{Q}^s$, and symmetric information, $\underline{Q}^e$. The distance between $\underline{Q}^e$ and $\underline{Q}^s$ is the standard screening distortion: it stems from asymmetric information exclusively and, therefore, it vanishes when $\gamma \geq \bar{\gamma}$ and information becomes symmetric in the optimal. The difference between $\underline{Q}^s$ and $\underline{Q}$, in turn, is the distortion given by information acquisition: the result of the agent's ability to acquire information that is not prescribed by the principal. This difference does not disappear when information is symmetric: the distortion persists and only vanishes asymptotically, as γ approaches infinity.

It is worth noting that three features of the optimal outcome for quadratic information costs do not hold in general. First, it is not the case that the optimal signal is either uninformative or reveals the high valuation—or any of the valuations, for that matter. Second, it is not always true that the information structure becomes progressively less informative as γ grows. Finally, because the high type is often not revealed, both qualities can be distorted relative to their efficient levels even when the optimal signal has two posteriors. The right panel on Figure 2 illustrates these observations with the optimal qualities for cubic production costs: $\kappa(q) = \frac{q^3}{3}$. Note that for $\gamma \in (\underline{\gamma}', \bar{\gamma}')$ the high type is not fully disclosed: indeed, the optimal quality for the high posterior is lower than the efficient one $\bar{Q} < \bar{Q}^e$. The low state is not perfectly revealed either. Moreover, by observing how $\underline{Q}^e$ varies, one can conclude that the low posterior becomes lower after $\underline{\gamma}'$, which implies that the quality of the low signal improves and the optimal information structure does not become less informative with γ .¹⁴ Finally, the qualities given to both types are strictly lower than in the symmetric information benchmark. ■

¹⁴Because the efficient allocation for posterior θ is defined by $\theta = \kappa'(Q)$, the only way that the allocation decreases is if θ itself is lower.

5. Comparative Statics

In this section we study how the scale of information costs affects consumer surplus and seller's profits. Several mechanisms influence the interplay between costs and surplus. First, higher information costs mechanically reduce surplus for any fixed signal. Second, because information allows the seller to better tailor her offers to the buyer, it has a productive role and affects total surplus. Finally, the value of different deviations for the buyer depends on information costs. Therefore, obtaining global comparative statics predictions that hold regardless of the shape of information and production costs is difficult. Nevertheless, the results below show that the third mechanism, how the value of deviations varies with information costs, dominates when information costs are very small or very large.

Formally, let information costs be

$$C(F) = \gamma \int c(\theta) F(d\theta),$$

where $\gamma > 0$ parametrizes the scale of information costs. The main results of this section describe how profits, U_S and consumer surplus U_B vary for small and large γ .

Theorem 3. *There exists γ_h such that:*

- (i) F_γ is uninformative for $\gamma \geq \gamma_h$ (i.e., $F_\gamma = \mathbf{1}_{[\theta_0, \infty)}$).
- (ii) U_S strictly increases in γ over $[\gamma_h, \infty)$.
- (iii) U_B strictly decreases in γ over $[\gamma_h, \infty)$.

Theorem 3 shows that, when information costs are large, the seller's profits and the consumer surplus move in opposite direction of γ . The intuition for this result follows from how the best deviation for the buyer changes with the level of information costs. First, part (i) shows that when costs are sufficiently high, the seller finds it optimal to induce no information acquisition. Therefore information is symmetric in equilibrium, and only one level of quality is chosen with positive probability. To induce no learning, the seller must dissuade the agent from acquiring any information by providing positive rents at the equilibrium contract. As the buyer's costs rise, the value of information acquisition drops, and his deviation becomes costlier. The rents required to prevent learning change accordingly: the seller benefits and the buyer loses.

We now establish comparative statics for low levels of information costs. Before stating the result, define the function $\tilde{c} : \Theta \rightarrow \mathbb{R}$, for any $\theta \in [\theta_J, \theta_{J+1}]$ as:

$$\tilde{c}(\theta) = c(\underline{\theta}) + \sum_{j < J} c'(\theta_j)(\theta_{j+1} - \theta_j) + c'(\theta_J)(\theta - \theta_J).$$

The function $\tilde{c}$ is a linear-by-parts approximation of the information cost, c , taking the slope of c at the points in the support of F_0 . Because information costs are convex, $\tilde{c}$ always lies below c . The finer the support of F_0 , the more precise this approximation is. Figure 3 illustrates these observations by plotting information costs against two approximations for different numbers of points in the support of F_0 . In the figure, c is quadratic, and we consider two distributions of types. The first one gives rise to $\tilde{c}_2$, with $\text{supp} F_0 = \{0, 1\}$. On its turn, $\tilde{c}_3$ is generated by a distribution F_0 with three points in the support, 0, .3 and 1. Our next result shows that, for low levels of information costs, changes in consumer surplus are connected to $\tilde{c}$.

For intuition about $\tilde{c}$, consider a distribution of types with binary support, $\text{supp} F_0 = \{\underline{\theta}, \bar{\theta}\}$. Start from the fully informative signal, which perfectly reveals whether the realization is $\underline{\theta}$ or $\bar{\theta}$. Consider a small deviation from full information: with a very small probability, the signal does not disclose some realizations of $\bar{\theta}$, and instead pools them with $\underline{\theta}$. We argue that $\tilde{c}$ is related to the reduction in information costs from implementing this deviation. Indeed, when the pooling probability is infinitesimal, it affects expected information costs through two distinct channels. First, pooling slightly raises the low posterior, so average costs change by $c'(\underline{\theta})$. Second, the posteriors must still average to the prior mean, so the change in the low posterior also changes the probabilities assigned to each posterior, which in turn move average costs in proportion to $-\frac{c(\bar{\theta}) - c(\underline{\theta})}{\bar{\theta} - \underline{\theta}}$. Putting the two effects together, the first-order change in costs relative to full revelation is proportional to

$$c(\underline{\theta}) - c(\bar{\theta}) + c'(\underline{\theta})(\bar{\theta} - \underline{\theta}) = \tilde{c}(\bar{\theta}) - c(\bar{\theta}).$$

For any finitely-supported distribution, F_0 , $\tilde{c}$ can be interpreted analogously at each support point: for every adjacent pair $\{\theta_j, \theta_{j+1}\}$ (with $J < L$), consider the same type of deviation that, with small probability, pools the higher type with the lower one, and define $\tilde{c}$ by the corresponding local first-order cost comparison. The next result shows that $\tilde{c}$ is central to how consumer surplus changes with information costs.

Theorem 4. *Let Condition 1 hold—that is, F_0 is FR. A $\gamma_l > 0$ exists such that:*

- (i) *For all $\gamma \in [0, \gamma_l]$, $F_\gamma = F_0$, and $Q_\gamma(\theta) = Q_0(\theta)$, for all $\theta \in \text{supp} F_0$*

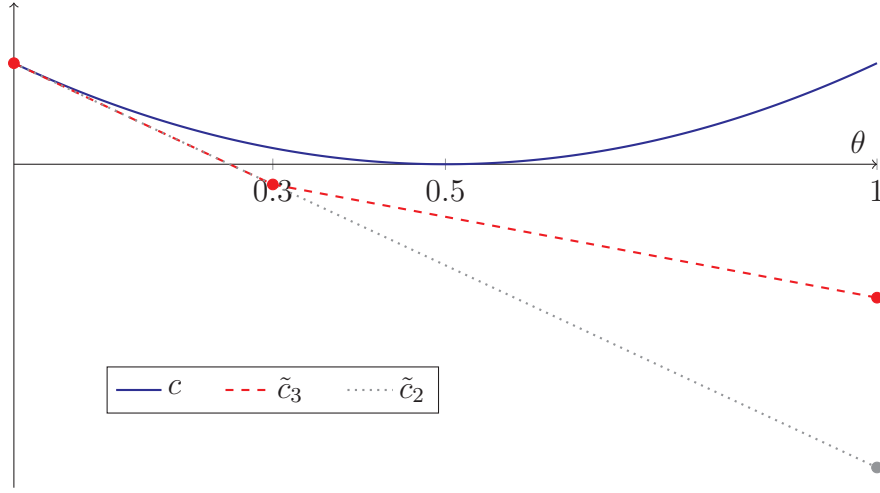


Figure 3: Illustration of $\tilde{c}$

(ii) U_S is strictly decreasing in γ over $[0, \gamma_l]$.

(iii) U_B is strictly increasing (decreasing) in γ over $[0, \gamma_l]$ whenever

$$\int \tilde{c}(\theta) F_0(d\theta) < (>) 0.$$

The first result in Theorem 4 is that, for small enough information costs, the seller finds it optimal to induce full information. Moreover, her menu has the same offered qualities as under exogenous information, with type distribution F_0 . Parts (ii) and (iii) establish comparative statics for profits and consumer surplus, respectively: profits are decreasing for small costs, whereas consumer surplus could move in either direction. The intuition for Theorem 4 is again that the attractiveness of the best deviation for the buyer changes with information costs. Because the seller offers the same quality and induces full information for $\gamma < \gamma_l$, the production surplus is kept fixed, and aggregate surplus decreases only due to information costs, as the buyer acquires the same signal, but now at a higher cost. Clearly, because acquiring signal F_0 is costly, the buyer must be discouraged from deviating to acquire less information. As costs increase, any deviation to acquire less information becomes more tempting, and therefore the seller must spend more resources to dissuade it: hence, seller's profits drop.

The effect on consumer surplus is more subtle. It turns out that the compensation that a buyer of type θ_i must receive so to find it optimal to acquire full information is proportional to the difference $c(\theta_i) - \tilde{c}(\theta_i)$. As we have previously argued, this compensation is the cost

differential associated from deviating from full learning towards a slight pooling signal for θ_i : a signal that, with small probability, pools θ_i with θ_{i-1} instead of disclosing it. The expected value of this compensation is always positive, because $\tilde{c}$ underestimates c , implying that gross consumer surplus grows. But when one discounts information costs, the consumer might be worse off. Intuitively, because the consumer benefits from information himself, the seller may be able to force him to share some of the cost of full learning.

Corollary 2. *Suppose F_0 has binary support. Then U_B is strictly increasing in γ for all sufficiently small γ .*

6. Steep Learning Costs and Simple Menus

If buyers are fully informed and F_0 has a continuous distribution, Mussa and Rosen's (1978) analysis shows the optimal menu is usually quite complicated, containing many alternatives. This section shows that the same is not generally true under costly learning. In particular, we show that whenever learning costs are steep, the optimal menu contains at most two purchasing options.

We refer to an outcome (Q, F) that is implementable with at most two purchasing options as a **binary-menu** outcome. Notice that (Q, F) is a binary-menu outcome whenever F has binary support. The reason is that, restricting the monopolist's menu to include only those alternatives that the buyer chooses with positive probability has no impact on the buyer's decisions as to what to learn and what to buy. Consequently, the monopolist can always maximize profits with a menu that contains weakly less options than the number of realizations in the support of the buyer's signal. More generally, (Q, F) is a binary-menu outcome whenever $Q(\text{supp } F) \setminus \{0\}$ has two or less elements.

Proposition 2 below shows that when learning costs are steep, the monopolist implements a binary-menu outcome. To precisely state what we mean by "steep learning-costs," we need to define the notion of a cutoff signal. A signal $F \in \mathcal{I}$ is a **cutoff signal** if it has one or two realizations in its support and $I_F(\theta^c) = 0$ for some $\theta^c \in (\underline{\theta}, \bar{\theta})$. Intuitively, a cutoff signal is induced by the buyer learning whether or not θ is above θ^c . We say learning costs satisfy the **steep-slope condition** if every cutoff signal F has $c'(\bar{\theta}_F) - c'(\underline{\theta}_F) > \bar{q}_e$

(recall that $\bar{q}_e$ is the efficient quality level for type $\bar{\theta}$).¹⁵

Proposition 2. *Suppose learning costs satisfy the steep-slope condition. Then some binary-menu outcome is monopolist optimal.*

We now describe the proposition's proof. First, by Theorem 1, a signal F can be made IC under some Q if and only if the allocation Q is F -almost surely equal to some F -CC allocation. Second, given an F -CC allocation Q , one must have $Q(\bar{\theta}_F) - Q(\underline{\theta}_F) \geq c'(\bar{\theta}_F) - c'(\underline{\theta}_F)$ —see equation 6. Third, Theorem 2 implies the monopolist optimal outcome (Q, F) never gives a quality above $\bar{q}_e$ to any realized type, meaning that $\bar{q}_e \geq Q(\bar{\theta}_F) - Q(\underline{\theta}_F)$. Therefore, no F for which $c'(\bar{\theta}_F) - c'(\underline{\theta}_F) > \bar{q}_e$ can be part of a monopolist optimal outcome. Fourth, mean preserving spreads make the most extreme signal realizations more extreme; that is, $F \succeq G$ only if $\underline{\theta}_F \leq \underline{\theta}_G \leq \bar{\theta}_G \leq \bar{\theta}_F$.¹⁶ Because c' and the efficient quality are both strictly increasing, we obtain that when learning costs satisfy the steep-slope condition, mean preserving spreads of cutoff signals must provide a quality that is above efficient, and hence cannot arise at the optimum. The fifth and final observation (which we prove in the appendix) is that any signal with more than two signal realizations is either a mean-preserving spread of a cutoff signal, or can be replaced with an optimal signal with binary support, thereby delivering a binary-menu outcome.

Of particular interest are outcomes that can be implemented via a menu with a single-purchasing option. We refer to such outcomes as **posted-price outcome**, because they can be implemented with a single quality sold at a fixed price. Note that a sufficient condition for (Q, F) to be a posted-price outcome is for $Q(\text{supp}(F))$ to have only one non-zero element—that is, for $\text{supp}(F) \setminus \{\underline{\theta}_Q\}$ to be a singleton.

Corollary 3 below provides conditions for optimality of a posted-price outcome. This condition relies on the following definition. For every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, let $\hat{\pi}_{\underline{\theta}^*} : [\underline{\theta}^*, \bar{\theta}] \rightarrow \mathbb{R}$ be the function given by

$$\hat{\pi}_{\underline{\theta}^*}(\theta) = \theta(c'(\theta) - c'(\underline{\theta}^*)) - \kappa(c'(\theta) - c'(\underline{\theta}^*)) - [c(\theta) - c(\underline{\theta}^*)].$$

To understand the motivation for this definition, suppose Q is an F -CC allocation whose

¹⁵We remark here that the steep-slope condition implies the condition Proposition 1 uses to establish strict distortion at the top, namely $c'(\bar{\theta}) - c'(\theta_0) \geq \bar{q}_e$. For an explanation, note that for any cutoff signal F with θ^c , has $\underline{\theta}_F \leq \theta_0$ and $\theta^c \leq \bar{\theta}_F$. Hence, one has $\bar{\theta}_{F_n} \rightarrow \bar{\theta}$ for any sequence of cutoff signals F_n with $\theta_n^c \rightarrow \theta^c$. Therefore, if c has steep slopes, one must have $\bar{q}_e \geq c'(\bar{\theta}_{F_n}) - c'(\underline{\theta}_{F_n}) \geq c'(\bar{\theta}_{F_n}) - c'(\theta_0) \rightarrow c'(\bar{\theta}) - c'(\theta_0)$. In other words, the steep slope condition is sufficient for existence of strict distortion at the top.

¹⁶See Lemma 9 in Ravid, Roesler, and Szentes (2022).

induced marginal price p_Q is constant. Then π_Q equals to $\hat{\pi}_{\underline{\theta}_Q} + c(\underline{\theta}_Q)$ on $[\underline{\theta}_Q, \bar{\theta}_Q]$. In other words, one can use the curvature of $\hat{\pi}$ to learn about the curvature of π_Q . Corollary 3 below utilizes this relationship.

Corollary 3. *A single-quality outcome is monopolist optimal if learning costs satisfy the high steepness condition and one of the following two conditions hold:*

- (i) *If $\hat{\pi}_{\underline{\theta}^*}$ is concave for every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, then a monopolist optimal outcome (Q, F) exists such that F is uninformative, and so (Q, F) is a single-quality outcome.*
- (ii) *If $\hat{\pi}_{\underline{\theta}^*}$ is convex for every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, then a monopolist optimal outcome (Q, F) exists such that $\text{supp}(F) = \{\underline{\theta}_Q, \bar{\theta}_Q\}$, and so (Q, F) is a single-quality outcome.*

Next we explain the corollary's proof. As discussed above, when learning costs are steep, all monopolist-optimal outcomes involve signals that do not separate any types—that is, (Q, F) is optimal only if $I_F(\theta) > 0$ for all $\theta \in (\underline{\theta}, \bar{\theta})$. Thus, if an optimal Q is F -CC, the induced marginal price of Q must be constant. Therefore, the monopolist per-realization expected profit, π_Q , would be equal to $\hat{\pi}_{\underline{\theta}_Q} + c(\underline{\theta}_Q)$ over the interval $[\underline{\theta}_Q, \bar{\theta}_Q]$. The desired result then follows from the information design problem in Corollary 1. Specifically, whenever the conditions of part (i) hold, the program (8) admits an uninformative solution, whereas the conditions of part (ii) imply (8) admits a binary solution supported on $\{\underline{\theta}_Q, \bar{\theta}_Q\}$. In both cases, one gets a single-quality monopolist-optimal outcome—provided it is optimal for the monopolist to serve some consumers, which is guaranteed by our assumption that $\kappa'(0) < \theta_0$.

The logic underlying Corollary 3 is reminiscent of the proof of Theorem 3 in Mensch (2022). Within our setting, one can interpret that theorem as providing sufficient conditions for a single-quality menu to be optimal when $\kappa(q) = 0$ for all q and the state is binary. And, indeed, the last step of our argument is very similar to the proof of Mensch's (2022) Theorem 3. Thus, one can interpret Corollary 3 as an extension of Mensch's (2022) Theorem 3 to case where costs are mean-measurable and have a steep-slope.

To better connect Corollary 3 to Mensch (2022) and the rest of the literature, comparing the corollary to the standard analysis with a fully-informed buyer is instructive. In that setup, the monopolist typically finds it optimal to offer a single-purchasing option in one of two fairly restrictive cases: either marginal quality costs κ' are constant, (e.g., Riley and Zeckhauser, 1983; Manelli and Vincent, 2007), or the marginal profit of serving a positive quality to any type other than the highest is negative, in which case the monopolist chooses

to only serve the highest type. The argument behind Corollary 3 is analogous to the second case, but where the type distribution is endogenous to the menu. Our result shows that, in such cases, posting a price can be optimal even if all virtual values are positive. Thus, Corollary 3 identifies a new set of conditions predicting the optimality of posted prices.

7. Concluding Remarks

We conclude our paper with a few brief remarks regarding our assumptions and results.

Support vs. positive probability. Our results show whenever learning costs are not too flat, the monopolist strictly distorts downward the quality she provides to *all* buyer types, including the one with maximal possible valuation according to the buyer’s signal. This result stands in contrast to the conclusion one obtains when information is exogenous, where the highest buyer type is allocated the efficient quality. As such, our paper suggests that an analyst who examines the market under the assumption that information is exogenous may come to erroneous conclusions regarding the efficiency of the market’s allocation. However, one might wonder whether this error actually occurs: since the buyer’s type distribution is endogenous, the buyer may choose an F that assigns zero probability to the top of its support. It turns out, however, that it is without loss for F to put positive probability on $\bar{\theta}_F$ whenever $\bar{\theta}_F < \bar{\theta}$. For an explanation, consider a monopolist optimal outcome (Q, F) in which $\bar{\theta}_F < \bar{\theta}$ and where F generates $\bar{\theta}_F$ with zero probability. We now sketch the construction of an alternative signal G that attains the same value as F .¹⁷ Because $\bar{\theta}_F < \bar{\theta}$, $\bar{\theta}_F$ must be a pooling type, and so F must have a pooling interval (θ_1, θ_2) that contains $\bar{\theta}_F$ at the top of its support. Let $\tilde{\theta}$ be the mean of F conditional on its realization being in that interval. Since $\bar{\theta}_F$ has zero probability, F must generate multiple signal realizations inside (θ_1, θ_2) . Some of these realizations are below $\tilde{\theta}$. Take l to be the highest one. Some other realizations are above $\tilde{\theta}$ —denote the lowest one by h . Take G to be the signal that is equal to F outside of (θ_1, θ_2) , but generates only l and h within (θ_1, θ_2) . One can show such a G exists, and attains the same value as F for the program (8). Moreover, G is feasible in that program. Since (Q, F) is monopolist optimal, (Q, G) must also be monopolist optimal. Moreover, G generates $\bar{\theta}_G = h$ with positive probability. Hence,

¹⁷When F_0 admits a density, one can get this result by observing that restricting attention to bipooling signals (Kleiner, Moldovanu, and Strack, 2021; Arieli et al., 2023) is without loss of optimality—see footnote 12.

whenever our model yields downward distortion at top, it also enables this distortion to occur with strictly positive probability.

Inefficiency of serving some types. Throughout, we assumed that $\kappa'(0) < \underline{\theta}$, meaning it is efficient to serve all types. We use this assumption to prove Theorem 2 and Theorem 3. In Theorem 2, we use the assumption that $\kappa'(0) < \underline{\theta}$ to show that excluding $\underline{\theta}_F$ is inefficient. Thus, the theorem continues to hold whenever it is efficient to serve all types that can arise in a signal that is incentive compatible for some menu. More specifically, suppose $C(F) = \gamma \int c(\theta) F(d\theta)$, and there is some $\hat{\theta}_\gamma$ such that $c'(\hat{\theta}_\gamma) = c'(\theta_0) - \frac{1}{\gamma} \bar{q}$. Then Theorem 2 holds as stated so long $\kappa'(0) < \hat{\theta}_\gamma$. The reason is simple: by Theorem 1, a signal F is IC for some allocation only if $\underline{\theta}_F \geq \hat{\theta}_\gamma$.¹⁸ Therefore, it is always optimal to serve the lowest realized buyer type. The rest of Theorem 2 goes through as before.

The assumption that $\kappa'(0) < \hat{\theta}_\gamma$ is also sufficient for extending Theorem 3. In fact, Theorem 3, requires an even weaker condition: $\kappa'(0) < \hat{\theta}_\gamma$ must hold for *some* γ . For an explanation, note that this weaker condition is equivalent to the assumption that $\kappa'(0) < \theta_0$, which in turn implies the monopolist can induce no learning while still making a positive profit. This implication is sufficient for Theorem 3: whenever the implication holds, the theorem is true regardless of whether it is efficient to serve the lowest possible type $\underline{\theta}$.

Without these alternative assumptions, Theorem 3, may fail, because it may be optimal for the monopolist to not serve anyone at the optimum. Theorem 2, however, remains true so long as it is efficient to give a strictly positive quality to $\bar{\theta}$. To prove this, one can show there is at most one signal realization $\theta^* \in \text{supp}(F)$ for which $\kappa'(Q(\theta^*)) > \theta^*$. Moreover, this realization must be excluded, meaning that $\theta^* = \underline{\theta}_Q$. So, in effect, the monopolist never provides an inefficiently high, positive quality to any realized θ . We provide a proof sketch in the online appendix.

Bounded Derivatives of c . Throughout the analysis we assumed that c has a bounded second derivative, which in turn implies a bounded first derivative. Most of our results do not depend on this assumption—their proof applies without change even without it. The only results whose proof does not go through are the results for the case with low learning costs. These are Part (ii) of Proposition 1 and Theorem 4. These results both rely on the fact that the seller induces full learning when costs are low. This fact fails when the first

¹⁸Formally, if F is IC for some allocation, it is also IC for some F -ICC allocation Q . Consequently, $\bar{q} \geq Q(\bar{\theta}_F) - Q(\underline{\theta}) \geq \gamma c'(\bar{\theta}_F) - \gamma c'(\underline{\theta}_F) \geq \gamma c'(\theta_0) - \gamma c'(\underline{\theta}_F)$. That $\underline{\theta}_F \geq \hat{\theta}_\gamma$ then follows from c' being strictly increasing.

derivative of c is unbounded: in that case, the monopolist cannot create a sufficient degree of product differentiation to make full-learning incentive compatible, even when γ is very low (see equation 6).

Our arguments also fail when the first derivative c' happens to be bounded but the second derivative c'' is not. The reason is the curvature of the seller's profits as a function of the buyer's realized type. When c'' is bounded, one can use Condition 1 to show that, when γ is low enough, the seller strictly benefits from having the buyer acquire additional information for all cost-canceling allocations. With unbounded c'' , this argument could fail. In particular, the monopolist could prefer to avoid revealing to the buyer that his true type is $\underline{\theta}$ for all values of γ . Such a preference would violate Theorem 4, which in turn invalidates our argument for Part (ii) of Proposition 1.

Turns out, however, that Proposition 1's Part (ii) holds as long as c' is bounded—even if c'' is not. While our current proof fails (since it relies on Theorem 4), one can make an alternative argument. Here is a rough sketch. If F_0 is finite and regular, one can show that for any optimal cost compensating (Q, F) , the associated F -marginal price p must be constant around any θ that is not in F_0 's support. Using this observation, one can find a $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$ that satisfies the following three properties for all sufficiently small γ . First, the highest realization of any monopolist optimal F must be in $(\hat{\theta}, \bar{\theta}]$. Second, every optimal F -CC allocation must be essentially flat on $(\hat{\theta}, \bar{\theta})$. And third, every optimal F -CC allocation has a strictly positive upward jump at $\bar{\theta}$ whose size is bounded away from zero. Together, these properties imply that π_Q must have a similar shape on $(\hat{\theta}, \bar{\theta}_F]$ as Q : π_Q must be nearly flat on $(\hat{\theta}, \bar{\theta}_F)$ and have a strictly positive upward jump at $\bar{\theta}$. This shape, in turn, implies that $\bar{\theta}_F$ must equal $\bar{\theta}$ —if not, the monopolist would strictly benefit from spreading $\bar{\theta}_F$ between $\bar{\theta}_F - \epsilon$ and $\bar{\theta}$ for some sufficiently small $\epsilon > 0$.

Affine Learning Costs. We assume the buyer's costs of learning to be affine in the distribution of her posterior estimate. As explained earlier, the assumption that costs depend only on the distribution of the buyer's posterior estimate is without loss. However, the assumption that costs are affine in this distribution is substantive. Next, we explain how our analysis changes if we instead assume the buyer's learning costs can be locally approximated by an affine function. Specifically, say C is **Gateaux differentiable** if every $F \in \mathcal{I}$ admits some twice differentiable strictly convex function $c_F : \Theta \rightarrow \mathbb{R}$ such that for

every $G \in \mathcal{I}$,

$$\lim_{\epsilon \searrow 0} \frac{1}{\epsilon} [C(F + \epsilon(G - F)) - C(F)] = \int c_F(\theta)(G - F)(d\theta).$$

Assumptions of this type were first introduced into information acquisition models by Ravid, Roesler, and Szentes (2022). Lipnowski and Ravid (2022) introduce a generalization of this class of cost functions that are not mean-measurable—i.e., costs that depend on the distribution of the buyer’s posterior *belief*.

Under Gateaux differentiability, our reduction to F -CC allocations is still without loss, though one must adjust the definition of CC allocations so that c'_F replaces c' . To get this result, one first applies Lemma 1 from Georgiadis, Ravid, and Szentes (2022) to get that F solves the buyers problem if and only if it solves the buyer’s problem when her costs are given by their affine approximation at F .¹⁹ Said differently, a given F is incentive compatible for the buyer if and only if it is buyer optimal when the buyer’s cost function is given by $G \mapsto \int c_F(\theta)G(d\theta)$. Theorem 1 therefore implies it is without loss to focus on (the properly defined) F -CC allocations.

The appropriate adjustment of Corollary 1 also continues to hold. In particular, it turns out the adjusted mechanism design problem (7) is amenable to the same techniques as the problem presented in this paper. Consequently, one can show that the monopolist optimal outcome (Q, F) distorts quality downward for any F -separating θ . The reason is that this observation relies only on perturbations of the F -marginal price.

The rest of our results, however, cannot be proven as is. The reason is that the affine approximation c_F depends on F , making the analogue of the information design problem (8) much less tractable. For an explanation, note that we prove our other results by fixing a CC allocation Q and varying F within the set of signals for which Q is CC. When learning costs are affine, this set contains any signal that separates any θ at which p_Q is strictly increasing. The same is not true when C is Gateaux differentiable: in that case, changing F typically involves changing c'_F , and therefore the allocation. Hence, the arguments for the rest of our results do not apply when costs are merely Gateaux differentiable. Developing a method for analyzing the case with Gateaux differentiable costs remains an open question.

¹⁹The lemma is stated for the case where the agent can induce any distribution, but applies as stated to any convex constraint set.

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A. Online Proofs Appendix

We introduce some notation that we use throughout the appendix. For every $\theta \in (\underline{\theta}, \bar{\theta}]$, let $\varphi_-(\theta) = \sup_{\theta' < \theta} \varphi(\theta')$ be the left limit of φ at θ , and set $\varphi_-(\underline{\theta}) = x$. Similarly, for $\theta \in [\underline{\theta}, \bar{\theta})$, define the right limit of φ and θ by $\varphi_+(\theta) = \inf_{\theta' > \theta} \varphi(\theta')$, while setting $\varphi_+(\bar{\theta}) = y$. Relatedly, for a convex function $\phi : \Theta \rightarrow [x, y]$, we let ϕ'_- and ϕ'_+ denote its left and right derivatives, respectively, whenever those exist (which is the case for every $\theta \in (\underline{\theta}, \bar{\theta})$).

Armed with the definitions above, we say that an allocation Q **jumps towards efficiency** if

$$Q(\theta) \in \operatorname{argmax}_{q \in [Q_-(\theta), Q_+(\theta)]} [\theta q - \kappa(q)]. \quad (9)$$

The following observations, proved next, simplify the analysis. First, one can take any allocation and replace it with an allocation that jumps towards efficiency without reducing the monopolist's profits. Consequently, focusing on allocations that jump towards efficiency is without loss of optimality. Second, allocations that jump towards efficiency convey a technical benefit: if Q jumps towards efficiency, then π_Q is an upper-semicontinuous function. Therefore, we restrict attention to allocations that jump towards efficiency in all the proofs.

A.1. Jumps Towards Efficiency

In this section we prove that jumps towards efficiency are without loss of optimality. Moreover, we show that whenever Q jumps towards efficiency, $\pi_{Q, \underline{u}}$ is upper-semicontinuous.

We begin with proving that jumping towards efficiency is without loss of optimality.

Lemma 3. *Suppose $(Q, \underline{u}, F)$ is IC. Then there is an allocation Q^* that jumps towards efficiency such that $(Q^*, \underline{u}, F)$ is IC, and $\int \pi_{Q^*, \underline{u}}(\theta) F(d\theta) \geq \int \pi_{Q, \underline{u}}(\theta) F(d\theta)$. Moreover, if $(Q, \underline{u}, F)$ is monopolist optimal and κ is strictly convex, Q^* equals Q F -almost surely.*

Proof of Lemma 3. Suppose Q is F -IC. Define the allocation Q^* via

$$Q^*(\theta) := \min \left\{ \tilde{q} : \tilde{q} \in \operatorname{argmax}_{q \in [Q_-(\theta), Q_+(\theta)]} [\theta q - \kappa(q)] \right\},$$

which is well-defined because κ is convex. Note that Q^* equals Q at any θ where Q is continuous. Since Q is discontinuous in at most a countable set of points, we get that $Q^*_+ = Q_+$ and $Q^*_- = Q_-$. It follows Q jumps towards efficiency. In addition, note that Q

and Q^* differ on at most a countable set of points, which has Lebesgue measure zero, and so $V_{Q,\underline{u}} = V_{Q^*,\underline{u}}$. It follows $(Q, \underline{u}, F)$ is IC if and only if $(Q^*, \underline{u}, F)$ is IC as well.

We now turn to showing the monopolists profit under $(Q^*, \underline{u}, F)$ is no lower than it is under $(Q, \underline{u}, F)$. To see this, note that

$$\int [\pi_Q(\theta) - \pi_{Q^*}(\theta)] F(d\theta) = \int [(\theta Q(\theta) - \kappa \circ Q(\theta)) - (\theta Q^*(\theta) - \kappa \circ Q^*(\theta))] F(d\theta) \leq 0, \quad (10)$$

where the first equality follows from $V_{Q,\underline{u}} = V_{Q^*,\underline{u}}$.

We now conclude the proof by arguing that if (Q, F) is monopolist optimal, and κ is strictly convex, then $Q^* = Q$ F -almost surely. For this purpose, note that strict convexity of κ means that Q^* is the unique maximizer of $q \mapsto \theta[q - \kappa(q)]$ over $[Q_-(\theta), Q_+(\theta)]$. Consequently, (10) must hold with strict inequality whenever $Q^* \neq Q$ over a positive F -measure set. But $(Q^*, \underline{u})$ is F -IC, and so $(Q, \underline{u}, F)$ being monopolist optimal means that the inequality in (10) must hold with equality. It follows that $Q^* = Q$ F -almost surely. $\square$

Next, we show that Q jumping towards efficiency has a useful technical benefit: it makes π_Q upper semicontinuous.

Lemma 4. *If an allocation Q jumps towards efficiency, π_Q is upper semicontinuous.*

Proof of Lemma 4. Suppose Q jumps towards efficiency. In what follows, define

$$Q_-^e(\theta) = \min\{q : q \in \operatorname{argmax}_{\tilde{q} \in [0, \bar{q}]} [\theta \tilde{q} - \kappa(\tilde{q})]\}, \quad \text{and} \quad Q_+^e(\theta) = \max\{q : q \in \operatorname{argmax}_{\tilde{q} \in [0, \bar{q}]} [\theta \tilde{q} - \kappa(\tilde{q})]\}.$$

It is easy to verify that both Q_-^e and Q_+^e are increasing. Berge's Maximum Theorem then delivers that Q_-^e and Q_+^e are respectively lower and upper semicontinuous, which (combined with monotonicity of the two functions) delivers that the functions are respectively left and right continuous. Moreover, Berge's Maximum Theorem also delivers that $\lim_{\theta_n \nearrow \theta} Q_+^e(\theta_n) = Q_-^e(\theta)$ and $\lim_{\theta_n \searrow \theta} Q_-^e(\theta_n) = Q_+^e(\theta)$. Finally, $\operatorname{argmax}_{q \in [0, \bar{q}]} [\theta q - \kappa(q)] = [Q_-^e(\theta), Q_+^e(\theta)]$ holds because $q \mapsto (\theta q - \kappa(q))$ is a concave function.

Let $(\theta_n)_{n \in \mathbb{N}}$ be some convergent sequence, and take θ_∞ to be its limit. Our goal is to show that $\limsup_n \pi_Q(\theta_n) \leq \pi_Q(\theta_\infty)$. Since V_Q is continuous, it is sufficient to show that

$$\limsup_n [\theta_n Q(\theta_n) - \kappa \circ Q(\theta_n)] \leq \theta_\infty Q(\theta_\infty) - \kappa \circ Q(\theta_\infty). \quad (11)$$

Obviously, equation (11) holds if Q is continuous at θ_∞ . Thus, from now on we assume a discontinuity in Q at θ_∞ , so $Q_-(\theta_\infty) < Q_+(\theta_\infty)$. Moreover, since every sequence admits a monotone subsequence, to show the above inequality it is sufficient to show that it holds when $(\theta_n)_{n \in \mathbb{N}}$ is monotone. Without loss of generality, suppose θ_n is monotone increasing.

We proceed in cases.

Case 1. Suppose that $Q(\theta_\infty) \in [Q_-^e(\theta_\infty), Q_+^e(\theta_\infty)]$ (i.e., $Q(\theta_\infty)$ is efficient). Then for every n ,

$$\begin{aligned} \theta_n Q(\theta_n) - \kappa \circ Q(\theta_n) &\leq \theta_n Q_+^e(\theta_n) - \kappa \circ Q_+^e(\theta_n) \rightarrow \theta_\infty Q_-^e(\theta_\infty) - \kappa \circ Q_-^e(\theta_\infty) \\ &= \theta_\infty Q(\theta_\infty) - \kappa \circ Q(\theta_\infty), \end{aligned}$$

where convergence follows from Berge's Maximum Theorem.

Case 2. Suppose $Q(\theta_\infty) < Q_-^e(\theta_\infty)$. Because $q \mapsto \theta q - \kappa(q)$ is concave and Q jumps towards efficiency, $Q(\theta_\infty) = Q_+(\theta_\infty) < Q_-^e(\theta_\infty)$. In the next paragraph we argue that $Q(\theta_n) < Q_-^e(\theta_n)$ holds for all sufficiently large n . Taking this inequality as given, note that, because $\theta_n \nearrow \theta_\infty$, $Q_-^e(\theta_n) \rightarrow Q_-^e(\theta_\infty)$, and so for all sufficiently large n , $Q(\theta_n) \leq Q(\theta_\infty) < Q_-^e(\theta_n)$. Since $q \mapsto \theta q - \kappa(q)$ is concave, we get that

$$\theta_n Q(\theta_n) - \kappa \circ Q(\theta_n) \leq \theta_n Q(\theta_\infty) - \kappa \circ Q(\theta_\infty) \rightarrow \theta_\infty Q(\theta_\infty) - \kappa \circ Q(\theta_\infty),$$

as required.

Thus, to complete the proof of this case, it remains to show that $Q(\theta_n) < Q_-^e(\theta_n)$ holds for all sufficiently large n . Suppose otherwise. Then one can find a subsequence $(Q_{n_k})_{k \in \mathbb{N}}$ for some infinite $N \subseteq \mathbb{N}$ such that $Q(\theta_{n_k}) \geq Q_-^e(\theta_{n_k})$ for all $k \in \mathbb{N}$. Since Q_-^e is left continuous and $\theta_{n_k} \nearrow \theta_\infty$, one gets that

$$Q(\theta_\infty) \geq Q(\theta_{n_k}) \geq Q_-^e(\theta_{n_k}) \rightarrow Q_-^e(\theta_\infty),$$

meaning $Q(\theta_\infty) \geq Q_-^e(\theta_\infty)$, a contradiction. The proof of this case is therefore complete.

Case 3. Suppose $Q(\theta_\infty) > Q_+^e(\theta_\infty)$. Then concavity of $q \mapsto \theta q - \kappa(q)$ and Q jumping towards efficiency means that $Q(\theta_\infty) = Q_-(\theta_\infty)$. Since $\theta_n \nearrow \theta_\infty$ and Q is monotone, $Q(\theta_n) \rightarrow Q_-(\theta_\infty) = Q(\theta_\infty)$. It follows (11) holds.

□

A.2. Proof of Lemma 1

We begin by formally defining the buyer's maximization problem holding the monopolist's menu fixed. Let $X = [0, \bar{q}] \times \mathbb{R}_+$, and endow the set of Borel measures over $X \times \Theta$, $\Delta(X \times \Theta)$, with the weak* topology. Given a menu M , the buyer's program can be written as

$$\begin{aligned} & \max_{\xi \in \Delta(X \times \Theta)} \int (\theta q - t) \xi(d(q, t, \theta)) - C(\text{marg}_\Theta \xi) \\ & \text{s.t. } \text{supp } \xi \subseteq M \times \Theta, \\ & \text{marg}_\Theta \xi \preceq F_0. \end{aligned}$$

Observe the above program involves the maximization of a continuous objective over a compact constraint set, and so the set of solution, $\Xi(M)$, is non-empty for every compact M . Letting $\mathcal{M}$ be the collection of compact subsets of X that contain the non-participation option $(0, 0)$, the monopolist's program can be written as

$$\begin{aligned} & \max_{(M, \xi) \in \mathcal{M} \times \Delta(X \times \Theta)} \int (t - \kappa(q)) \xi(d(q, t, \theta)) \\ & \text{s.t. } \xi \in \Xi(M). \end{aligned}$$

Notice it is without loss to assume $M \subseteq \bar{X} = [0, \bar{q}] \times [0, \bar{\theta}\bar{q}]$, because the buyer strictly prefers $(0, 0)$ to any menu item that includes a transfer strictly above $\bar{\theta}\bar{q}$. Let $\mathcal{K}(\bar{X})$ be the set of all compact non-empty subsets of $\bar{X}$ endowed with the Hausdorff metric,

$$d(A, B) = \max \left\{ \max_{b \in B} \min_{a \in A} d(b, a), \max_{a \in A} \min_{b \in B} d(a, b) \right\},$$

and take $\bar{\mathcal{M}}$ to be the elements of $\mathcal{K}(\bar{X})$ that contain $(0, 0)$. Taking $\bar{\Xi}$ to be the restriction of Ξ to $\bar{\mathcal{M}}$, and letting

$$\text{Gr } \bar{\Xi} = \left\{ (M, \xi) \in \bar{\mathcal{M}} \times \Delta(X \times \Theta) : \xi \in \bar{\Xi}(M) \right\}$$

denote the restriction's graph, we get that the monopolist's problem can be rewritten as

$$\max_{(M, \xi) \in \text{Gr } \bar{\Xi}} \int (t - \kappa(q)) \xi(d(q, t, \theta)). \quad (12)$$

Observe $\bar{\mathcal{M}}$ is a closed subset of $\mathcal{K}(\bar{X})$, and so because $\mathcal{K}(\bar{X})$ is compact (Aliprantis and Border (2006), Theorem 3.85), $\bar{\mathcal{M}}$ must be compact as well. It follows, by Berge's theorem of the maximum, that $\bar{\Xi}$ is upper-hemicontinuous and has a closed graph (Aliprantis and Border (2006), Theorem 17.10). Hence, this graph must be compact because it is a subset of $\bar{\mathcal{M}} \times \Delta(\bar{X} \times \Theta)$, which is compact. That (12) admits a solution follows.

A.3. Proofs from Section 3

We begin with stating Dizdar and Kováč's (2020) generalization of Dworczak and Martini's (2019) duality result. Towards this goal, for every $F \in \mathcal{I}$, define an F -**price** to be a Lipschitz continuous, convex function that is affine on any interval of F -pooling types, that is, over any interval $(\underline{\theta}_0, \bar{\theta}_0) \subseteq \{\theta : I_F(\theta) > 0\}$. The following result relates this concept to the buyer's problem.

Theorem 5 (Dizdar and Kováč, 2020). *Let $\Phi : \Theta \rightarrow \mathbb{R}$ be a bounded, upper semicontinuous function that admits some $\epsilon > 0$ and $\bar{L} \in \mathbb{R}_+$ such that $\Phi(\theta) - \Phi(\theta') \geq \bar{L}(\theta' - \theta)$ holds for $\underline{\theta} \leq \theta' \leq \theta \leq \underline{\theta} + \epsilon$, and $\Phi(\theta) - \Phi(\theta') \leq \bar{L}(\theta - \theta')$ holds for $\bar{\theta} - \epsilon \leq \theta' \leq \theta \leq \bar{\theta}$. Then,*

$$F \in \operatorname{argmax} \int \Phi(\theta) F(d\theta)$$

if and only if an F -price P exists such that $P(\theta) \geq \Phi(\theta)$ for θ , where the inequality holds with equality for $\theta \in \operatorname{supp} F$.²⁰

We now use this theorem to prove Lemma 2.

Proof of Lemma 2. We begin by arguing that $V_Q - c$ satisfies the prerequisites of Theorem 5. By definition, $V_Q - c$ is continuous, and therefore bounded and upper-semicontinuous. Now pick any $\epsilon > 0$. Then for any $\theta' < \theta$ such that $(\theta', \theta) \in [\underline{\theta}, \underline{\theta} + \epsilon]$ we have

$$V_Q(\theta) - c(\theta) - V_Q(\theta') - c(\theta') = \int_{\tilde{\theta} \in (\theta', \theta)} [Q(\tilde{\theta}) - c'(\tilde{\theta})] d\tilde{\theta} \geq -c'(\underline{\theta} + \epsilon)(\theta - \theta') = c'(\underline{\theta} + \epsilon)(\theta' - \theta),$$

where the inequality follows from $Q \geq 0$ and c' being increasing. Similarly, for every $\theta' < \theta$ such that $(\theta', \theta) \in [\bar{\theta} - \epsilon, \bar{\theta}]$,

$$V_Q(\theta) - c(\theta) - V_Q(\theta') - c(\theta') = \int_{\tilde{\theta} \in (\theta', \theta)} [Q(\tilde{\theta}) - c'(\tilde{\theta})] d\tilde{\theta} \leq (\bar{q} - c'(\bar{\theta} - \epsilon))(\theta - \theta'),$$

²⁰The statement of the theorem here is slightly more general than the one stated by Dizdar and Kováč (2020). However, the exact same steps as in their proof hold, with $\bar{L}$ replacing L in equation (8) of their paper.

where the inequality follows from $Q \leq \bar{q}$ and c' being increasing. It follows that, by taking $\Phi = V_Q - c$, the assumptions of Theorem 5 are satisfied. Because the lemma's conditions imply P is an F -price, Theorem 5 delivers that F is buyer optimal.

For the “only if” part, suppose F solves the buyer's problem. Let P be the F -price delivered by Theorem 5. We then have $P \geq V_Q - c(Q)$, with the inequality holding with equality over the support of F .

□

Next we prove a lemma that recovers an F -marginal price from an F -price that certifies F as buyer optimal given Q . Moreover, we show one can choose p such that $p + c'$ coincide with Q on the support of F .

Lemma 5. *Fix an outcome (Q, F) and let P be an F -price with $P \geq V_Q - c$, with equality in $\text{supp} F$. Then, there exists an F -marginal price p with $Q = p + c'$ in $\text{supp} F$, and:*

$$P(\theta) = P(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} p(x) dx, \quad \theta \in [\underline{\theta}, \bar{\theta}].$$

Proof of Lemma 5. Since P is Lipschitz, there is a function $p : \Theta \rightarrow \mathbb{R}$ such that $P(\theta) = P(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} p(\tilde{\theta}) d\tilde{\theta}$ for all θ .

Next, we claim one can take p such that $p(\theta) = Q(\theta) - c'(\theta)$ for all $\theta \in \text{supp } F$. To do so, we show below that for each such θ ,

$$p_+(\theta) \geq Q_+(\theta) - c'(\theta) \geq Q_-(\theta) - c'(\theta) \geq p_-(\theta). \quad (13)$$

Equation 13 immediately implies the desired equality whenever $p_+(\theta) = p_-(\theta)$. Moreover, since P is convex, we can take p to be increasing, and so $p_+(\theta) > p_-(\theta)$ holds over a Lebesgue-null set, meaning one can edit p so that it satisfies the desired equality without impacting its integral. Hence, to prove that one can take p such that $p(\theta) = Q(\theta) - c'(\theta)$ for all $\theta \in \text{supp } F$, it is sufficient to show that (13) holds for all such θ . To show this inequality, notice that for any $\epsilon > 0$,

$$\begin{aligned} 0 &\leq \frac{1}{\epsilon} [P(\theta - \epsilon) - (V_Q - c)(\theta - \epsilon)] \\ &= \frac{1}{\epsilon} [P(\theta - \epsilon) - P(\theta) + (V_Q - c)(\theta) - (V_Q - c)(\theta - \epsilon)] \\ &\xrightarrow{\epsilon \searrow 0} -p_-(\theta) + Q_-(\theta) - c'(\theta), \end{aligned}$$

where the first inequality follows from $P \geq V_Q - c$, and the second equality from P being equal to $V_Q - c$ over the support of F (and θ being in that support). It follows $Q_-(\theta) - c'(\theta) \geq p_-(\theta)$. An analogous argument delivers that $p_+(\theta) \geq Q_+(\theta) - c'(\theta)$.

Thus, all that remains is to argue that we can take p so that it satisfies the conditions of an F -marginal-price. Towards this goal, recall we already argued we can take p to be increasing. Moreover, because P is affine on any interval over which I_F is strictly positive, p is constant over any such interval. Continuity of I_F then implies that if $I_F(\theta)$ is strictly positive, I_F must be strictly positive over a neighborhood of θ , and so p must be constant around θ .

Finally, we need to argue that $p \in [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$. To do so, notice first it is without loss to choose p so that $p(\bar{\theta}) = p_-(\bar{\theta}_F)$: either $\bar{\theta}_F = \bar{\theta}$, or $\bar{\theta}_F < \bar{\theta}$, in which case I_F must be strictly positive over $(\bar{\theta}_F - \epsilon, \bar{\theta})$ for some small $\epsilon > 0$, meaning p must be constant over the same, and so setting $p(\bar{\theta}) = p(\bar{\theta}_F) = p_-(\bar{\theta}_F)$ is without loss. An analogous argument delivers it is without loss to have $p(\underline{\theta}) = p_+(\underline{\theta}_F)$. Combined with equation 13, we get that

$$p(\bar{\theta}) = p(\bar{\theta}_F) = Q(\bar{\theta}_F) - c(\bar{\theta}_F) \leq \bar{q} - c(\bar{\theta}_F).$$

An analogous argument delivers $p(\underline{\theta}) = p(\underline{\theta}_F) \geq -c'(\underline{\theta}_F)$. Since p is increasing, we get $p \in [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$ as required. $\square$

Finally, we prove that focusing on F -CC mechanisms is without loss of generality.

Proof of Theorem 1. As a preliminary step, suppose F is buyer-optimal for some mechanism, and that we have some F -CC allocation Q . Let $p = p_Q$ be the F -marginal price associated with Q . In the case where $\lim_{\theta \searrow \underline{\theta}} c'(\theta) = -\infty$ and $\lim_{\theta \nearrow \bar{\theta}} c'(\theta) = \infty$, because c' is continuous and strictly increasing, $\underline{\theta}_Q \in (\underline{\theta}, \underline{\theta}_F]$ is the unique solution to $p(\underline{\theta}_F) + c'(\underline{\theta}_Q) = 0$, and $\bar{\theta}_Q \in [\bar{\theta}_F, \bar{\theta})$ is the unique solution to $p(\bar{\theta}_F) + c'(\bar{\theta}_Q) = \bar{q}$. If the slope of c at the boundary is finite and no such respective solution exists in the interval $[\underline{\theta}, \bar{\theta}]$, we set $\underline{\theta}_Q = \underline{\theta}$ and $\bar{\theta}_Q = \bar{\theta}$, respectively.

We first show F is buyer-optimal for every F -CC allocation. Let Q be an F -CC allocation, with associated F -price P_Q . To show that F is buyer-optimal given Q , by Lemma 2, it suffices to show that $P_Q(\theta) \geq V_Q(\theta) - c(\theta)$ holds for all θ , and that $P_Q(\theta) = V_Q(\theta) - c(\theta)$ for all $\theta \in \text{supp } F$. But by the definition of F -CC allocations, $P_Q \geq V_Q - c$, with $P_Q = V_Q - c$ in $[\underline{\theta}_Q, \bar{\theta}_Q]$. Because $[\underline{\theta}_F, \bar{\theta}_F] \subset [\underline{\theta}_Q, \bar{\theta}_Q]$, the result is proved.

Next, we argue every allocation $\tilde{Q}$ for which F is buyer-optimal admits an equivalent F -CC allocation. Let F be buyer-optimal for an allocation $\tilde{Q}$. Let $P_{\tilde{Q}}$ be an F -price that certifies buyer-optimality of $\tilde{Q}$. We shall prove there is an F -CC allocation, Q such that, in the support of F , $Q = \tilde{Q}$, and $V_Q \leq V_{\tilde{Q}}$. By Lemma 5, we can find an F -marginal price $p_{\tilde{Q}}$ such that:

$$P_{\tilde{Q}}(\theta) = P_{\tilde{Q}}(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} p_{\tilde{Q}}(x) dx, \quad \theta \in [\underline{\theta}, \bar{\theta}],$$

and $p = \tilde{Q} - c'$ in the support of F . Define:

$$Q(\theta) = \min\{\max\{p_{\tilde{Q}}(\theta) + c'(\theta), 0\}, \bar{q}\}.$$

Also define $\underline{\theta}_Q = \inf[\{\bar{\theta}\} \cup Q^{-1}(0, \bar{q})]$ and $\bar{\theta}_Q = \sup[\{\underline{\theta}\} \cup Q^{-1}[0, \bar{q})]$. Note that, because $p_{\tilde{Q}} \in [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$, one has $\underline{\theta}_Q \leq \underline{\theta}_F$ and $\bar{\theta}_F \leq \bar{\theta}_Q$, and so $Q = p(\theta) + c'(\theta) = \tilde{Q}(\theta)$ for all $\theta \in \text{supp } F$.

We now find an F -price that shows that Q is an F -CC allocation and that certifies F as optimal for Q . For $\theta \in [\underline{\theta}_Q, \bar{\theta}_Q]$:

$$V_Q(\theta) = \int_{\underline{\theta}}^{\theta} Q(x) dx = \int_{\underline{\theta}_Q}^{\theta} (p_{\tilde{Q}}(x) + c'(x)) dx = P_{\tilde{Q}}(\theta) - P_{\tilde{Q}}(\underline{\theta}_Q) + c(\theta) - c(\underline{\theta}_Q).$$

Then, letting $\eta = -P_{\tilde{Q}}(\underline{\theta}_Q) - c(\underline{\theta}_Q)$, every $\theta \in [\underline{\theta}_Q, \bar{\theta}_Q]$ has

$$V_Q(\theta) - c(\theta) = P_{\tilde{Q}}(\theta) + \eta.$$

We show now that $P_Q := P_{\tilde{Q}} + \eta$ is an F -price and, therefore, Q is F -CC. Because P_Q differs from $P_{\tilde{Q}}$ only by a constant, it inherits convexity and Lipschitz continuity. It must also be affine for any F -pooling intervals. The condition above shows that $P_Q = V_Q - c$ for $\theta \in [\underline{\theta}_Q, \bar{\theta}_Q]$. All that is left is to show Q is F -CC with F -price P_Q is that $P_Q \geq V_Q - c$ for all θ . We show this fact next.

For $\theta \leq \underline{\theta}_Q$, we have

$$0 \geq \int_{\theta}^{\underline{\theta}_Q} (p_{\tilde{Q}}(x) + c'(x)) dx = P_{\tilde{Q}}(\underline{\theta}_Q) - P_{\tilde{Q}}(\theta) + c(\underline{\theta}_Q) - c(\theta),$$

where the first inequality holds because, by definition of $\underline{\theta}_Q$, $p_{\tilde{Q}}(x) + c'(x) \leq 0$ for $x \leq \underline{\theta}_Q$. Reorganizing that expression, we obtain $V_Q(\theta) - c(\theta) = -c(\theta) \leq P_{\tilde{Q}}(\theta) + \eta$.

For $\theta \geq \bar{\theta}_Q$,

$$\begin{aligned} V_Q(\theta) - c(\theta) &= V_Q(\bar{\theta}_Q) + \bar{q}(\theta - \bar{\theta}_Q) - c(\theta) = P_{\tilde{Q}}(\bar{\theta}_Q) + \eta + c(\bar{\theta}_Q) + \bar{q}(\theta - \bar{\theta}_Q) - c(\theta) \\ &= P_{\tilde{Q}}(\bar{\theta}_Q) + \int_{\bar{\theta}_Q}^{\theta} (\bar{q} - c'(x)) dx + \eta \leq P_{\tilde{Q}}(\theta) + \eta, \end{aligned}$$

where the first equality follows from envelope and the definition of $\bar{\theta}_Q$, the second equality from $V_Q = P_{\tilde{Q}} + \eta$ at $\bar{\theta}_Q$, and the inequality follows from $p_{\tilde{Q}}(x) + c'(x) \geq \bar{q}$ for $x \geq \bar{\theta}_Q$, by definition of $\bar{\theta}_Q$. We then conclude $P_Q = P_{\tilde{Q}} + \eta$ is an F -price certifying optimality of F -CC allocation Q .

What is left to prove is $V_Q \leq V_{\tilde{Q}}$ for $\theta \in \text{supp} F$. Because, in the support of F , $P_{\tilde{Q}} + \eta = P_Q = V_Q + c$, and $P_{\tilde{Q}} = V_{\tilde{Q}} + c$, it suffices to show $\eta \leq 0$. But this follows from:

$$0 = V_Q(\underline{\theta}_Q) = P_Q(\underline{\theta}_Q) + c(\underline{\theta}_Q) = P_{\tilde{Q}}(\underline{\theta}_Q) + c(\underline{\theta}_Q) + \eta \geq V_{\tilde{Q}}(\underline{\theta}_Q) + \eta,$$

where the inequality follows from $P_{\tilde{Q}}$ certifying optimality of $\tilde{Q}$ as in Lemma 2. It follows that $-\eta \geq V_{\tilde{Q}}(\underline{\theta}_Q) \geq 0$ —that is, $\eta \leq 0$, as required. $\square$

A.4. Proof of Theorem 2

The ultimate goal of this section is to prove Theorem 2. En-route, we prove several auxiliary results about the programs (7) and (8).

A.4.1. Allocation Perturbations

We first consider the program (7). We begin with two lemmas. The first lemma notes the set of allocations that are F -IC is convex. The second lemma uses this convexity to derive a necessary first order condition for an allocation to be part of a monopolist optimal allocation.

Lemma 6. *Suppose Q and $\tilde{Q}$ are both F -IC. Then, $(1 - \beta)Q + \beta\tilde{Q}$ is also F -IC for all $\beta \in [0, 1]$.*

Proof. Note that for any two allocations $Q, \tilde{Q}$, and any $\beta \in [0, 1]$,

$$\begin{aligned} V_{(1-\beta)Q+\beta\tilde{Q}}(\theta) &= \int_{\underline{\theta}}^{\theta} \left((1-\beta) Q(\tilde{\theta}) + \beta \tilde{Q}(\tilde{\theta}) \right) d\tilde{\theta} \\ &= (1-\beta) \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta}) d\tilde{\theta} + \beta \int_{\underline{\theta}}^{\theta} \tilde{Q}(\tilde{\theta}) d\tilde{\theta} = (1-\beta) V_Q(\theta) + \beta V_{\tilde{Q}}(\theta). \end{aligned}$$

Therefore, if both $Q, \tilde{Q}$ are F -IC, one obtains the following inequality for all $\tilde{F}$:

$$\begin{aligned} \int \left(V_{(1-\beta)Q+\beta\tilde{Q}} - c \right) (\theta) F(d\theta) &= (1-\beta) \int (V_Q - c) (\theta) F(d\theta) + \beta \int (V_{\tilde{Q}} - c) (\theta) F(d\theta) \\ &\geq (1-\beta) \int (V_Q - c) (\theta) d\tilde{F}(\theta) + \beta \int (V_{\tilde{Q}} - c) (\theta) d\tilde{F}(\theta) \\ &= \int \left(V_{(1-\beta)Q+\beta\tilde{Q}} - c \right) (\theta) d\tilde{F}(\theta), \end{aligned}$$

meaning $(1-\beta)Q + \beta\tilde{Q}$ is also F -IC. $\square$

Next, we obtain a necessary first-order condition for the monopolist's optimal outcome by perturbing the allocation while keeping the buyer's information fixed.

Lemma 7. *Let (Q^*, F^*) be monopolist optimal. Suppose Q also incentivizes F^* . Then,*

$$\int [(\theta - \kappa'(Q^*(\theta))) (Q - Q^*)(\theta) - (V_Q - V_{Q^*})(\theta)] F^*(d\theta) \leq 0.$$

Proof. Suppose (Q^*, F^*) is monopolist optimal, and let Q be any other F^* -IC allocation. Defining the allocation $Q_\epsilon := Q^* + \epsilon(Q - Q^*)$ for every $\epsilon \in (0, 1)$, it follows from the previous lemma that Q_ϵ is also F^* -IC. Therefore, it must be that (Q_ϵ, F^*) is weakly worse for the monopolist than (Q^*, F^*) . In other words, we must have

$$\int (\pi_{Q_\epsilon}(\theta) - \pi_Q(\theta)) F^*(d\theta) \leq 0$$

for all ϵ . Dividing this inequality by $\epsilon > 0$, and taking the limit as $\epsilon \searrow 0$, gives

$$\begin{aligned}
0 &\geq \frac{1}{\epsilon} \int (\pi_{Q_\epsilon}(\theta) - \pi_Q(\theta)) F^*(d\theta) \\
&= \int \theta (Q - Q^*)(\theta) - (V_Q - V_{Q^*})(\theta) F^*(d\theta) \\
&\quad - \int \frac{1}{\epsilon} (\kappa(Q^*(\theta) + \epsilon(Q - Q^*)(\theta)) - \kappa(Q^*(\theta))) F^*(d\theta) \\
&\rightarrow \int \theta (Q - Q^*)(\theta) - (V_Q - V_{Q^*})(\theta) F^*(d\theta) \\
&\quad - \int \kappa'(Q^*(\theta)) (Q - Q^*)(\theta) F^*(d\theta),
\end{aligned}$$

where convergence follows from Beppo Levi's Theorem (e.g., Aliprantis and Border (2006) Theorem 11.18).²¹ The lemma follows. $\square$

We now work towards the main result of this section. This result establishes conditions under-which the following inequalities hold:

$$\int_{\theta \geq \theta^*} \kappa'(Q(\theta)) F(d\theta) \geq \theta^*(1 - F_-(\theta^*)), \quad (14)$$

$$\int_{\theta \geq \theta^*} \kappa'(Q(\theta)) F(d\theta) \leq \theta^*(1 - F_-(\theta^*)). \quad (15)$$

These inequalities compare the average marginal cost of the allocation provided to every θ above θ^* to θ^* . Mussa and Rosen (1978) provide conditions under which θ^* is larger or smaller than this average marginal production costs when information is exogenous.

Before discussing these inequalities for the endogenous information case, we first show that the following inequality implies (15):

$$\int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(d\theta) \leq (1 - F(\theta^*)) \theta^*. \quad (16)$$

This inequality is the same as (15), except that the integral on the left hand side excludes θ^* , and the right hand side has $1 - F(\theta^*)$ instead of $1 - F_-(\theta^*)$.

Lemma 8. *If (16) holds for $\theta^* \in [\underline{\theta}_F, \bar{\theta}_F)$, then (15) also holds*

Proof. We first argue (16) implies $\kappa' \circ Q(\theta^*) < \theta^*$. To do so, assume (16) holds, and suppose $\kappa' \circ Q(\theta^*) \geq \theta^*$ for a contradiction. Because Q is F -ICC, the allocation Q is

²¹Because κ is convex, the function $\epsilon \mapsto \frac{1}{\epsilon} (\kappa(q + \epsilon(\tilde{q} - q)) - \kappa(q))$ is decreasing in ϵ for all $\tilde{q}$ and q .

strictly increasing on $[\underline{\theta}_F, \bar{\theta}_F] \supseteq [\theta^*, \bar{\theta}_F]$, and so $\kappa' \circ Q(\theta) > \kappa' \circ Q(\theta^*)$ for all $\theta > \theta^*$, because κ' is strictly increasing. Therefore,

$$\theta^* < \int \kappa' \circ Q(\theta) F(d\theta | \theta \in (\theta^*, \bar{\theta}_F]) = \int \kappa' \circ Q(\theta) F(d\theta | \theta > \theta^*) = \frac{\int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(d\theta)}{1 - F(\theta^*)},$$

contradicting (16). Thus, we have shown that $\kappa' \circ Q(\theta^*) < \theta^*$. Using this inequality, we now show that (16) implies (15). Specifically, (15) follows from the inequality chain

$$\begin{aligned} \int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(d\theta) &= \int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(d\theta) + (F(\theta^*) - F_-(\theta^*)) \kappa' \circ Q(\theta^*) \\ &\leq (1 - F(\theta^*)) \theta^* + (F(\theta^*) - F_-(\theta^*)) \theta^* = (1 - F_-(\theta^*)) \theta^*. \end{aligned}$$

The proof is now complete. $\square$

We now state Lemma 9, which is the main result of this section.

Lemma 9. *Suppose (Q, F) is monopolist optimal, and that Q is F -ICC. Fix any $\hat{\theta} \in \Theta$, and let*

$$\theta^* = \max\{\hat{\theta}, \underline{\theta}_Q\}.$$

Then:

- (i) *If $I_F(\hat{\theta}) = 0$ and $Q(\bar{\theta}_F) < \bar{q}$, the inequality (14) holds for θ^* .*
- (ii) *If $\hat{\theta} = \underline{\theta}$ and $\underline{\theta}_F > \underline{\theta}_Q$, the inequality (15) holds for θ^* .*
- (iii) *If p_Q strictly increases at $\hat{\theta}$ and $\hat{\theta} > \underline{\theta}_Q$, the inequality (15) holds for θ^* .*

To understand our proof, revisiting Mussa and Rosen (1978) is helpful. To show (14), Mussa and Rosen (1978) consider the change in the monopolist's profit due to a slight increase in the quality given to all types weakly above θ^* . If one starts from an optimal allocation, the change in the monopolist's revenues—represented by equation (14)'s right hand side—must be below the change in the monopolist's production costs, which are given by the left hand side of (14). The opposite inequality (15) is derived in a similar fashion by noting the monopolist cannot benefit from slightly *reducing* the quality given to all types above θ^* —provided that the allocation Q is strictly increasing at θ^* . The reason for this caveat is that, if Q is constant around θ^* , reducing the quality given to θ^* would result in non-monotone allocation, which would violate the buyer's incentive constraint.

It turns out that, by moving p , one can apply Mussa and Rosen's (1978) perturbation arguments when reasoning about the solution to (7), subject to two caveats. First, one needs to take care to perturb p in a way that results in a new F -marginal price. In particular, one cannot increase p at F -pooling points, and one must make sure Q_p remains in $[0, \bar{q}]$ over F 's support. Second, shifting p only affects the induced allocation Q_p for types at which $p + c'$ is in the interval $[0, \bar{q}]$. Thus, a marginal increase in p for all types above θ only changes the allocation for types above $\underline{\theta}_{Q_p}$. Taking these caveats into account and applying the ideas of Mussa and Rosen (1978) delivers the result.

Proof of Lemma 9-(i). Suppose (Q, F) is monopolist optimal, and that $Q(\bar{\theta}_F) < \bar{q}$. The following inequality holds for every θ^* such that $I_F(\theta^*) = 0$:

$$\int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(d\theta) \geq (1 - F_-(\theta^*)) \max\{\theta^*, \underline{\theta}_Q\}. \quad (17)$$

Let p be the F -marginal price associated with the allocation Q . For any $\epsilon > 0$ such that $p(\bar{\theta}_F) + \epsilon < \bar{q}$, define

$$p_\epsilon(\theta) = \begin{cases} p(\theta) & \text{if } \theta < \hat{\theta}, \\ p(\theta) + \epsilon & \text{otherwise.} \end{cases}$$

Because $I_F(\hat{\theta}) = 0$ and p is an F -marginal price, p_ϵ is also an F -marginal price. Note that $Q^{p_\epsilon} \geq Q_p$ because $p_\epsilon \geq p$, meaning that $\underline{\theta}_{Q^{p_\epsilon}} \leq \underline{\theta}_Q \leq \underline{\theta}_F$. Therefore, every $\theta \geq \underline{\theta}_F$ has $Q^{p_\epsilon}(\theta) = Q_p(\theta) + \epsilon \mathbf{1}_{\theta \geq \hat{\theta}}$. Below we prove that

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [V_{Q^{p_\epsilon}}(\theta) - V_{Q_p}(\theta)] = [\theta - \max\{\hat{\theta}, \underline{\theta}_Q\}]_+. \quad (18)$$

Using this convergence, we can appeal to Lemma 7 to get

$$\begin{aligned} 0 &\geq \frac{1}{\epsilon} \left\{ \int [(\theta - \kappa' \circ Q_p(\theta))(Q^{p_\epsilon}(\theta) - Q_p(\theta)) - (V_{Q^{p_\epsilon}}(\theta) - V_{Q_p}(\theta))] F(d\theta) \right\} \\ &= \frac{1}{\epsilon} \left\{ \int [(\theta - \kappa' \circ Q_p(\theta))\epsilon \mathbf{1}_{\theta \geq \hat{\theta}} - (V_{Q^{p_\epsilon}}(\theta) - V_{Q_p}(\theta))] F(d\theta) \right\} \\ &= \int_{\theta \geq \hat{\theta}} (\theta - \kappa' \circ Q_p(\theta)) F(d\theta) - \frac{1}{\epsilon} \int (V_{Q^{p_\epsilon}}(\theta) - V_{Q_p}(\theta)) F(d\theta) \\ &\rightarrow \int_{\theta \geq \hat{\theta}} (\theta - \kappa' \circ Q_p(\theta)) F(d\theta) - \int [\theta - \max\{\hat{\theta}, \underline{\theta}_Q\}]_+ F(d\theta) \\ &= \int_{\theta \geq \hat{\theta}} [\max\{\hat{\theta}, \underline{\theta}_Q\} - \kappa' \circ Q_p(\theta)] F(d\theta), \end{aligned}$$

where the convergence follows from Beppo Levi's Theorem (e.g., Aliprantis and Border, 2006, Theorem 11.18), and the last equality follows from observing that $\underline{\theta}_Q \leq \underline{\theta}_F$ implies that $\max\{\hat{\theta}, \underline{\theta}_Q\} > \underline{\theta}_F$ if and only if $\hat{\theta} > \underline{\theta}_F$. Rearranging the above inequality then delivers part (i) of the lemma.

Thus, to complete the proof of this part, arguing that (18) holds suffices. We divide our argument into two cases.

Case 1 Suppose $\hat{\theta} \geq \underline{\theta}_Q$. Direct computation reveals

$$V_{Q^{p^\epsilon}}(\theta) - V_{Q^p}(\theta) = \int_{\hat{\theta}}^{\theta \vee \hat{\theta}} \epsilon d\tilde{\theta} = \epsilon[\theta \vee \hat{\theta} - \hat{\theta}] = \epsilon[\theta - \hat{\theta}]_+ = \epsilon \left[\theta - \max\{\hat{\theta}, \underline{\theta}_p\} \right]_+.$$

where

$$\underline{\theta}_p := \underline{\theta}_{Q^p}.$$

The equality (18) immediately follows.

Case 2 Suppose $\hat{\theta} < \underline{\theta}_p$. If $\theta < \underline{\theta}_{Q^{p^\epsilon}}$, then $\theta \leq \underline{\theta}_Q$, and so $V_{Q^{p^\epsilon}}(\theta) - V_{Q^p}(\theta) = 0 = [\theta - \max\{\hat{\theta}, \underline{\theta}_Q\}]_+$, as required. The same equality also holds if $\theta < \hat{\theta}$. Consider then a $\theta \geq \max\{\underline{\theta}_{Q^{p^\epsilon}}, \hat{\theta}\}$. Then,

$$\begin{aligned} V_{Q^{p^\epsilon}}(\theta) - V_{Q^p}(\theta) &= \int_{\underline{\theta}_Q}^{\underline{\theta}_Q \vee \theta} \epsilon d\tilde{\theta} + \int_{\underline{\theta}_{Q^{p^\epsilon}}}^{\underline{\theta}_Q \wedge \theta} \epsilon \mathbf{1}_{\tilde{\theta} \geq \hat{\theta}} d\tilde{\theta} \\ &= \epsilon[\underline{\theta}_Q \vee \theta - \underline{\theta}_Q] + \epsilon[\underline{\theta}_Q \wedge \theta - \underline{\theta}_{Q^{p^\epsilon}} \vee \hat{\theta}]. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{\epsilon} [V_{Q^{p^\epsilon}}(\theta) - V_{Q^p}(\theta)] &= [\underline{\theta}_p \vee \theta - \underline{\theta}_p] + [\underline{\theta}_p \wedge \theta - \underline{\theta}_{p^\epsilon} \vee \hat{\theta}] \\ &\xrightarrow{\epsilon \rightarrow 0} [\underline{\theta}_p \vee \theta - \underline{\theta}_p] = [\theta - \underline{\theta}_p]_+ = [\theta - \max\{\hat{\theta}, \underline{\theta}_Q\}]_+, \end{aligned}$$

where convergence follows from Q_{p^ϵ} uniformly converging to Q , and the last equality from the current case's assumption that $\hat{\theta} < \underline{\theta}_Q$.

□

Proof of Lemma 9-(ii). Since $\underline{\theta}_F > \underline{\theta}_Q$, $Q(\underline{\theta}_F) > 0$. For any $\varepsilon \in (0, Q(\underline{\theta}_F))$, let $\theta_\varepsilon = \inf \{\theta : Q(\theta) \geq \varepsilon\}$, and define $p_\varepsilon(\cdot) := p(\cdot) - \varepsilon$. Observe p_ε is an F -marginal price because p is an F -marginal price, and $\varepsilon < Q(\underline{\theta}_F) = p(\underline{\theta}_F) + c'(\underline{\theta}_F)$, meaning $p_\varepsilon(\underline{\theta}_F) > -c'(\underline{\theta}_F)$ (which is sufficient for $p_\varepsilon \geq -c'(\underline{\theta}_F)$ to hold).

Let $Q_\varepsilon := Q_{p_\varepsilon}$ be the F -ICC allocation induced by p_ε . Obviously, $\theta_\varepsilon \leq \underline{\theta}_F$, and

$$Q_\varepsilon(\theta) = p_Q(\theta) + c'(\theta) - \varepsilon = Q(\theta) - \varepsilon$$

for all $\theta \in [\theta_\varepsilon, \bar{\theta}_F]$. Noting that for all $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$,

$$V_{Q_\varepsilon}(\theta) - V_Q(\theta) = \int_{\theta_\varepsilon}^{\theta} -\varepsilon d\tilde{\theta} + \int_{\underline{\theta}}^{\theta_\varepsilon} -Q(\tilde{\theta}) d\tilde{\theta} = \varepsilon(\theta_\varepsilon - \theta) - V_Q(\theta_\varepsilon),$$

and so by Lemma 7, we have

$$\begin{aligned} 0 &\geq \int [(\theta - \kappa' \circ Q(\theta))(-\varepsilon) - \varepsilon(\theta_\varepsilon - \theta) + V_Q(\theta_\varepsilon)] F(d\theta) \\ &= \int [(\kappa' \circ Q(\theta) - \theta_\varepsilon)\varepsilon + V_Q(\theta_\varepsilon)] F(d\theta). \end{aligned}$$

This inequality, however, implies that

$$\varepsilon \int_{\theta \geq \underline{\theta}_Q} \kappa' \circ Q(\theta) F(d\theta) = \varepsilon \int \kappa' \circ Q(\theta) F(d\theta) \leq \int [\varepsilon\theta_\varepsilon - V_Q(\theta_\varepsilon)] F(d\theta) \leq \varepsilon\theta_\varepsilon.$$

Dividing both sides by ε , taking $\varepsilon \searrow 0$, and noting that $\theta_\varepsilon \searrow \underline{\theta}_Q$ completes the proof. $\square$

Proof of Lemma 9-(iii) The proof of this part proceeds in 3 cases:

1. p_Q has an upward jump at θ^* .
2. p_Q strictly increases immediately below θ^* .
3. p_Q strictly increases immediately above θ^* .

We begin with the case in which p_Q jumps at θ^* .

Case 1: $p_{Q-}(\theta^*) < p_{Q+}(\theta^*)$. Observe Q being F -ICC and $p_{Q-}(\theta^*) < p_{Q+}(\theta^*)$ means $I_F(\theta^*) = 0$, and so $\theta^* > \underline{\theta}_F$. For any $\varepsilon \in (0, p_{Q+}(\theta^*) - p_{Q-}(\theta^*))$, define

$$p_\varepsilon(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta < \theta^*, \\ p_Q(\theta^*) \wedge (p_{Q+}(\theta^*) - \varepsilon) & \text{if } \theta = \theta^*, \\ p_Q(\theta) - \varepsilon & \text{if } \theta > \theta^*. \end{cases}$$

It is easy to verify that p_ε is an F -marginal price. Let $Q_\varepsilon := Q_{p_\varepsilon}$ be the F -ICC mechanism associated with p_ε . It follows Q_ε is F -IC, and thus one can apply Lemma 7 to get the following inequality for every $\varepsilon \in (0, p_{Q+}(\theta^*) - p_{Q-}(\theta^*))$,

$$\begin{aligned} 0 &\geq \int_{\theta > \theta^*} [(\theta - \kappa' \circ Q(\theta))(-\varepsilon) - \varepsilon(\theta^* - \theta)] F(d\theta) \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*)) (p_\varepsilon(\theta^*) - p_Q(\theta^*)) \\ &= \int_{\theta > \theta^*} (\kappa' \circ Q(\theta) - \theta^*) \varepsilon F(d\theta) \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*)) (p_\varepsilon(\theta^*) - p_Q(\theta^*)). \end{aligned}$$

Rearranging gives

$$\begin{aligned} \int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(d\theta) &\leq (1 - F(\theta^*)) \theta^* \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*)) \left(\frac{p_\varepsilon(\theta^*) - p_Q(\theta^*)}{\varepsilon} \right). \end{aligned} \tag{19}$$

We now distinguish between two cases. Suppose first $p_Q(\theta^*) < p_{Q+}(\theta^*)$. Then for all small enough $\varepsilon > 0$, $p_\varepsilon(\theta^*) - p_Q(\theta^*) = 0$, and so equation (19) is equivalent to (15).

Suppose then $p_Q(\theta^*) = p_{Q+}(\theta^*)$. Then $p_Q(\theta^*) - p_\varepsilon(\theta^*) = \varepsilon$. Substituting into (19) and rearranging gives (15), as desired. $\square$

Case 2: $p_{Q-}(\theta^*) = p_{Q+}(\theta^*)$ **and** $p_{Q-}(\theta^*) > p_Q(\theta)$ **for all** $\theta < \theta^*$: We begin by arguing that we can find a sequence $(\theta_n)_{n \in \mathbb{N}}$ in $[\underline{\theta}_F, \bar{\theta}_F]$ such that $\theta_n \nearrow \theta^*$, $I_F(\theta_n) = 0$ for all n , and $p_Q(\theta_n) < p_Q(\theta_{n+1})$ for all n . We then use this sequence to construct a sequence of allocations that keep F incentive compatible. This allocation sequence, combined with Lemma 7, delivers a sequence of first-order conditions whose limit delivers (15).

Let us find the sequence $(\theta_n)_{n \in \mathbb{N}}$. For every $\delta > 0$, p_Q is non-constant on $[\theta^* - \delta, \theta^*]$, because if it were, $p_{Q-}(\theta^*) = p_Q(\theta^* - \delta) < p_{Q-}(\theta^*)$. It follows we can find a sequence $\{\tilde{\theta}_n\}_{n \in \mathbb{N}}$ in $(\underline{\theta}_F, \theta^*)$ with $\tilde{\theta}_n \nearrow \theta^*$ such that $p_Q(\tilde{\theta}_n) < p_Q(\tilde{\theta}_{n+1})$ for all n . It follows p_Q is non-constant on $[\tilde{\theta}_m, \tilde{\theta}_n]$ for any $m < n$, and so every $m < n$ admits some $\theta_{m,n} \in [\tilde{\theta}_m, \tilde{\theta}_n]$ for which $I_F(\theta_{m,n}) = 0$. Choosing $\theta_n := \theta_{2n,2n+1}$, we have $\theta_n \nearrow \theta^*$, and

$$p_Q(\theta_n) = p_Q(\theta_{2n,2n+1}) \leq p_Q(\tilde{\theta}_{2n+1}) < p_Q(\tilde{\theta}_{2n+2}) \leq p_Q(\theta_{2(n+1),2(n+1)+1}) = p_Q(\theta_{n+1}),$$

meaning $(\theta_n)_{n \in \mathbb{N}}$ is as desired.

We now construct an F -ICC allocation for every θ_n in the above sequence. For this purpose, let $\delta_n = p_Q(\theta^*) - p_Q(\theta_n) > 0$,

$$p_n(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta \leq \theta_n \\ p_Q(\theta) - \delta_n & \text{if } \theta \geq \theta^* \\ p_Q(\theta_n) & \text{if } \theta \in [\theta_n, \theta^*]. \end{cases}$$

Since p_Q is an F -marginal price, and because $I_F(\theta^*) = 0$ (because p_Q is not constant around θ^*), the function p_n is an F -marginal price for every n , and so the allocation $Q_n := Q_{p_n}$ is an F -ICC allocation. Thus, Q_n is F -ICC.

Next, we apply Lemma 7 to get a first-order condition indexed by n . For this purpose, observe that for all $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$,

$$V_{Q_n}(\theta) - V_Q(\theta) = \int_{\theta_n \wedge \theta}^{\theta^* \wedge \theta} (p_Q(\theta_n) - p_Q(\tilde{\theta})) d\tilde{\theta} - \delta_n (\theta - \theta^* \wedge \theta).$$

Therefore, Lemma 7 delivers the following inequality for all n ,

$$\begin{aligned} 0 &\geq \int_{\theta \geq \theta^*} (\theta - \kappa' \circ Q(\theta)) (-\delta_n) F(d\theta) + \int_{\theta \in [\theta_n, \theta^*]} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta_n) - p_Q(\theta)) F(d\theta) \\ &\quad - \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} (p_Q(\theta_n) - p_Q(\tilde{\theta})) d\tilde{\theta} F(d\theta) - \int_{\theta \geq \theta^*} -\delta_n (\theta - \theta^*) F(d\theta) \\ &= \int_{\theta \geq \theta^*} (\kappa' \circ Q(\theta) - \theta^*) \delta_n F(d\theta) + \int_{\theta \in [\theta_n, \theta^*]} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta_n) - p_Q(\theta)) F(d\theta) \\ &\quad - \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} (p_Q(\theta_n) - p_Q(\tilde{\theta})) d\tilde{\theta} F(d\theta). \end{aligned}$$

Rearranging and noting that $p_Q(\theta_n) \leq p_Q(\theta)$ for all $\theta \geq \theta_n$ delivers

$$\begin{aligned} \int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(d\theta) &\leq \int_{\theta \geq \theta^*} \theta^* F(d\theta) - \int_{\theta \in [\theta_n, \theta^*]} (\theta - \kappa' \circ Q(\theta)) \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) F(d\theta) \\ &\quad + \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\tilde{\theta})) d\tilde{\theta} F(d\theta) \\ &\leq \int_{\theta \geq \theta^*} \theta^* F(d\theta) - \int_{\theta \in [\theta_n, \theta^*]} (\theta - \kappa' \circ Q(\theta)) \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) F(d\theta) \\ &\leq \int_{\theta \geq \theta^*} \theta^* F(d\theta) + \int_{\theta \in [\theta_n, \theta^*]} |\theta - \kappa' \circ Q(\theta)| \left| \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) \right| F(d\theta). \end{aligned} \tag{20}$$

We now show taking the limit of equation (20) as $n \rightarrow \infty$ delivers equation (15). To do so, observe $|p_Q(\theta_n) - p_Q(\theta)| \leq \delta_n$ for all $\theta \in [\theta_n, \theta^*)$, and that $\theta - \kappa' \circ Q(\theta) \leq \bar{\theta}_F + \kappa' \circ Q(\bar{\theta}_F)$ for all $\theta \in [\theta_n, \theta^*)$. Therefore, an M exists such that $|\theta - \kappa' \circ Q(\theta)| \left| \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) \right| \leq M$ for all n . Substituting back into (20) and taking limit with n delivers

$$\int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(d\theta) \leq \int_{\theta \geq \theta^*} \theta^* F(d\theta) + M(F_-(\theta^*) - F(\theta_n)) \rightarrow \int_{\theta \geq \theta^*} \theta^* F(d\theta).$$

Hence (15) holds at θ^* . $\square$

Case 3: $p_{Q-}(\theta^*) = p_{Q+}(\theta^*)$ **and** $p_{Q+}(\theta^*) < p_Q(\theta)$ **holds for all** $\theta > \theta^*$. We begin by finding a sequence $(\theta_n)_{n \in \mathbb{N}}$ in $[\underline{\theta}_F, \bar{\theta}_F]$ such that $\theta_n \searrow \theta^*$, $I_F(\theta_n) = 0$ for all n , and $p_Q(\theta_n) > p_Q(\theta_{n+1})$ for all n . We then construct a corresponding sequence of allocations that keep F incentive compatible for the buyer. This allocation sequence, combined with Lemma 7, delivers a sequence of first-order conditions whose limit delivers (15).

Let us find the sequence $(\theta_n)_{n \in \mathbb{N}}$. Observe that for every $\delta > 0$, p_Q is non-constant on $[\theta^*, \theta^* + \delta]$, because if it were constant, $p_{Q+}(\theta^*) = p_Q(\theta^* + \delta) > p_{Q+}(\theta^*)$. It follows we can find a sequence $(\tilde{\theta}_n)_{n \in \mathbb{N}}$ in $(\theta^*, \bar{\theta})$ with $\tilde{\theta}_n \searrow \theta^*$ such that $p_Q(\tilde{\theta}_n) > p_Q(\tilde{\theta}_{n+1})$ for all n . To define $(\theta_n)_{n \in \mathbb{N}}$, observe that p_Q is non-constant on $[\tilde{\theta}_m, \tilde{\theta}_n]$ for any $m < n$, and so every $m < n$ admits some $\theta_{m,n} \in [\tilde{\theta}_m, \tilde{\theta}_n]$ for which $I_F(\theta_{m,n}) = 0$. Choosing $\theta_n := \theta_{2n, 2n+1}$, we have $\theta_n \searrow \theta^*$, and

$$p_Q(\theta_n) = p_Q(\theta_{2n, 2n+1}) \geq p_Q(\tilde{\theta}_{2n+1}) > p_Q(\tilde{\theta}_{2n+2}) \geq p_Q(\theta_{2(n+1), 2(n+1)+1}) = p_Q(\theta_{n+1}).$$

Finally, observe $I_F(\theta) = 0$ and $\theta < \bar{\theta}$ implies $\theta < \bar{\theta}_F$. Hence, because $(\theta_n)_{n \in \mathbb{N}}$ is a strictly decreasing sequence, it has at most one element weakly above $\bar{\theta}_F$, and so it is without loss to take $(\theta_n)_{n \in \mathbb{N}}$ to be strictly below $\bar{\theta}_F$, as desired.

We now construct an F -ICC mechanism for every θ_n in the above sequence. Let $\delta_n := p_Q(\theta_n) - p_Q(\theta^*) > 0$. Define

$$p_n(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta \leq \theta^* \\ p_Q(\theta) - \delta_n & \text{if } \theta \geq \theta_n \\ p_Q(\theta^*) & \text{if } \theta \in [\theta^*, \theta_n], \end{cases}$$

Note $I_F(\theta^*) = 0$ because p is not constant around θ^* . Using this fact and the fact that p is an F -marginal price, it is straightforward to verify that p_n is an F -marginal price as well

for every n . We let $Q_n := Q_{p_n}$ be the F -ICC allocation induced by p_n .

Our next goal is to apply Lemma 7 to get a first-order condition indexed by n . For this purpose, observe that for $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$,

$$V_{Q_n}(\theta) - V_Q(\theta) = \int_{\theta^* \wedge \theta}^{\theta_n \wedge \theta} (p_Q(\theta^*) - p_Q(\tilde{\theta})) d\tilde{\theta} - \delta_n (\theta - \theta_n \wedge \theta).$$

Therefore, Lemma 7 delivers the following inequality for all n :

$$\begin{aligned} 0 &\geq \int_{\theta \geq \theta_n} (\theta - \kappa' \circ Q(\theta)) (-\delta_n) F(d\theta) + \int_{\theta \in [\theta^*, \theta_n]} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta^*) - p_Q(\theta)) F(d\theta) \\ &\quad - \int_{\theta \geq \theta^*} \int_{\theta^*}^{\theta_n \wedge \theta} (p_Q(\theta^*) - p_Q(\tilde{\theta})) d\tilde{\theta} F(d\theta) - \int_{\theta \geq \theta_n} -\delta_n (\theta - \theta_n) F(d\theta) \\ &= \int_{\theta \geq \theta_n} (\kappa' \circ Q(\theta) - \theta_n) \delta_n F(d\theta) + \int_{\theta \in [\theta^*, \theta_n]} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta^*) - p_Q(\theta)) F(d\theta) \\ &\quad - \int_{\theta \geq \theta^*} \int_{\theta^*}^{\theta_n \wedge \theta} (p_Q(\theta^*) - p_Q(\tilde{\theta})) d\tilde{\theta} F(d\theta). \end{aligned}$$

Dividing both sides by δ_n and noting that $p_Q(\theta^*) \leq p_Q(\theta)$ for all $\theta \geq \theta^*$ delivers

$$\begin{aligned} 0 &\geq \int_{\theta \geq \theta_n} (\kappa' \circ Q(\theta) - \theta_n) F(d\theta) - \int_{\theta \in [\theta^*, \theta_n]} (\kappa' \circ Q(\theta) - \theta) \left(\frac{p_Q(\theta^*) - p_Q(\theta)}{\delta_n} \right) F(d\theta) \\ &\geq \int_{\theta \geq \theta_n} (\kappa' \circ Q(\theta) - \theta_n) F(d\theta) - \int_{\theta \in [\theta^*, \theta_n]} |\kappa' \circ Q(\theta) - \theta| \left| \frac{p_Q(\theta^*) - p_Q(\theta)}{\delta_n} \right| F(d\theta). \end{aligned} \tag{21}$$

We now show taking the limit of equation (21) as $n \rightarrow \infty$ delivers equation (15). To do so, observe first $\mathbf{1}_{[\theta_n, \infty)}(\theta) (\kappa' \circ Q(\theta) - \theta_n)$ converges pointwise to $\mathbf{1}_{(\theta^*, \infty)}(\theta) (\kappa' \circ Q(\theta) - \theta^*)$. Second, notice $|p_Q(\theta^*) - p_Q(\theta)| \leq \delta_n$ for all $\theta \in [\theta_n, \theta^*)$, and that $\theta - \kappa' \circ Q(\theta) \leq \bar{\theta}_F + \kappa' \circ Q(\bar{\theta}_F)$ for all $\theta \in [\theta_n, \theta^*)$. Therefore, an M exists such that

$$|\theta - \kappa' \circ Q(\theta)| \left| \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) \right| \leq M$$

for all n . Substituting these facts back into (21) gives

$$\begin{aligned}
0 &\geq \int_{\theta \geq \theta_n} (\kappa' \circ Q(\theta) - \theta_n) F(d\theta) - \int_{\theta \in [\theta^*, \theta_n]} |\kappa' \circ Q(\theta) - \theta| \left| \frac{p_Q(\theta^*) - p_Q(\theta)}{\delta_n} \right| F(d\theta) \\
&\geq \int \mathbf{1}_{[\theta_n, \infty)}(\theta) (\kappa' \circ Q(\theta) - \theta_n) F(d\theta) - M(F_-(\theta_n) - F(\theta^*)) \\
&\rightarrow \int \mathbf{1}_{(\theta^*, \infty)}(\theta) (\kappa' \circ Q(\theta) - \theta^*) F(d\theta) = \int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(d\theta) - \theta^* (1 - F(\theta^*)),
\end{aligned}$$

where convergence follows from right continuity of F and the Lebesgue dominated convergence theorem. Hence, we've shown (16) holds for θ^* . Lemma 8 then implies (15) also holds, as desired. $\square$

We have now completed the proof of Lemma 9.

A.4.2. Information Perturbations

In this section we discuss the consequences of applying a perturbation based approach for analyzing the information design program (8). Such perturbations must satisfy two broad restrictions. The first restriction is that the perturbation must result in a signal—that is, a mean-preserving contraction of F_0 . This restriction is satisfied whenever the perturbation creates a mean-preserving contraction of the original distribution, or when the perturbation creates a (small) mean-preserving spread over a set of F -pooling types. The second restriction is that the original allocation must be an ICC allocation with respect to the perturbed signal. This requirement means one can only alter F in regions where p is constant.

Lemma 10 outlines the consequence of two perturbation satisfying the above-mentioned restrictions. The lemma's first part identifies situations in which one can conduct mean-preserving contractions in the buyer's signal. Since such contractions cannot be profitable for the seller, they imply π_Q must satisfy the convex inequality (23). The lemma's second part identifies situations where one can spread the buyer's signal in mean-preserving manner. Profit maximization then delivers that π_Q must satisfy the concave inequality (22).

Lemma 10. *Let (Q^*, F^*) be monopolist-optimal, and suppose Q^* is F^* -ICC. Suppose p_{Q^*} is constant over $(\theta_*, \theta^*) \subseteq [\underline{\theta}_Q, \bar{\theta}_Q]$, and $F(\theta^*) > F_-(\theta_*)$. Then,*

(i) *For every $\theta_1, \theta_2 \in \text{supp } F^*(\cdot | \theta \in [\theta_*, \theta^*])$ and every $\alpha \in [0, 1]$,*

$$\pi_{Q^*}(\alpha\theta_1 + (1 - \alpha)\theta_2) \leq \alpha\pi_{Q^*}(\theta_1) + (1 - \alpha)\pi_{Q^*}(\theta_2) \quad (22)$$

(ii) If $\theta_1 < \theta_2$ are such that θ_1 satisfies either $I_{F^*}(\theta_1) > 0$ or $F_0(\theta_1) > F^*(\theta_1)$, θ_2 satisfies either $I_{F^*}(\theta_2) > 0$ or $F_-^*(\theta_2) > F_{0-}(\theta_2)$, and $I_{F^*}(\theta) > 0$ for all $\theta \in (\theta_1, \theta_2) \subseteq [\theta_*, \theta^*]$, then

$$\pi_{Q^*}(\alpha\theta_1 + (1 - \alpha)\theta_2) \geq \alpha\pi_{Q^*}(\theta_1) + (1 - \alpha)\pi_{Q^*}(\theta_2) \quad (23)$$

for all $\alpha \in [0, 1]$ such that $\alpha\theta_1 + (1 - \alpha)\theta_2 \in \text{supp } F^*(\cdot | \theta \in [\theta_*, \theta^*])$.

The lemma's first part says that, for appropriately chosen θ_1 and θ_2 , the monopolist cannot benefit from contracting the mass on θ_1 and θ_2 into $\alpha\theta_1 + (1 - \alpha)\theta_2$. Similarly, the lemma's second condition says the monopolist cannot benefit from the spreading mass on $\alpha\theta_1 + (1 - \alpha)\theta_2$ across θ_1 and θ_2 . For a rough proof sketch, consider the lemma's part (i), and suppose that F^* has atoms at θ_1 and θ_2 . As explained above, that p_{Q^*} is constant on $[\theta_*, \theta^*]$ means one can pool together some mass from θ_1 and θ_2 without violating the buyer's incentive constraints. It follows that such pooling cannot benefit the monopolist; that is, (22) must hold. To prove the result without atoms, we approximate θ_1 and θ_2 with a shrinking neighborhood. The intuition for part (ii) of the lemma is similar: if equation (23) did not hold, the monopolist would strictly benefit from having the buyer spread the mass he puts on (a small neighborhood around) $\alpha\theta_1 + (1 - \alpha)\theta_2$ across θ_1 and θ_2 , thereby violating optimality of F^* .

Before discussing the formal proof, observe first that both parts of the lemma trivially hold when $\theta_1 = \theta_2$ or when $\alpha \in \{0, 1\}$. Therefore, suppose (without loss of generality) that $\theta_1 < \theta_2$.

Broadly speaking, the formal proof of the lemma proceeds as follows. Using that p_{Q^*} is constant on $[\theta_*, \theta^*]$, we construct a family of informational deviations which are incentive compatible for the buyer and that are indexed by $\epsilon > 0$. As ϵ vanishes, the difference between these deviations and F^* converges to the difference between an atom at $\alpha\theta_1 + (1 - \alpha)\theta_2$ and a split of that atom's mass between an atom on θ_1 and an atom on θ_2 for the first part, and vice-versa for the second part. Then, we show the desired inequality using optimality of (Q^*, F^*) and continuity of $\pi_{Q^*}|_{[\theta_*, \theta^*]}$ (where the latter is implied by continuity of c' and p_{Q^*} being constant over $[\theta_*, \theta^*]$).

We now proceed with the actual proof. As a preliminary step, let $G = F^*(\cdot | \theta \in [\theta_*, \theta^*])$, $H = F^*(\cdot | \theta \notin [\theta_*, \theta^*])$, and $\beta = F^*(\theta^*) - F_-^*(\theta_*)$, and observe $F^* = \beta G + (1 - \beta) H$.

In addition, notice that N_{Q^*} is affine on $[\theta_*, \theta^*]$, because for any $\theta \in [\theta_*, \theta^*] \subseteq [\underline{\theta}_{Q^*}, \bar{\theta}_{Q^*}]$,

$$\begin{aligned} N_{Q^*} &= N_{Q^*}(\theta_*) + \int_{\theta_*}^{\theta} (Q^* - c')(\theta) d\theta = N_{Q^*}(\theta_*) + \int_{\theta_*}^{\theta} p_{Q^*}(\theta) d\theta \\ &= N_{Q^*}(\theta_*) + p_{Q^*}(\theta_*) (\theta - \theta_*), \end{aligned}$$

where the last equality follows from p_{Q^*} being constant on $[\theta_*, \theta^*] \subseteq [\underline{\theta}_{Q^*}, \bar{\theta}_{Q^*}]$.

Proof of Part (i). We begin by constructing the above-mentioned class of informational deviations. Take any $\epsilon \in (0, \frac{1}{2}(\theta_2 - \theta_1))$ (which implies $[\theta_1 - \epsilon, \theta_1 + \epsilon] \cap [\theta_2 - \epsilon, \theta_2 + \epsilon] = \emptyset$), and define the following objects:

$$\begin{aligned} G_{0,\epsilon} &= G(\cdot | \theta \notin [\theta_1 - \epsilon, \theta_1 + \epsilon] \cup [\theta_2 - \epsilon, \theta_2 + \epsilon]), \\ G_{1,\epsilon} &= G(\cdot | \theta \in [\theta_1 - \epsilon, \theta_1 + \epsilon]), \\ G_{2,\epsilon} &= G(\cdot | \theta \in [\theta_2 - \epsilon, \theta_2 + \epsilon]), \\ \gamma_{1,\epsilon} &= G(\theta_1 + \epsilon) - G_-(\theta_1 - \epsilon), \\ \gamma_{2,\epsilon} &= G(\theta_2 + \epsilon) - G_-(\theta_2 - \epsilon), \\ \gamma_{0,\epsilon} &= 1 - \gamma_{1,\epsilon} - \gamma_{2,\epsilon}. \end{aligned}$$

Clearly, $G = \sum_{i=0}^2 \gamma_{i,\epsilon} G_{i,\epsilon}$. Moreover, since $\theta_1, \theta_2 \in \text{supp } G$, both $\gamma_{1,\epsilon}$ and $\gamma_{2,\epsilon}$ are strictly positive for all $\epsilon > 0$. For any $\epsilon \in (0, \frac{1}{2}(\theta_2 - \theta_1))$, define

$$\begin{aligned} \theta_\epsilon &= \int \theta d(\alpha G_{1,\epsilon} + (1 - \alpha) G_{2,\epsilon}), \\ \tilde{\gamma}_\epsilon &= \min\{\gamma_{1,\epsilon}, \gamma_{2,\epsilon}\} > 0, \text{ and} \\ G_\epsilon &= \gamma_{0,\epsilon} G_{0,\epsilon} + \tilde{\gamma}_\epsilon \mathbf{1}_{[\theta_\epsilon, \infty)} + (\gamma_{1,\epsilon} - \alpha \tilde{\gamma}_\epsilon) G_{1,\epsilon} + (\gamma_{2,\epsilon} - (1 - \alpha) \tilde{\gamma}_\epsilon) G_{2,\epsilon}. \end{aligned}$$

In words, G_ϵ alters G by pooling $\alpha \tilde{\gamma}_\epsilon$ mass from the ϵ -ball around θ_1 and $(1 - \alpha) \tilde{\gamma}_\epsilon$ mass from the ϵ -ball around θ_2 and pooling them to create an $\tilde{\gamma}_\epsilon > 0$ mass on θ_ϵ ; that is,

$$G_\epsilon - G = \tilde{\gamma}_\epsilon \left(\mathbf{1}_{[\theta_\epsilon, \infty)} - (\alpha G_{1,\epsilon} + (1 - \alpha) G_{2,\epsilon}) \right).$$

With the above in hand, we can finally define our informational perturbation: specifically, take $F_\epsilon = \beta G_\epsilon + (1 - \beta) H$.

Next, we argue $F_\epsilon \in \mathcal{I}$ and that Q^* is F_ϵ -IC. For the first claim, observe that because $\alpha G_{1,\epsilon} + (1 - \alpha) G_{2,\epsilon} \succ \mathbf{1}_{[\theta_\epsilon, \infty)}$, G_ϵ is less informative than G , and so $F_\epsilon \preceq F^* \preceq F_0$. That

$F_\epsilon \in \mathcal{I}$ follows from $\preceq$ being transitive. To see Q^* is F_ϵ -IC for all $\epsilon \in (0, \frac{1}{2}(\theta_2 - \theta_1))$, observe that

$$\begin{aligned} \int N_{Q^*}(\theta) (F_\epsilon - F^*) (d\theta) &= \beta \int N_{Q^*}(\theta) (G_\epsilon - G) (d\theta) \\ &= \beta \tilde{\gamma}_\epsilon \int N_{Q^*}(\theta) [\mathbf{1}_{[\theta_\epsilon, \infty)} - (\alpha G_{1,\epsilon} + (1 - \alpha)G_{2,\epsilon})] (d\theta) = 0, \end{aligned}$$

where the last equality follows from $\alpha G_{1,\epsilon} + (1 - \alpha)G_{2,\epsilon} \succ \mathbf{1}_{[\theta_\epsilon, \infty)}$, the support of $\alpha G_{1,\epsilon} + (1 - \alpha)G_{2,\epsilon}$ being contained in $[\theta_*, \theta^*] \subseteq [\underline{\theta}_Q, \bar{\theta}_Q]$, and N_{Q^*} being affine on $[\theta_*, \theta^*]$.

Now, because (Q^*, F^*) is monopolist-optimal, that Q^* is F_ϵ -IC all small $\epsilon > 0$ means that $\int \pi_{Q^*} dF_\epsilon \leq \int \pi_{Q^*} dF$. Rearranging this inequality, dividing by $\beta \tilde{\gamma}_\epsilon$, and taking ϵ to zero delivers

$$\begin{aligned} 0 &\leq \frac{1}{\beta \tilde{\gamma}_\epsilon} \int \pi_{Q^*}(\theta) (F^* - F_\epsilon) (d\theta) = \frac{1}{\tilde{\gamma}_\epsilon} \int \pi_{Q^*}(\theta) (G - G_\epsilon) (d\theta) \\ &= \left[\alpha \int \pi_{Q^*}(\theta) G_{1,\epsilon} (d\theta) + (1 - \alpha) \int \pi_{Q^*}(\theta) G_{2,\epsilon} (d\theta) \right] - \pi_{Q^*}(\theta_\epsilon) \\ &\rightarrow (\alpha \pi_{Q^*}(\theta_1) + (1 - \alpha) \pi_{Q^*}(\theta_2)) - \pi_{Q^*}(\alpha \theta_1 + (1 - \alpha) \theta_2), \end{aligned}$$

where convergence follows from continuity of $\pi_{Q^*}|_{[\theta_*, \theta^*]}$, convergence of $G_{1,\epsilon}$ and $G_{2,\epsilon}$ to $\mathbf{1}_{[\theta_1, \infty)}$ and $\mathbf{1}_{[\theta_2, \infty)}$ respectively, and $\theta_\epsilon \rightarrow \alpha \theta_1 + (1 - \alpha) \theta_2$.

Proof of Part (ii). Suppose now $\theta_1 < \theta_2$ are such that θ_1 satisfies either $I_F(\theta_1) > 0$ or $F_0(\theta_1) > F(\theta_1)$, θ_2 satisfies $I_F(\theta_2) > 0$ or $F_-(\theta_2) > F_{0-}(\theta_2)$, and $I_F(\theta) > 0$ for all $\theta \in (\theta_1, \theta_2) \subseteq [\theta_*, \theta^*]$. Let $\alpha \in (0, 1)$ is such that $\theta_\alpha := \alpha \theta_1 + (1 - \alpha) \theta_2 \in \text{supp } G$. We begin by defining the following family of deviations. For any strictly positive $\epsilon < \min \{\theta_* - \theta_1, \theta_2 - \theta^*\}$, define

$$\begin{aligned} G_{0,\epsilon}(\cdot) &:= G(\cdot | \theta \notin [\theta_\alpha - \epsilon, \theta_\alpha + \epsilon]), \\ G_{1,\epsilon}(\cdot) &:= G(\cdot | \theta \in [\theta_\alpha - \epsilon, \theta_\alpha + \epsilon]), \\ \theta_\epsilon &:= \int \theta G_{1,\epsilon} (d\theta) \\ \gamma_\epsilon &:= G(\theta_\alpha + \epsilon) - G_-(\theta_\alpha - \epsilon). \end{aligned}$$

Clearly, $G = (1 - \gamma_\epsilon)G_{0,\epsilon} + \gamma_\epsilon G_{1,\epsilon}$. Observe $\gamma_\epsilon > 0$, because $\theta_\alpha \in \text{supp } G$, and that an $\alpha_\epsilon \in (0, 1)$ exists such that

$$\theta_\epsilon = \alpha_\epsilon \theta_1 + (1 - \alpha_\epsilon) \theta_2,$$

by our choice of ϵ . Obviously, $\theta_\epsilon \rightarrow \theta_\alpha$ and $\alpha_\epsilon \rightarrow \alpha$ as $\epsilon \rightarrow 0$. For a given $\tilde{\gamma} \in (0, \gamma_\epsilon)$,

define

$$G_{\tilde{\gamma},\epsilon} = (1 - \gamma_\epsilon)G_{0,\epsilon} + (\gamma_\epsilon - \tilde{\gamma})G_{1,\epsilon} + \tilde{\gamma} \left(\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2,\infty)} \right).$$

Clearly, $G_{\tilde{\gamma},\epsilon}$ is a CDF.

We now construct our informational deviation: set $F_{\tilde{\gamma},\epsilon} := \beta G_{\tilde{\gamma},\epsilon} + (1 - \beta) H$ for all ϵ and $\tilde{\gamma}$ satisfying the above conditions. We begin by arguing that this deviation is a signal—that is, $F_{\tilde{\gamma},\epsilon} \in \mathcal{I}$ —whenever $\tilde{\gamma}$ is sufficiently small (holding ϵ fixed). To do so, observe that the function $F \mapsto I_F(\theta)$ is affine for all θ , meaning that

$$I_{F_{\tilde{\gamma},\epsilon}} - I_{F^*} = \tilde{\gamma}\beta \left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}} + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}} - I_{G_{1,\epsilon}} \right) \leq 0, \quad (24)$$

where the inequality follows from $G_{1,\epsilon}$ being a mean preserving contraction of $\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2,\infty)}$. This latter fact also implies that the above inequality holds with equality whenever $\theta \leq \theta_1$ or $\theta \geq \theta_2$. It follows that for all such θ , $I_{F_{\tilde{\gamma},\epsilon}}(\theta) = I_{F^*}(\theta) \geq 0$, with equality whenever $\theta \in \{\underline{\theta}, \bar{\theta}\}$.

Next we show $I_{F_{\tilde{\gamma},\epsilon}}(\theta) \geq 0$ for all $\theta \in (\theta_1, \theta_2)$ whenever $\tilde{\gamma}$ is sufficiently small. Towards this goal, we first argue one can find $\tilde{\theta}_1, \tilde{\theta}_2$ such that the following properties hold: (a) $\theta_1 \leq \tilde{\theta}_1 < \tilde{\theta}_2 \leq \theta_2$; (b) $I_{F^*}(\theta) > 0$ for all $\theta \in [\tilde{\theta}_1, \tilde{\theta}_2]$; (c) $\zeta_1 := \inf_{\theta \in (\theta_1, \tilde{\theta}_1)} (F_0 - F^*)(\theta) > 0$ and $\zeta_2 := \inf_{\theta \in (\tilde{\theta}_2, \theta_2)} (F^* - F_0)(\theta) > 0$. We begin with defining $\tilde{\theta}_1$. If $I_{F^*}(\theta_1) > 0$, set $\tilde{\theta}_1 = \theta_1$. If $I_{F^*}(\theta_1) = 0$, then $F^*(\theta_1) < F_0(\theta_1)$ but the Lemma's assumptions. By right continuity of F^* and F_0 , a small $\nu > 0$ exists such that $F_0(\theta) - F^*(\theta) > \nu$ for all $\theta \in (\theta_1, \theta_1 + \nu)$: otherwise, one could find a decreasing sequence $(\theta_n)_{n \in \mathbb{N}}$ that converges to θ_1 and has $F_0(\theta_n) \leq F^*(\theta_n) + 1/n$, delivering $F_0(\theta_1) \leq F^*(\theta_1)$, which is a contradiction. Defining $\tilde{\theta}_2$ analogously (and using left continuity of F_{0-} and F^*) delivers a $\tilde{\theta}_1$ and $\tilde{\theta}_2$ as desired (where property (b) is guaranteed by $I_{F^*} > 0$ over (θ_1, θ_2)).

Armed with $\tilde{\theta}_1$ and $\tilde{\theta}_2$ as above, we now find $\tilde{\gamma} > 0$ small such that $I_{F_{\tilde{\gamma},\epsilon}}(\theta) \geq 0$ holds for all $\theta \in (\theta_1, \theta_2)$. Define

$$\zeta_3 := \frac{\min_{\theta \in [\tilde{\theta}_1, \tilde{\theta}_2]} I_{F^*}(\theta)}{\max_{\theta \in [\tilde{\theta}_1, \tilde{\theta}_2]} \left[I_{G_{1,\epsilon}}(\theta) - \alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}}(\theta) + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}}(\theta) \right]}.$$

Note ζ_3 is well-defined because $\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2,\infty)}$ is a strict mean-preserving spread of $G_{1,\epsilon}$, and so the denominator is strictly positive. By choice of $\tilde{\theta}_1$ and $\tilde{\theta}_2$ and continuity of I_{F^*} , the numerator is also strictly positive, meaning $\zeta_3 > 0$. Take any $\tilde{\gamma} <$

$\min\{\zeta_1, \zeta_2, \zeta_3\}/\beta$. We argue $I_{F_{\tilde{\gamma},\epsilon}}(\theta) \geq 0$ for all $\theta \in (\theta_1, \theta_2)$ by dividing (θ_1, θ_2) into three subintervals.

Consider first $\theta \in (\tilde{\theta}_1, \tilde{\theta}_2)$. For such θ ,

$$\begin{aligned} I_{F_{\tilde{\gamma},\beta}}(\theta) &= I_{F^*}(\theta) + \tilde{\gamma}\beta \left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}}(\theta) + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}}(\theta) - I_{G_{1,\epsilon}}(\theta) \right) \\ &\geq I_{F^*}(\theta) - \min_{\theta \in [\tilde{\theta}_1, \tilde{\theta}_2]} I_{F^*}(\theta) \geq 0. \end{aligned}$$

Second, take any $\theta \in [\theta_1, \tilde{\theta}_1]$. For such θ ,

$$(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}} + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}} - I_{G_{1,\epsilon}})(\theta) \leq \alpha_\epsilon(\theta - \theta_1) \leq \theta - \theta_1.$$

Moreover, by definition of ζ_1 , $I_{F^*}(\theta) \geq I_{F^*}(\theta_1) + \zeta_1(\theta - \theta_1)$. Therefore, by choice of $\tilde{\gamma}$,

$$\begin{aligned} I_{F_{\tilde{\gamma},\beta}}(\theta) &= I_{F^*}(\theta) + \tilde{\gamma}\beta \left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}}(\theta) + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}}(\theta) - I_{G_{1,\epsilon}}(\theta) \right) \\ &\geq I_{F^*}(\theta_1) + (\zeta_1 - \tilde{\gamma}\beta)(\theta - \theta_1) \geq 0. \end{aligned}$$

Third, take any $\theta \in [\tilde{\theta}_2, \theta_2]$. Some simple algebra reveals that for such θ ,

$$(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}} + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}} - I_{G_{1,\epsilon}})(\theta) \leq \alpha_\epsilon(\theta_2 - \theta) \leq \theta_2 - \theta.$$

Moreover,

$$I_{F^*}(\theta) = I_{F^*}(\theta_2) - \int_{\theta}^{\theta_2} (F_{0-} - F_{-}^*)(\hat{\theta}) d\hat{\theta} \geq I_{F^*}(\theta_2) + \zeta_2(\theta_2 - \theta).$$

Thus,

$$\begin{aligned} I_{F_{\tilde{\gamma},\beta}}(\theta) &= I_{F^*}(\theta) + \tilde{\gamma}\beta \left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}}(\theta) + (1 - \alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}}(\theta) - I_{G_{1,\epsilon}}(\theta) \right) \\ &\geq I_{F^*}(\theta_2) + (\zeta_2 - \tilde{\gamma}\beta)(\theta_2 - \theta) \geq 0. \end{aligned}$$

It follows that $F_{\tilde{\gamma},\epsilon} \in \mathcal{I}$ for all sufficiently small $\tilde{\gamma}$.

We now argue Q^* is $F_{\tilde{\gamma},\epsilon}$ -IC for all above-mentioned ϵ and all sufficiently small $\tilde{\gamma}$. To see this, observe that

$$\begin{aligned} \int N_{Q^*}(\theta) (F_{\tilde{\gamma},\epsilon} - F^*) (d\theta) &= \tilde{\gamma}\beta \int N_{Q^*}(\theta) \left(\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2,\infty)} - G_{1,\epsilon} \right) (d\theta) \\ &= \tilde{\gamma}\beta (\alpha_\epsilon N_{Q^*}(\theta_1) + (1 - \alpha_\epsilon) N_{Q^*}(\theta_2) - N_{Q^*}(\theta_\epsilon)) = 0, \end{aligned}$$

where the last equality follows from $\alpha_\epsilon \mathbf{1}_{[\theta_1, \infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2, \infty)} \succeq G_{1, \epsilon}$, the support of $\alpha_\epsilon \mathbf{1}_{[\theta_1, \infty)} + (1 - \alpha_\epsilon) \mathbf{1}_{[\theta_2, \infty)}$ and $G_{1, \epsilon}$ being contained in $[\theta_*, \theta^*]$, and N_{Q^*} being affine on $[\theta_*, \theta^*]$.

For the proof's last step, observe that because Q^* is $F_{\tilde{\gamma}, \epsilon}$ -IC for the buyer for all small ϵ and $\tilde{\gamma}$, monopolist optimality of (Q^*, F^*) implies

$$0 \geq \frac{1}{\tilde{\gamma}} \int \pi_{Q^*}(\theta) (F_{\tilde{\gamma}, \epsilon} - F^*) (d\theta) = \alpha_\epsilon \pi_{Q^*}(\theta_1) + (1 - \alpha_\epsilon) \pi_{Q^*}(\theta_2) - \pi(\theta_\epsilon) \\ \xrightarrow{\epsilon \rightarrow 0} \alpha \pi_{Q^*}(\theta_1) + (1 - \alpha) \pi_{Q^*}(\theta_2) - \pi(\theta_\alpha),$$

where convergence follows from $\theta_\epsilon \rightarrow \theta_\alpha$, $\alpha_\epsilon \rightarrow \alpha$, and π_{Q^*} being continuous on $[\theta_*, \theta^*]$. The desired inequality follows.

A.4.3. Proof of Theorem 2

Without loss, we can assume (Q, F) is a monopolist optimal outcome with the property that Q is F -ICC.

We first argue that if (15) holds at $\theta^* \in [\underline{\theta}_F, \bar{\theta}_F)$, then $\kappa'(Q(\theta^*)) < \theta^*$. To do so, note that because the allocation Q is F -ICC, the allocation is strictly increasing over $[\underline{\theta}_F, \bar{\theta}_F)$. When κ is strictly convex, the marginal cost for quality provision is strictly increasing as well, and so equation (15) implies that

$$\kappa'(Q(\theta^*)) < \int_{\theta \geq \theta^*} \kappa'(Q(\theta)) F^*(d\theta | \theta \geq \theta^*) \leq \theta^*;$$

that is, $Q(\theta^*)$ lies strictly below its efficient level.

Next, we argue the inequality (15) holds for every $\theta \in [\underline{\theta}_F, \bar{\theta}_F)$ at which p_Q is strictly increasing, and so $Q(\theta)$ lies below its efficient level. If $\theta > \underline{\theta}_F$, this claim directly follows from Lemma 9-(iii). For $\theta = \underline{\theta}_F$, distinguish two cases: either $\underline{\theta}_F > \underline{\theta}_Q$ and so we can apply Lemma 9-(ii), or $\underline{\theta}_F = \underline{\theta}_Q$, in which case $\kappa' \circ Q(\underline{\theta}_Q) = \kappa'(0) < \underline{\theta} \leq \underline{\theta}_F$. Either way, the inequality (15) applies. It follows quality is inefficiently low (strictly) at all such θ .

Next, we argue that quality is inefficiently low for any $\theta^* \in [\underline{\theta}_F, \bar{\theta})$ around which p_Q is constant. Thus suppose $\theta^* \in \text{supp } F$ is such that p_Q is constant on $[\theta^* - \delta, \theta^* + \delta]$ for some $\delta > 0$. Our goal is to show $\kappa' \circ Q(\theta^*) < \theta^*$.

Let $\theta_* = \inf \{\theta \geq \underline{\theta}_F : p_{Q+}(\theta) = p_Q(\theta^*)\}$. We now argue (15) holds at θ_* . There are three cases to consider: $\theta_* = \underline{\theta}_F = \underline{\theta}_Q$, $\theta_* = \underline{\theta}_F > \underline{\theta}_Q$, and $\theta_* > \underline{\theta}_F$. In the first case we have $Q(\theta_*) = 0$ and so $\kappa' \circ Q(\theta_*) < \underline{\theta} \leq \underline{\theta}_F$. In the second case, the desired inequality

follows from Lemma 9-(ii). In the third case, p_Q must be strictly increasing at θ_* , and so (15) must hold by Lemma 9-(iii), and so $\kappa' \circ Q(\theta_*) < \theta_*$.

Define $\bar{\theta}^* = (\theta^* + \delta) \wedge \bar{\theta}_F$, and let $G := F(\cdot | \theta \in [\theta_*, \bar{\theta}^*])$. We claim (15) holds for

$$\theta' := \min(\text{supp } G).$$

Clearly, we are done if $\theta' = \theta_*$. If $\theta' > \theta_*$, then $\theta_* \notin \text{supp } G$, and so $F_-(\theta_*) = F(\theta_*) = F_-(\theta')$. We therefore have the following inequality chain:

$$\begin{aligned} (1 - F_-(\theta')) \theta' &> (1 - F_-(\theta')) \theta_* = (1 - F_-(\theta_*)) \theta_* \\ &\geq \int_{\theta \geq \theta_*} \kappa' \circ Q(\theta) F(d\theta) = \int_{\theta \geq \theta'} \kappa' \circ Q(\theta) F(d\theta), \end{aligned}$$

where the weak inequality follows from (15) holding at θ_* . Thus, we have shown (16) holds at θ' , and so (15) holds as well (see Lemma 8).

If $\theta^* = \theta'$, then (15) holds for θ^* , and so $\kappa' \circ Q(\theta^*) < \theta^*$, as explained after Lemma 9. Hence, there is nothing left to prove in this case. Thus, hereafter, we suppose $\theta^* \neq \theta'$. Since $\theta^* \in \text{supp } G$, we must have $\theta^* > \theta'$.

We now argue p_Q is constant on $[\theta', \bar{\theta}^*]$. To do so, notice $\kappa' \circ Q(\theta') < \theta'$ implies $Q(\theta') = Q_+(\theta')$ since Q jumps towards efficiency. Hence,

$$p_Q(\theta') = Q(\theta') - c'(\theta') = Q_+(\theta') - c'(\theta') = p_{Q_+}(\theta') = p_Q(\theta^*),$$

where the last equality follows from $\theta' \geq \theta_*$. It follows p_Q is constant on $[\theta', \theta^*] \cup [\theta^* - \delta, \bar{\theta}^*] = [\theta', \bar{\theta}^*]$. Recalling $\bar{\theta}^* = \min\{\bar{\theta}_F, \theta^* + \delta\}$, it follows $p_Q(\bar{\theta}^*) = p_Q(\theta^*)$. Thus, we have shown p_Q is constant on $[\theta', \bar{\theta}^*]$.

Consider now the line segment connecting $(\theta', \pi_Q(\theta'))$ with $(\theta^*, \pi_Q(\theta^*))$,

$$\begin{aligned} \varphi : [\theta', \theta^*] &\rightarrow \mathbb{R}, \\ \theta &\mapsto \pi_Q(\theta') + \left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'} \right) (\theta - \theta'). \end{aligned}$$

We claim $\varphi(\theta) \geq \pi_Q(\theta)$ for all $\theta \in [\theta', \theta^*]$. Obviously, $\varphi(\theta) = \pi_Q(\theta)$ whenever $\theta \in$

$\{\theta', \theta^*\}$. For $\theta \in (\theta', \theta^*)$, we get the following inequality:

$$\begin{aligned}\varphi(\theta) &= \left(\frac{\theta^* - \theta}{\theta^* - \theta'}\right) \varphi(\theta') + \left(\frac{\theta - \theta'}{\theta^* - \theta'}\right) \varphi(\theta^*) \\ &= \left(\frac{\theta^* - \theta}{\theta^* - \theta'}\right) \pi_Q(\theta') + \left(\frac{\theta - \theta'}{\theta^* - \theta'}\right) \pi_Q(\theta^*) \geq \pi_Q(\theta),\end{aligned}$$

where the inequality follows from Lemma 10 part (i), which applies because p_Q is constant on $[\theta', \bar{\theta}^*]$.

Next, we show $\left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right)$ is strictly positive. For this purpose, fix any $\epsilon \in (0, \theta^* - \theta')$. Observe p_Q is constant on $[\theta', \bar{\theta}^*]$ means that

$$Q(\theta' + \epsilon) - Q_+(\theta') = Q(\theta' + \epsilon) - Q(\theta') = c'(\theta' + \epsilon) - c'(\theta').$$

It follows $Q'_+(\theta') = c''(\theta')$, delivering the following inequality chain,

$$\begin{aligned}\left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right) &= \frac{1}{\epsilon} [\varphi(\theta' + \epsilon) - \varphi(\theta')] \\ &\geq \frac{1}{\epsilon} [\pi_Q(\theta' + \epsilon) - \pi_Q(\theta')] \\ &= \frac{1}{\epsilon} (\theta' (Q(\theta' + \epsilon) - Q(\theta'))) - \frac{1}{\epsilon} [\kappa \circ Q(\theta' + \epsilon) - \kappa \circ Q(\theta')] \\ &\quad + Q(\theta' + \epsilon) - \frac{1}{\epsilon} [V_Q(\theta' + \epsilon) - V_Q(\theta')] \\ &\rightarrow (\theta' - \kappa' \circ Q(\theta')) c''(\theta') > 0,\end{aligned}$$

where convergence follows from the chain rule and $V'_{Q+}(\theta') = Q_+(\theta')$, and the strict inequality from c being strictly convex.

We now turn to establishing $\kappa' \circ Q(\theta^*) < \theta^*$, thereby concluding the proof. Toward this goal, notice again that for any $\epsilon \in (0, \theta^* - \theta')$,

$$Q(\theta^*) - Q(\theta^* - \epsilon) = c'(\theta^*) - c'(\theta^* - \epsilon),$$

because p_Q is constant on $[\theta', \bar{\theta}^*]$. Therefore, $Q'_-(\theta^*) = c''(\theta')$. So, we obtain the follow-

ing inequality chain:

$$\begin{aligned}
0 &< \left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'} \right) = \frac{1}{\epsilon} [\varphi(\theta^*) - \varphi(\theta^* - \epsilon)] \\
&\leq \frac{1}{\epsilon} [\pi_Q(\theta^*) - \pi_Q(\theta^* - \epsilon)] \\
&= \frac{1}{\epsilon} \theta^* (Q(\theta^*) - Q(\theta^* - \epsilon)) - \frac{1}{\epsilon} [\kappa \circ Q(\theta^*) - \kappa \circ Q(\theta^* - \epsilon)] \\
&\quad + Q(\theta^* - \epsilon) - \frac{1}{\epsilon} [V_Q(\theta^*) - V_Q(\theta^* - \epsilon)] \\
&\rightarrow (\theta^* - \kappa' \circ Q(\theta^*)) c''(\theta^*),
\end{aligned}$$

where convergence follows from the chain rule and $V'_{Q-}(\theta^*) = Q_-(\theta^*)$. Since $c''(\theta^*) > 0$, the above inequality implies $\kappa' \circ Q(\theta^*) < \theta^*$, as required.

To conclude the proof, it remains to show that quality is efficient at $\bar{\theta}_F$ whenever $\bar{\theta}_F = \bar{\theta}$. Suppose $\bar{\theta}_F = \bar{\theta}$. By the above, one can approach $\bar{\theta}_F$ from below with a sequence of types that receive inefficiently low quality, meaning $Q_-(\bar{\theta}_F)$ must be weakly below the efficient level. Since $Q_+(\bar{\theta}_F) = Q_+(\bar{\theta}) = \bar{q}$, and Q jumps towards efficiency, it must then be that $\bar{\theta}_F$ gets the efficient quality. The result follows. $\square$

A.5. Proofs from Section 5 and Proposition 1

A.5.1. Proof of Theorem 3

Proof of Theorem 3-(i). It is straightforward to show that some $\hat{\gamma}$ exists such that the steep slope condition holds for all $\gamma > \hat{\gamma}$. For such γ , the MPS constraint only binds at $\underline{\theta}$ and $\bar{\theta}$, and only binary signals can be incentive compatible. We consider $\gamma \geq \hat{\gamma}$, and start by showing that one can restrict attention to a compact strict subset of Θ , $[\tilde{\theta}_L, \tilde{\theta}_H] \subseteq (\underline{\theta}, \bar{\theta})$, with $\tilde{\theta}_L < \theta_0 < \tilde{\theta}_H$.

For any $\gamma \geq \hat{\gamma}$, let (F_γ, Q_γ) be an optimal outcome with Q_γ F_γ -CC. Define $\text{supp} F_\gamma = \{\theta_L^\gamma, \theta_H^\gamma\}$, with $\underline{\theta} \leq \theta_L^\gamma \leq \theta_0 \leq \theta_H^\gamma \leq \bar{\theta}$. F -CC implies:

$$\gamma (c'(\theta_H^\gamma) - c'(\theta_L^\gamma)) \leq Q_\gamma(\theta_H) - Q_\gamma(\theta_L) \leq \bar{q},$$

where the first inequality follows from F -CC and p_γ increasing, and the second inequality from $Q_\gamma \in [0, \bar{q}]$.

Thus:

$$c'(\theta_H^\gamma) \leq \frac{\bar{q}}{\gamma} + c'(\theta_L^\gamma) \leq \frac{\bar{q}}{\gamma} + c'(\theta_0),$$

where the second inequality follows from convexity of c and $\theta_L^\gamma \leq \theta_0$. Thus, one can choose a sufficiently high γ so that $\theta_H^\gamma \leq \tilde{\theta}_H < \bar{\theta}$. Similarly:

$$c'(\theta_L^\gamma) \geq c'(\theta_H^\gamma) - \frac{\bar{q}}{\gamma} \geq c'(\theta_0) - \frac{\bar{q}}{\gamma},$$

implying that we can choose sufficiently large γ for $\theta_L^\gamma \geq \tilde{\theta}_L > \underline{\theta}$. Therefore, we can find γ' such that, for $\gamma \geq \max\{\gamma', \tilde{\gamma}\}$, we can focus on $[\tilde{\theta}_L, \tilde{\theta}_H]$, where c', c'', c''' —which are assumed to be continuous—are bounded.

We now move on to the main result, which consists of showing that there exists a $\bar{\gamma}$ such that the principal's objective function is concave for large $\gamma \geq \bar{\gamma}$, for all feasible F -marginal prices, p . For that, start by noticing that, for $\gamma \geq \max\{\gamma', \tilde{\gamma}\}$, the optimal signal is binary, and therefore any $\theta \in (\underline{\theta}, \bar{\theta})$ is F -pooling. As a consequence, we restrict attention to marginal price functions p which are constant in $[\underline{\theta}, \bar{\theta}]$. For any such p , recall the profit function $\pi_p(\theta) = \theta Q_p(\theta) - \kappa(Q_p(\theta)) - V_p(\theta)$. Define $\Theta_p^\gamma = \{\theta : \gamma c'(\theta) + p(\theta) \in [0, \bar{q}]\}$

Because p is constant on the interval Θ_p^γ , π_p is twice differentiable in that interval. We consider the second derivative of π_p with respect to $\theta \in (\tilde{\theta}_L, \tilde{\theta}_H)$:

$$\pi_p''(\theta) = (\theta - \kappa'(Q_p(\theta)))\gamma c'''(\theta) + (1 - \kappa''(Q_p(\theta))\gamma c''(\theta))\gamma c''(\theta),$$

Dividing this expression by $\gamma c''(\theta)$, which is strictly positive by strong convexity of c :

$$\begin{aligned} \frac{\pi_p''(\theta)}{\gamma c''(\theta)} &= (\theta - \kappa'(Q_p(\theta)))\frac{c'''(\theta)}{c''(\theta)} + 1 - \kappa''(Q_p(\theta))\gamma c''(\theta) \\ &\leq \max_{q \in [0, \bar{q}], x \in [\tilde{\theta}_L, \tilde{\theta}_H]} |x - \kappa'(q)| \left| \frac{c'''(x)}{c''(x)} \right| + 1 - \gamma \min_q \kappa''(q) \min_{x \in [\tilde{\theta}_L, \tilde{\theta}_H]} c''(x), \end{aligned}$$

where the first term is bounded because c is strongly convex, and the term multiplying γ is strictly positive, as c and κ are both strongly convex, ensuring that c'' and κ'' are positive and bounded away from zero. Thus, by choosing high enough γ we obtain that π_p'' is negative on Θ_p^γ . Because the bound is uniform across p and θ , we conclude that there exists $\bar{\gamma}$ such that, for $\gamma \geq \bar{\gamma}$, the principal's objective is concave for $\theta \in \Theta_p^\gamma$, for any p . Finally, because $\text{supp} F \subset \Theta_p^\gamma$ for any p , the principal benefits from pooling together all types, regardless of the marginal price function. Therefore, $\text{supp} F_\gamma = \{\theta_0\}$.

□

Proof of Theorem 3-(ii). We now consider $\gamma \geq \bar{\gamma}$, so $\text{supp} F_\gamma = \{\theta_0\}$. For any marginal price $\hat{p}$, re-normalize it as $\hat{p} = \gamma p$. Notice that any feasible $\hat{p}$ for F_γ is constant in $[\underline{\theta}, \bar{\theta})$ and therefore so is the associated normalization p . We use $p(\underline{\theta})$ to denote that constant value.

Define $\underline{\theta}_p^\gamma = \arg \min_{v \in [\underline{\theta}, \bar{\theta}]} \{\gamma c(v) - p(\underline{\theta})(\theta_0 - v)\}$. Note that, if $\gamma c'(\theta_0) + p(\underline{\theta}) = 0$ for any $\theta \in [\underline{\theta}, \bar{\theta}]$, $\underline{\theta}_p^\gamma = \theta_0$. Otherwise, $\underline{\theta}_p^\gamma = \underline{\theta}$. In other words, $\underline{\theta}_p^\gamma$ is the largest type such that the $Q_p(\theta) = 0$ —i.e. for a fixed γ , $\underline{\theta}_p^\gamma = \underline{\theta}_{Q_p}$.

We now rewrite the gross value for buyer θ_0 using the observation above:

$$\begin{aligned} V_p^\gamma(\theta_0) &= \gamma c(\theta_0) - \gamma c(\underline{\theta}_p^\gamma) + p(\underline{\theta})(\theta_0 - \underline{\theta}_p^\gamma) \\ &= \gamma c(\theta_0) - \min_{v \in [\underline{\theta}, \bar{\theta}]} \{\gamma c(v) - p(\underline{\theta})(\theta_0 - v)\} \\ &= \gamma c(\theta_0) - \min_{v \in [\underline{\theta}, \bar{\theta}]} \{\gamma c(v) + \gamma c'(\theta_0)(\theta_0 - v) - Q_p(\theta_0)(\theta_0 - v)\}, \end{aligned} \quad (25)$$

where the last equality follows from noticing that, by F_γ -CC, $Q_p(\theta_0) = \gamma c'(\theta_0) + p(\underline{\theta})$. Two observations about V_p^γ are in order. First, it depends of p only through $Q_p(\theta_0)$. Second, it is decreasing in γ . Indeed, by the envelope theorem, the derivative of V_p^γ with respect to γ is $c(\theta_0) - c(\underline{\theta}_p^\gamma) - c'(\theta_0)(\theta_0 - \underline{\theta}_p^\gamma)$, which is negative by strong convexity of c .

By the first observation, we can write the problem of the seller, given $\text{supp} F_\gamma = \{\theta_0\}$, as an optimization by direct choice of $Q_p(\theta_0)$:

$$U_S := \max_{Q_p(\theta_0) \in [0, \bar{q}]} \theta_0 Q_p(\theta_0) - \kappa(Q_p(\theta_0)) - V_p^\gamma(\theta_0).$$

By the second observation, this objective function is strictly increasing in γ . Thus, the optimal profits must strictly increase in γ . □

Proof of Theorem 3-(iii). Consider $\gamma \geq \bar{\gamma}$ again such that, by Part (i), $\text{supp} F = \{\theta_0\}$, and define $\underline{\theta}_p$ as in the proof of Part (ii). Recall that $c'(\theta_0) = 0$, so we omit it moving forward to save notation, observing that $Q_p(\theta_0) = p(\underline{\theta})$.

We can write the seller's problem, given $\text{supp} F_\gamma = \{\theta_0\}$ as follows:

$$\max_{p(\underline{\theta}) \geq 0} \theta_0 p(\underline{\theta}) - \kappa(p(\underline{\theta})) - \left(\gamma c(\theta_0) - \min_{v \in [\underline{\theta}, \bar{\theta}]} \{\gamma c(v) - p(\underline{\theta})(\theta_0 - v)\} \right).$$

We first argue that, if $\text{supp} F_\gamma = \{\theta_0\}$, then $p(\underline{\theta}) > 0$. Indeed, if $p(\underline{\theta}) = 0$, the seller makes zero profits. However, the seller can always guarantee positive profits by offering any contract that is accepted by all types—for example, offering only the efficient contract of type $\underline{\theta}$, at a price that extracts all rents from that type. Therefore, $p(\underline{\theta}) > 0$.

We can then apply the envelope theorem, to immediately obtain the following interior solution to the seller's problem:

$$\kappa'(p(\underline{\theta})) = \underline{\theta}_p^\gamma.$$

By total differentiation of the equation, we obtain the derivative of the optimal $p(\underline{\theta})$ with respect to γ :

$$\frac{dp(\underline{\theta})}{d\gamma} = -\frac{c'(\underline{\theta}_p^\gamma)}{\gamma\kappa''(\hat{p})c''(\underline{\theta}_p^\gamma) + 1} > 0.$$

Because no information is acquired in the optimal, the consumer surplus is:

$$U_B := V(\theta_0) = \gamma c(\theta_0) - \min_{v \in [\underline{\theta}, \bar{\theta}]} \{\gamma c(v) - p(\underline{\theta})(\theta_0 - v)\}.$$

Taking derivatives with respect to γ we obtain:

$$\begin{aligned} \frac{dU_B}{d\gamma} &= c(\theta_0) - c(\underline{\theta}_p^\gamma) + \frac{dp(\underline{\theta})}{d\gamma}(\theta_0 - \underline{\theta}_p^\gamma) \\ &= c(\theta_0) - c(\underline{\theta}_p^\gamma) - \frac{c'(\underline{\theta}_p^\gamma)}{\gamma\kappa''(\hat{p})c''(\underline{\theta}_p^\gamma) + 1}(\theta_0 - \underline{\theta}_p^\gamma). \end{aligned}$$

We now apply two observations. First, by the mean value theorem, there exists some $x^\gamma \in [\underline{\theta}_p^\gamma, \theta_0]$, such that $-c'(\underline{\theta}_p^\gamma) = c''(x^\gamma)(\theta_0 - \underline{\theta}_p^\gamma)$. Second, by strong convexity of c , there exists some $\alpha > 0$ such that $c(\underline{\theta}_p^\gamma) \geq c(\theta_0) + c'(\theta_0)(\theta_0 - \underline{\theta}_p^\gamma) + \alpha(\theta_0 - \underline{\theta}_p^\gamma)^2$. These two observations imply, in the formula above:

$$\frac{dU_B}{d\gamma} \leq \left(-\alpha + \frac{c''(x^\gamma)}{\gamma\kappa''(p(\underline{\theta}))c''(\underline{\theta}_p^\gamma) + 1} \right) (\theta_0 - \underline{\theta}_p^\gamma)^2$$

Because for $\gamma > \bar{\gamma}$, c'' is bounded by the proof Part (i), and because κ, c are strongly convex, there exists $\bar{\bar{\gamma}} \geq \hat{\gamma}$ such that the term in the first parentheses is negative, which completes the proof.

□

A.5.2. Proof of Theorem 4

Proof of Theorem 4-(i). Below, we establish the existence of $\underline{\gamma} > 0$, such that for all $\gamma \leq \underline{\gamma}$, the optimal signal is $F_\gamma = F_0$. This is done in a sequence of Lemmas. Lemmas 11-14 establish that, when $\gamma = 0$, the full learning outcome— $F = F_0$ and Q being the optimal screening allocation for F_0 —is uniquely optimal for the seller. Then, Lemmas 15-17 establish, through an approximation argument, that there exists an information cost scale that is low enough that guarantees F_0 to be optimal.

Lemma 11. *Let Condition 1 hold. Let $F \in \mathcal{I}$ be such that $I_F(\theta) = 0$ for some $\theta \in (\theta_i, \theta_{i+1})$ for some i . Then $I_F([\theta_i, \theta_{i+1}]) = \{0\}$ and $F(\theta') = F_0(\theta')$ for all $\theta' \in [\theta_i, \theta_{i+1})$.*

Proof. Since $\theta \notin \text{supp} F_0$, F_0 is continuous at θ , and so the argument of Lemma 3 part (i) in Ravid, Roesler, and Szentes (2022) delivers that $F(\theta) = F_0(\theta)$. Moreover, the same argument delivers that $I_F([\theta_i, \theta_{i+1}]) = \{0\}$ implies that $F(\theta') = F_0(\theta')$ for all $\theta' \in [\theta_i, \theta_{i+1})$. Hence, proving that $I_F([\theta_i, \theta_{i+1}]) = \{0\}$ is sufficient.

We first argue that $I_F(\theta') = 0$ for all $\theta' \in (\theta, \theta_{i+1}]$. This follows from noting that

$$\begin{aligned} 0 \leq I_F(\theta') &= I_F(\theta) + \int_{\theta}^{\theta'} (F_0 - F)(\tilde{\theta}) d\tilde{\theta} = \int_{\theta}^{\theta'} (F_0 - F)(\tilde{\theta}) d\tilde{\theta} \\ &\leq \int_{\theta}^{\theta'} (F_0(\tilde{\theta}) - F(\theta)) d\tilde{\theta} \\ &= (\theta' - \theta)(F_0(\theta) - F(\theta)) = 0, \end{aligned}$$

where the second equality follows from $I_F(\theta) = 0$, the third equality from F being increasing, and the fourth equality from F_0 being constant on $[\theta_i, \theta_{i+1}) \supseteq [\theta, \theta']$.

To conclude the lemma, we show that $I_F(\theta') = 0$ for all $\theta' \in [\theta_i, \theta]$. We do so by way of contradiction—hence, suppose $I_F(\theta') > 0$. Then,

$$0 < I_F(\theta') = I_F(\theta) - \int_{\theta'}^{\theta} (F_0 - F)(\tilde{\theta}) d\tilde{\theta} = \int_{\theta'}^{\theta} (F - F_0)(\tilde{\theta}) d\tilde{\theta}.$$

In particular, a $\theta'' \in (\theta', \theta)$ must exist such that $F(\theta'') > F_0(\theta'')$. To get a contradiction, notice that

$$F(\theta) \geq F(\theta'') > F_0(\theta'') = F_0(\theta) = F(\theta),$$

where the first inequality follows from F being increasing, and the second equality from F_0 being constant on $[\theta_i, \theta_{i+1}) \supseteq [\theta'', \theta)$. $\square$

Lemma 12. *Let Condition 1 hold. Assume (Q, F) is monopolist optimal and Q is F -CC. Then p_Q is constant on (θ_i, θ_{i+1}) for all $i = 1, \dots, L - 1$.*

Proof. Suppose by way of a contradiction that p_Q is strictly increasing at some $\hat{\theta} \in (\theta_i, \theta_{i+1})$. We argue that $\hat{\theta} > \underline{\theta}_Q$. To do so, note that $I_F(\hat{\theta}) = 0$, and so $F(\theta') = F_0(\theta') = F_0(\theta_i) > 0$ for all $\theta' \in [\theta_i, \theta_{i+1})$. Hence, $\hat{\theta} > \theta_i \geq \underline{\theta}_F \geq \underline{\theta}_Q$, as required.

Thus, $\hat{\theta} > \underline{\theta}_Q$. Moreover, $I_F(\theta_{i+1}) = 0$ by Lemma 11. Notice that Theorem 2 implies $Q(\bar{\theta}_F) < \bar{q}$. Hence, we can apply Lemma 9 to obtain the following contradiction:

$$\begin{aligned} \int_{\theta \geq \hat{\theta}} \kappa'(Q(\theta)) F(d\theta | \theta \geq \hat{\theta}) &\leq \hat{\theta} < \theta_{i+1} \leq \int_{\theta \geq \theta_{i+1}} \kappa'(Q(\theta)) F(d\theta | \theta \geq \theta_{i+1}) \\ &= \int_{\theta \geq \hat{\theta}} \kappa'(Q(\theta)) F(d\theta | \theta \geq \hat{\theta}), \end{aligned}$$

where the first inequality follows from Lemma 9 part (iii), the third inequality from Lemma 9 part (i), and the last inequality from F being constant on $[\theta_i, \theta_{i+1})$ (by Lemma 11). $\square$

Lemma 13. *Suppose Condition 1 holds, $\gamma = 0$, and let Q be an F -CC allocation that is (weakly) below efficient for all $\theta \in \text{supp } F$. Then,*

$$\int \pi_Q(\theta) F_0(d\theta) \geq \int \pi_Q(\theta) F(d\theta).$$

Moreover, the inequality is strict whenever $F_{0-}(\theta) \neq F(\theta)$ at some $\theta \in \{\underline{\theta}, \bar{\theta}\}$ or some θ at which either p is strictly increasing.

Proof. Let $\tilde{\Theta}_1$ be the set of θ at which p is strictly increasing, and $\tilde{\Theta} = \tilde{\Theta}_1 \cup \{\underline{\theta}, \bar{\theta}\}$. By Lemma 12, $\tilde{\Theta} \subseteq \text{supp } F_0$, and so we can order its elements, $\tilde{\Theta} = \{\tilde{\theta}_1, \dots, \tilde{\theta}_K\}$, where $\theta_i < \theta_j$ for all $i < j$. Because Q is F -CC and $\gamma = 0$, π_Q is constant on $[\tilde{\theta}_i, \tilde{\theta}_{i+1})$ (that $\tilde{\theta}_i$ is in the interval follows from Q jumping towards efficiency and quality being underprovided). Therefore, for every $G \in \mathcal{I}$,

$$\int \pi_Q(\theta) G(d\theta) = \sum_i \pi_Q(\tilde{\theta}_i) [G_-(\tilde{\theta}_{i+1}) - G_-(\tilde{\theta}_i)] = \sum_{i=2}^K [\pi_Q(\tilde{\theta}_{i-1}) - \pi_Q(\tilde{\theta}_i)] G_-(\tilde{\theta}_i) + \pi_Q(\tilde{\theta}_K),$$

where $G_-(\tilde{\theta}_{K+1}) := 1$ (recall $\tilde{\theta}_1 = \underline{\theta}$ and $\tilde{\theta}_K = \bar{\theta}$). Therefore,

$$\int \pi_Q(\theta)(F - F_0)(d\theta) = \sum_{i=2}^K [\pi_Q(\tilde{\theta}_{i-1}) - \pi_Q(\tilde{\theta}_i)][F_-(\tilde{\theta}_i) - F_{0-}(\tilde{\theta}_i)].$$

Now, since $I_F(\tilde{\theta}_i) = 0$ for all i , the logic of Ravid, Roesler, and Szentes (2022), Lemma 3 implies $F_-(\tilde{\theta}_i) \geq F_{0-}(\tilde{\theta}_i)$.

Therefore, showing $\pi_Q(\tilde{\theta}_i) < \pi_Q(\tilde{\theta}_j)$ for all $i < j$ is sufficient for proving the lemma. We prove this inequality by showing it holds for $j = i + 1$. Since Q is constant on $[\tilde{\theta}_i, \tilde{\theta}_{i+1})$, we have $Q_-(\tilde{\theta}_{i+1}) = Q(\tilde{\theta}_i)$ and $T_{Q-}(\tilde{\theta}_{i+1}) = T_Q(\tilde{\theta}_i)$. We can therefore derive the following inequality chain,

$$\pi_Q(\tilde{\theta}_{i+1}) = \Pi(Q(\tilde{\theta}_{i+1}), T_Q(\tilde{\theta}_{i+1})) > \Pi(Q(\tilde{\theta}_i), T_Q(\tilde{\theta}_i)) = \pi_Q(\tilde{\theta}_i),$$

where the inequality follows from Q jumping towards efficiency. The lemma follows. $\square$

In what follows, let $Q_0 \in \mathcal{Q}$ be the unique optimal allocation when the buyer is restricted to full learning. This allocation is defined by two properties: it is constant around every $\theta \notin \text{supp } F_0$, and has $Q_0(\theta_i) = (\kappa')^{-1}(\theta_i)$ for all i .

Lemma 14. *Suppose Condition 1 holds and $\gamma = 0$. Then (Q_0, F_0) is the unique monopolist optimal outcome.*

Proof. Note that Q_0 is F_0 -CC. Moreover, by standard arguments, Q_0 is the unique allocation in $\mathcal{Q}$ that maximizes $\int \pi_Q(\theta)F_0(d\theta)$. Take the monopolist optimal CC (Q, F) . By Theorem 2, Q is below efficient for all $\theta \in \text{supp } F$. Therefore, by Lemma 13, every CC (Q, F) has

$$\int \pi_Q(\theta)F(d\theta) \leq \int \pi_Q(\theta)F_0(d\theta) \leq \int \pi_{Q_0}(\theta)F_0(d\theta),$$

where the second inequality is strict whenever $Q \neq Q_0$.

Thus, it remains to show that F_0 is the unique maximizer of $\int \pi_{Q_0}(\theta)F(d\theta)$ across all $F \in \mathcal{I}$ for which Q_0 is F -CC. For this purpose, note that every $\theta \in \Theta$ has $Q_0(\theta) = Q_0(\theta_i)$ for some $\theta_i \leq \theta$. Since the efficient quality is increasing in type and Q_0 provides quality weakly below efficiency to all types in $\text{supp } F_0$, it follows Q_0 is below efficient for all types. We can therefore apply Lemma 13 to get that every F for which Q_0 is F -CC has $\int \pi_{Q_0}(\theta)F(d\theta) < \int \pi_{Q_0}(\theta)F_0(d\theta)$ whenever $F_-(\theta) \neq F_{0-}(\theta)$ for some $\theta \in \text{supp } F_0$. Moreover, since p_{Q_0} is strictly increasing at every $\theta \in \{\theta_2, \dots, \theta_L\}$, Q_0 is CC for some

$F \in \mathcal{I}$ if and only if $I_F(\theta) = 0$ for all $\theta \in \text{supp } F_0$. Next, we argue that the unique $F \in \mathcal{I}$ for which this equality holds and which has $F_-(\theta) = F_{0-}(\theta)$ for all $\theta \in \text{supp } F_0$ is F_0 .

Towards this goal, take any $i = 1, \dots, L - 1$, and note that every $\theta \in [\theta_i, \theta_{i+1})$ has $F(\theta) \leq F_-(\theta_{i+1}) = F_{0-}(\theta_{i+1}) = F_0(\theta)$. In other words, $F \leq F_0$ —that is, F first order stochastically dominates F_0 . Since F is also a mean-preserving contraction of F_0 (since $F \in \mathcal{I}$), we must have $F = F_0$, as required. $\square$

Lemma 15. *Suppose Condition [1](#) holds. For every $\epsilon > 0$, a $\gamma_\epsilon > 0$ exists such that, for every CC (Q, F) that is monopolist optimal for $\gamma < \gamma_\epsilon$ one has $|Q(\theta) - Q_0(\theta)| < \epsilon$ for all θ , and $|F(\theta) - F_0(\theta)| < \epsilon$ for every θ at which F_0 is continuous.*

Proof. We prove the lemma by way of contradiction. If the lemma is false, one can find a sequence $(\gamma_n, Q_n, F_n)_{n \in \mathbb{N}}$ such that $\gamma_n \rightarrow 0$, (Q_n, F_n) is a monopolist optimal CC outcome for γ_n , and either Q_n does not uniformly converge to Q_0 or F_n does not converge in distribution to F_0 . We obtain a contradiction by arguing that every subsequence $(\gamma_n, Q_n, F_n)_{n \in N}$ for some countably infinite $N \subseteq \mathbb{N}$ admits a further subsequence $(\gamma_n, Q_n, F_n)_{n \in K}$, $K \subseteq N$ where both convergences hold. We make this argument by passing to an appropriately convergent subsubsequence and showing the limit must be an optimal CC outcome at $\gamma = 0$. Lemma [14](#) then implies the limit is (Q_0, F_0) .

We begin by finding an appropriately convergent subsubsequence. Let $p_n := p_{Q_n}$ be the marginal price function associated with Q_n and γ_n . Note $[-\gamma_n c'(\underline{\theta}_{F_n}), \bar{q} - c'(\bar{\theta}_{F_n})] \subseteq [0, \bar{q}]$. Thus, by Helly's Selection Theorem, we can choose the subsubsequence such that p_n converges pointwise to some p_∞ , and F_n converges in distribution to some F_∞ . By Lemma [12](#), each p_n is completely determined by its values on $\text{supp } F_0$, which is finite. It follows p_n converges to p_∞ uniformly. Thus, as c' is bounded, $p_n + \gamma_n c'$ also converges uniformly, and so $Q_n = \min\{(p_n + \gamma_n c')_+, \bar{q}\}$ also converges uniformly to the CC allocation (for $\gamma = 0$) defined by p_∞ , Q_∞ . Note this uniform convergence also implies that π_{Q_n} uniformly converges to π_{Q_∞} .

Next we show that the limit outcome (Q_∞, F_∞) is an CC outcome for $\gamma = 0$. We achieve this goal by arguing that p_∞ is an F_∞ -marginal price. For this purpose, note first that $p_n(\Theta) \subseteq [-\gamma_n c'(\underline{\theta}_{F_n}), \bar{q} - \gamma_n c'(\bar{\theta}_{F_n})] \subseteq [-0c'(\underline{\theta}_{F_\infty}), \bar{q} - 0c'(\bar{\theta}_{F_\infty})]$. Second, observe Lemma [12](#) means that p_∞ strictly increases at some θ only if $\theta \in \text{supp } F_0$, and p_n is strictly increasing at θ for all sufficiently large n . Consequently, p_∞ strictly increases at θ only if $I_{F_n}(\theta) = 0$ holds for all large n , and so $I_{F_\infty}(\theta) = 0$ must hold as well.

Now we prove a helpful intermediate step by showing that Q_∞ jumps towards efficiency. This property follows from noting that Q_∞ can only have jumps on $\text{supp } F_0$, which is finite. Moreover, if Q_∞ jumps at $\theta \in \text{supp } F_0$, then Q_n must jump at θ as well for all sufficiently large n . Therefore, p_n is strictly increasing at each θ where Q_∞ jumps. By Lemma 9, $Q_n(\theta)$ must be below efficient, and so $Q_\infty(\theta)$ must be weakly below efficient, too. Hence, Q_∞ jumps towards efficiency if and only if it is right continuous at every $\theta \in \text{supp } F_0$. All that remains is to observe that Q_n is increasing, right continuous, and constant on $[\theta_i, \theta_{i+1})$ for every $i = 1, \dots, L-1$. We conclude Q_∞ jumps towards efficiency.

We now claim (Q_∞, F_∞) is monopolist optimal at $\gamma = 0$. Note this claim completes the proof, since then $(Q_\infty, F_\infty) = (Q_0, F_0)$ holds by Lemma 14. Let $\tilde{Q}_n$ be the F_0 -CC allocation for γ_n with the property that $\tilde{Q}_n(\theta) = Q_0(\theta)$ for all $\theta \in \text{supp } F_0$ (such an allocation exists whenever γ is sufficiently small). To define this allocation, define $\tilde{p}_n : \Theta \rightarrow [0, \bar{q}]$ by setting it $\tilde{p}_n(\theta) = Q_0(\theta_i) - \gamma_n c'(\theta_i)$ for every $\theta \in [\theta_i, \theta_{i+1})$, $i = 1, \dots, L-1$. Observe $\tilde{Q}_n$ converges to Q_0 , and so $\pi_{\tilde{Q}_n}$ converges pointwise to π_{Q_0} . By the Bounded Convergence Theorem, $\int \pi_{\tilde{Q}_n}(\theta) F_0(d\theta) \rightarrow \int \pi_{Q_0}(\theta) F_0(d\theta)$. Together with the above observations, we obtain that, for every $\epsilon > 0$, the following inequality chain holds for all sufficiently large $n \in K$:

$$\begin{aligned} \int \pi_{Q_0}(\theta) F_0(d\theta) - \frac{\epsilon}{2} &\leq \int \pi_{\tilde{Q}_n}(\theta) F_0(d\theta) \leq \int \pi_{Q_n}(\theta) F_n(d\theta) \\ &\leq \int \pi_{Q_\infty}(\theta) F_n(d\theta) + \frac{\epsilon}{3} \\ &\leq \int \pi_{Q_\infty}(\theta) F_\infty(d\theta) + \frac{2\epsilon}{3}, \end{aligned}$$

where the first inequality is due to $\pi_{\tilde{Q}_n}$ converging pointwise to π_{Q_0} , the second inequality from (Q_n, F_n) being monopolist optimal for γ_n and $\tilde{Q}_n$ being F_n -CC for γ_n , the third from uniform convergence of π_{Q_n} to π_{Q_∞} , and the last inequality from F_n converging to F_∞ and π_{Q_∞} being upper semicontinuous (due to Q_∞ jumping towards efficiency, see Lemma 4). Since $\epsilon > 0$ is arbitrary, it follows the monopolist's profit from (Q_∞, F_∞) is the same as from (Q_0, F_0) . Since (Q_0, F_0) is optimal at $\gamma = 0$ (Lemma 14, and (Q_∞, F_∞) is IC for $\gamma = 0$ (because it is CC), (Q_∞, F_∞) must be monopolist optimal as well. Thus, (Q_∞, F_∞) must equal the unique monopolist optimal outcome at $\gamma = 0$, which is (Q_0, F_0) . $\square$

Lemma 16. *Suppose Condition 1 holds. Then there exists $\tilde{\gamma} > 0$ such that for all $\gamma \leq \tilde{\gamma}$, $\text{supp } F = \{\tilde{\theta}, \theta_2, \dots, \theta_L\}$, for some $\tilde{\theta} \leq \theta_2$, and $I_F(\theta) = 0$ for all $\theta \geq \theta_2$.*

Proof. Fix some $0 < \delta < \min_{i \geq 2} \{Q_0(\theta_i) - Q_0(\theta_{i-1})\}$, and any $\frac{\min_{i \geq 2} \{Q_0(\theta_i) - Q_0(\theta_{i-1})\} - \delta}{3} >$

$\epsilon > 0$, and consider $\gamma \leq \gamma_\epsilon$, where γ_ϵ is defined as in Lemma 15. First, we prove that $\theta_i \in \text{supp} F$ for all $i \geq 2$. By Lemma 15, we know that $Q(\theta_{i-1}) \leq Q_0(\theta_{i-1}) + \epsilon$, and $Q(\theta_i) \geq Q_0(\theta_i) - \epsilon$, and thus:

$$Q(\theta_i) - Q(\theta_{i-1}) \geq Q_0(\theta_i) - Q_0(\theta_{i-1}) - 2\epsilon$$

By F-CC:

$$Q(\theta_i) - Q(\theta_{i-1}) = p(\theta_i) - p(\theta_{i-1}) + \gamma (c'(\theta_i) - c'(\theta_{i-1})).$$

By choosing $\gamma \leq \frac{\epsilon}{c'(\theta) - c'(\theta)}$, and putting these two observations together we get:

$$p(\theta_i) - p(\theta_{i-1}) \geq Q_0(\theta_i) - Q_0(\theta_{i-1}) - 3\epsilon > \delta > 0,$$

implying, through Lemma 12, that p jumps at θ_i and thus, that $\theta_i \in \text{supp} F$.

To show that there is no other $\theta_2 < \theta \notin \text{supp} F_0$ with $\theta \in \text{supp} F$, note that p can only jump at θ_i if $I_F(\theta_i) = 0$, which means that, for all θ_i , $i \geq 2$, $I_F(\theta_i) = 0$. Therefore, by Lemma 11, $F(\theta) = F_0(\theta)$ for all $\theta \geq \theta_2$, which is what we wanted to prove.

It remains to show that there is exactly one point $\tilde{\theta} \leq \theta_2$, with $\tilde{\theta} \in \text{supp} F$. For that, consider any $F \in \mathcal{I}$ such that $F(\theta) = F_0(\theta)$ for all $\theta \geq \theta_2$, and an F-CC allocation Q . Let $\underline{\theta}_Q := \tilde{\theta}$. We shall find a $\tilde{\gamma}$ such that, for all $\gamma \leq \tilde{\gamma}$, the principal prefers to concentrate all the support in $\{\tilde{\theta}, \theta_2\}$, regardless of what F and $\tilde{\theta}$ are. Recall that p must be constant in (θ_1, θ_2) . Therefore, notice that, for any type $\theta \in [\tilde{\theta}, \theta_2)$, the profits obtained by the seller from that type are:

$$\pi(\theta) = \pi(\tilde{\theta}) + \gamma \int_{\tilde{\theta}}^{\theta} (x - \kappa'(Q(x))) c''(x) dx \leq \pi(\tilde{\theta}) + \gamma \max_{\theta \in [\underline{\theta}, \theta_2]} \{c''(\theta)\} (\theta - \tilde{\theta}),$$

Moreover:

$$\begin{aligned} \pi(\theta_2) &= \pi(\theta_2-) + \theta_2 (Q(\theta_2) - Q(\theta_2-)) + \kappa(Q(\theta_2-)) - \kappa(Q(\theta_2)) \\ &\geq \pi(\theta_2-) + (\theta_2 - \kappa'(Q(\theta_2))) (p(\theta_2) - p(\theta_1)) \\ &\geq \pi(\theta_2-) + (\theta_2 - \kappa'(Q(\theta_2))) \delta, \end{aligned}$$

where the first inequality used convexity of κ , and the second inequality used the fact that,

at any optimum, $\theta_2 \geq \kappa'(Q(\theta_2))$. Additionally, we must have:

$$\begin{aligned}\theta_2 - \kappa'(Q(\theta_2)) &= \underbrace{\theta_2 - \kappa'(Q_0(\theta_2))}_{v_2} + \kappa'(Q_0(\theta_2)) - \kappa'(Q(\theta_2)) \\ &\geq v_2 + \min_{x \leq Q_0(\theta_2) + \epsilon} \{\kappa''(x)\} \epsilon,\end{aligned}$$

where the inequality follows from the mean-value theorem.

Now, define $\tilde{\gamma} = \frac{\min_{x \leq Q_0(\theta_2) + \epsilon} \{\kappa''(x)\}}{\kappa'(Q_0(\theta_2) + \epsilon)(c'(\theta_2) - c'(\underline{\theta}))}$, and let $\gamma \leq \tilde{\gamma}$. Then:

$$\begin{aligned}\pi(\theta_2) - \pi(\tilde{\theta}) &\geq \gamma \int_{\tilde{\theta}}^{\theta_2} (x - \kappa'(Q(x))) c''(x) dx + v_2 \delta + \min_{x \leq Q_0(\theta_2) + \epsilon} \{\kappa''(x)\} \epsilon \delta \\ &> \gamma \int_{\underline{\theta}}^{\theta_2} (-\kappa'(Q(x))) c''(x) dx + v_2 \delta + \min_{x \leq Q_0(\theta_2) + \epsilon} \{\kappa''(x)\} \epsilon \delta \\ &\geq v_2 \delta > 0,\end{aligned}$$

where the first inequality simply used the bounds derived for $\pi(\theta_2)$ above, the second inequality makes the integral term as negative as possible and the third uses the fact that $\gamma \leq \tilde{\gamma}$. We can now conclude that the line segment connecting $\pi(\theta_2)$ and $\pi(\tilde{\theta})$ is everywhere above π . Indeed, for all $\theta \in [\tilde{\theta}, \theta_2]$:

$$\begin{aligned}&\pi(\tilde{\theta}) + \frac{\pi(\theta_2) - \pi(\tilde{\theta})}{\theta_2 - \tilde{\theta}}(\theta - \tilde{\theta}) \\ &\geq \pi(\tilde{\theta}) + \frac{v_2 \delta}{\theta_2 - \underline{\theta}}(\theta - \tilde{\theta}) \geq \pi(\tilde{\theta}) + \gamma \max_{\theta \in [\underline{\theta}, \theta_2]} \{c''(\theta)\}(\theta - \tilde{\theta}) \geq \pi(\theta),\end{aligned}$$

for $\gamma \leq \tilde{\gamma}$. Therefore, the seller always prefer to concentrate any mass available for $[\theta_1, \theta_2]$ in $\{\tilde{\theta}, \theta_2\}$. $\square$

Lemma 17. *Suppose Condition 1 holds. There exists $\underline{\gamma} > 0$ such that for $\gamma \leq \underline{\gamma}$, $F = F_0$.*

Proof. By the previous result, $\text{supp} F = \{\tilde{\theta}, \theta_2, \dots, \theta_L\}$, with $\tilde{\theta} < \theta_2$, $F(\theta) = F_0(\theta)$ for $\theta \geq \theta_2$, and $I_F(\theta_2) = I_{F_0}(\theta_2)$. We assume $\tilde{\theta} > \underline{\theta}$ and show a contradiction. By Lemma 15, for any ε there is small enough γ_ε such that $\tilde{\theta} - \underline{\theta} < \varepsilon$ for $\gamma < \gamma_\varepsilon$.

Let $\mu_1 := \int \theta F(d\theta | \theta \in [\theta_1, \theta_2])$. Because F is a mean-preserving-contraction of F_0 , $I_F(\theta_1) = I_F(\theta_2) = 0$, and, by the proof of Lemma 16, $F_0(\theta) = F(\theta)$ for $\theta \geq \theta_2$, we must

have:

$$F(\tilde{\theta})\tilde{\theta} + (F_0(\theta_2) - F(\tilde{\theta}))\theta_2 = \mu_1 F_0(\theta_2).$$

Thus, $\frac{dF(\tilde{\theta})}{d\tilde{\theta}} = F_0(\theta_2) \frac{\theta_2 - \mu_1}{(\theta_2 - \tilde{\theta})^2}$, which is positive and bounded away from zero since $\tilde{\theta} \in [\underline{\theta}, \underline{\theta} + \varepsilon]$.

Lemma 9 Part (iii) implies that, for $i > 2$, $Q_\gamma(\theta_i) = Q_0(\theta_i)$, where $Q_0(\theta_i)$ is the optimal screening quality for type θ_i under F_0 . Lemma 9 Parts (i) and (iii) imply:

$$\begin{aligned} \theta_2 &= \int \kappa'(Q_\gamma(\theta)) F(d\theta | \theta \geq \theta_2) = \frac{F_0(\theta_2) - F(\tilde{\theta})}{1 - F(\tilde{\theta})} \kappa'(Q_\gamma(\theta_2)) + \\ &\quad \frac{1 - F_0(\theta_2)}{1 - F(\tilde{\theta})} \int \kappa'(Q_\gamma(\theta)) F(d\theta | \theta > \theta_2) \\ &= \frac{F_0(\theta_2) - F(\tilde{\theta})}{1 - F(\tilde{\theta})} \kappa'(Q_\gamma(\theta_2)) + \frac{1 - F_0(\theta_2)}{1 - F(\tilde{\theta})} \theta_3, \end{aligned} \quad (26)$$

where the second equality follows from applying Lemma 9 Parts (i) and (iii) to θ_3 . Reorganizing the equation above obtains:

$$\kappa'(Q_\gamma(\theta_2)) = \theta_2 - \frac{1 - F_0(\theta_2)}{F_0(\theta_2) - F(\tilde{\theta})} (\theta_3 - \theta_2),$$

whenever the right-hand-side of the equation above is positive. By choosing ε small enough, we can make sure that this is the case since, by Condition 1 the right-hand-side is positive for $\tilde{\theta} = \underline{\theta}$, and it is continuous in $\tilde{\theta}$.

The above, in turn, implies (i) that $Q_\gamma(\theta_2)$ is the optimal screening quantity for θ_2 under F , and (ii) that $Q_\gamma(\theta_2)$ is decreasing with $\tilde{\theta}$, with $Q_\gamma(\theta_2) = Q_0(\theta_2)$, when $\tilde{\theta} = \underline{\theta}$, and it is bounded away from zero by choosing ε small enough. Define $Q_2(\tilde{\theta}) := Q_\gamma(\theta_2)$ to emphasize the dependence on $\tilde{\theta}$.

Let S_i be the optimal screening surplus obtained from an agent with type θ_i —that is, $S_i = \theta_i Q_0(\theta_i) - \kappa(Q_0(\theta_i))$. Noting that $Q_\gamma(\tilde{\theta}) = 0$, and therefore the seller makes no profits from $\tilde{\theta}$, the monopolist's profits are:

$$\begin{aligned}
& (F(\theta_2) - F(\tilde{\theta})) \left(\theta_2 Q_2(\tilde{\theta}) - \kappa(Q_2(\tilde{\theta})) - \gamma \left(c(\theta_2) - c(\tilde{\theta}) - c'(\tilde{\theta}) (\theta_2 - \tilde{\theta}) \right) \right) + \\
& \sum_{i>2} (F(\theta_i) - F(\theta_{i-1})) \left\{ S_i - Q_2(\tilde{\theta})(\theta_3 - \theta_2) + \sum_{2<j<L} q_j(\theta_{j+1} - \theta_j) \right. \\
& \quad \left. - \gamma \left(c(\theta_i) - c(\tilde{\theta}) - \sum_{j<L} c'(\theta_j) (\theta_{j+1} - \theta_j) \right) \right\},
\end{aligned}$$

where the first line describes the profits obtained from type θ_2 . Regrouping the terms that depend on $\tilde{\theta}$ and dropping the terms that do not, we obtain:

$$\begin{aligned}
& (F(\theta_2) - F(\tilde{\theta})) \left(\theta_2 Q_2(\tilde{\theta}) - \kappa(Q_2(\tilde{\theta})) \right) - (1 - F(\theta_2)) Q_2(\tilde{\theta})(\theta_3 - \theta_2) \\
& - \sum_{i>1} (F(\theta_i) - F(\theta_{i-1})) \gamma \left(c(\theta_i) - c(\tilde{\theta}) - c'(\tilde{\theta}) (\theta_2 - \tilde{\theta}) \right).
\end{aligned}$$

We can then take derivatives on the expression above with respect to $\tilde{\theta}$ to find:

$$\begin{aligned}
& (F(\theta_2) - F(\tilde{\theta})) \underbrace{\left(\theta_2 - \kappa'(Q_2(\tilde{\theta})) - \sum_{i>2} (F(\theta_i) - F(\theta_{i-1}))(\theta_3 - \theta_2) \right)}_{=0} \frac{dQ_2(\tilde{\theta})}{d\tilde{\theta}} \\
& - \frac{dF(\tilde{\theta})}{d\tilde{\theta}} \left(\theta_2 Q_2(\tilde{\theta}) - \kappa(Q_2(\tilde{\theta})) \right) + (1 - F_0(\tilde{\theta})) \gamma c''(\tilde{\theta}),
\end{aligned}$$

where the first term equals zero because of [26](#). Now, the second term is negative and bounded away from zero due to $\frac{dF(\tilde{\theta})}{d\tilde{\theta}}$ being positive and bounded away from zero, and due to $Q_2(\tilde{\theta}) > 0$. The third term can be made arbitrarily small for small γ since c'' is bounded. We then find $\underline{\gamma}$ such that the expression is strictly decreasing in $\tilde{\theta}$, proving that it is optimal for the monopolist to choose $\tilde{\theta} = \underline{\theta}$ for sufficiently low γ . $\square$

$\square$

Proof of Theorem 4-(ii), (iii) and Corollary 2. We now consider $\gamma \leq \underline{\gamma}$ which, by Part (i), implies $F_\gamma = F_0$. By the argument in Step 1 of the proof of Part (i), $p(\theta)$ is constant around any $\theta \notin \text{supp} F$. Moreover, by CC: $\gamma p(\theta_i) = Q_0(\theta_i) - \gamma c'(\theta_i)$, for all $i = 1, \dots, L$. We can then write the consumers' gross value as a function of his interim-type as:

$$\begin{aligned}
V_\gamma(\theta_i) &= \gamma c(\theta_i) - \gamma c(\underline{\theta}) + \gamma \int_{\underline{\theta}}^{\theta} p(x) dx = \gamma \left(c(\theta_i) - c(\underline{\theta}) + \sum_{j < i} p(\theta_j)(\theta_{j+1} - \theta_j) \right) \\
&= \sum_{j < i} q_j(\theta_{j+1} - \theta_j) + \gamma \left(c(\theta_i) - c(\underline{\theta}) - \sum_{j < i} c'(\theta_j)(\theta_{j+1} - \theta_j) \right),
\end{aligned} \tag{27}$$

where the first equality uses the fact that p is constant outside the support of F_0 .

First, note that because the qualities in the contract are fixed for $\gamma \leq \underline{\gamma}$, the change in the monopolist's profit U_S is the negative of the change in the expected gross consumer value, $\int V_\gamma(\theta) F_0(d\theta)$. Furthermore, for $\gamma < \gamma' \leq \underline{\gamma}$:

$$V_{\gamma'}(\theta_i) - V_\gamma(\theta_i) = (\gamma' - \gamma) \left(c(\theta_i) - c(\underline{\theta}) - \sum_{j < i} c'(\theta_j)(\theta_{j+1} - \theta_j) \right) = (\gamma' - \gamma) (c(\theta_i) - \tilde{c}(\theta_i)) > 0,$$

for all $i > 1$. Therefore, gross surplus is increasing type by type for $\gamma \leq \underline{\gamma}$. Therefore, expected gross surplus is also increasing, which implies that profits are decreasing, proving part (ii) of the Theorem.

We now proceed to studying how consumer surplus, $U_B := \int [V_\gamma(\theta) - \gamma c(\theta)] F_0(d\theta)$, changes with γ . Assuming again $\gamma < \gamma' \leq \underline{\gamma}$ we have:

$$\begin{aligned}
& \int [V_{\gamma'}(\theta) - \gamma' c(\theta) - (V_\gamma(\theta) - \gamma c(\theta))] F_0(d\theta) \\
&= (\gamma' - \gamma) \sum_{i=1}^L (F_0(\theta_i) - F_0(\theta_{i-1})) \left\{ \left(c(\theta_i) - c(\underline{\theta}) - \sum_{j < i} c'(\theta_j)(\theta_{j+1} - \theta_j) \right) - c(\theta_i) \right\} \\
&= -(\gamma' - \gamma) \sum_{i=1}^L (F_0(\theta_i) - F_0(\theta_{i-1})) \left(c(\underline{\theta}) + \sum_{j < i} c'(\theta_j)(\theta_{j+1} - \theta_j) \right) \\
&= -(\gamma' - \gamma) \int \tilde{c}(\theta) F_0(d\theta),
\end{aligned} \tag{28}$$

proving the claim about consumer surplus.

We finish by proving Corollary 2. Assume $L = 2$. In that case, $\tilde{c}$ is a linear function, and therefore:

$$\begin{aligned}
& \int [V_{\gamma'}(\theta) - \gamma'c(\theta) - (V_{\gamma}(\theta) - \gamma c(\theta))] F_0(d\theta) \\
&= -(\gamma' - \gamma) \int \tilde{c}(\theta) F_0(d\theta) = -(\gamma' - \gamma) \tilde{c}(\theta_0) \\
&> -(\gamma' - \gamma) c(\theta_0) = 0,
\end{aligned}$$

where the second equality follows from linearity of $\tilde{c}$, the inequality from $\tilde{c}(\theta) < c(\theta)$ for $\theta \neq \underline{\theta}$, and the last equality from $c(\theta_0) = 0$. $\square$

A.5.3. Proof of Proposition 1

Proof of Proposition 1-(i). Take $\gamma \geq \bar{\gamma}$ as defined in Theorem 3. Part (i) of the theorem says $\text{supp} F = \{\theta_0\}$, so $\bar{\theta} \notin \text{supp} F$, and $\theta_0 = \underline{\theta}_F$. Moreover, $\underline{\theta}_F > \underline{\theta}_Q \geq \underline{\theta}$, since otherwise the monopolist's profits would be zero, which is suboptimal.

$$\theta_0 - \kappa'(Q^s(\theta_0)) = 0 < \theta_0 - \underline{\theta}_Q = \theta_0 - \int \kappa'(Q(\theta)) F(d\theta),$$

where the first equality follows from efficiency of the screening contract when information is symmetric, the inequality follows from the observation that $\theta_0 > \underline{\theta}_Q$, and the final equality from parts (i) and (ii) of Lemma 9. This proves the strict inequality for τ . $\square$

Proof of Proposition 1-(ii). Take $\gamma \leq \underline{\gamma}$, as defined in Theorem 4. Part (i) of the Theorem says $F = F_0$, so $\bar{\theta} \in \text{supp} F$. The proof of Part (i) also shows that the outcome $Q = Q^s$, therefore efficiency is exactly the same as in the pure screening problem. $\square$

A.6. Proofs from Section 6

A.6.1. Proof of Proposition 2

Note that the main text explanations are sufficient for the proof of Proposition 2, conditional on showing that if there is a signal with more than two realizations, then either (i) the mean-preserving spread constraint is binding at some $\theta \in (\underline{\theta}, \bar{\theta})$, and so it is a mean-preserving spread of a cutoff signal; or (ii) it is without loss of optimality to replace it with (at most) a binary signal.

For case (i), let θ^* be such that the mean-preserving spread constraint is binding at it. Let $\theta_-^* := E_{F_0}[\theta | \theta < \theta^*]$ and $\theta_+^* := E_{F_0}[\theta | \theta \geq \theta^*]$. Then the binary distribution which has

support on $\{\theta_-^*, \theta_+^*\}$ is a mean-preserving contraction of F and is a cutoff signal. As this is prohibitively costly, F cannot be optimal.

For case (ii), either $\theta_0 \in \text{supp}(F)$, or $\theta_0 \notin \text{supp}(F)$. Notice that by Lemma 10 and the upper-semicontinuity of π_{Q^*} , all points $\theta \in \text{supp}(F)$ lie along the same affine function. In particular, in the former case, $\pi_{Q^*}(\theta_0) = E_F[\pi_{Q^*}(\theta)]$. In this case, δ_{θ_0} is an optimal, feasible distribution with a single signal realization (with posterior at p_0). In the latter case, let $\theta_- := \sup\{\theta < \theta_0 : \theta \in \text{supp}(F)\}$ and $\theta_+ := \inf\{\theta > \theta_0 : \theta \in \text{supp}(F)\}$. Then there exists a mean-preserving contraction of F which has support at $\{\theta_-, \theta_+\}$, and preserves the expected profit of the seller. Thus there exists a binary optimal signal in this case as well. $\square$

A.7. Extending Theorem 2 to Allow Efficient Exclusion

In this section we explain the argument that extends Theorem 2 to the case where $\kappa'(0) \geq \underline{\theta}$. Note that the argument establishing that $\bar{\theta} = \kappa'(Q(\bar{\theta}))$ whenever $\bar{\theta} \in \text{supp}(F)$ easily extends to this case. Thus, all we need to do is argue that any $\theta \in \text{supp}(F)$ such that $Q(\theta) > 0$ and $\theta < \bar{\theta}$, $\theta > \kappa'(Q(\theta))$. For brevity, we provide the major steps only, and skip some of the details. So, suppose (Q, F) is a monopolist optimal outcome in which Q is F -ICC. Our goal is to argue that $\theta \in \text{supp}(F) \setminus \{\bar{\theta}\}$ has $\kappa'(Q(\theta)) \geq \theta$ only if $\theta = \underline{\theta}_Q$. Suppose then for a contradiction that another $\tilde{\theta} \in \text{supp}(F) \setminus \{\underline{\theta}_Q, \bar{\theta}\}$ exists for which said inequality holds.

First, recall that (as shown in Section A.5.3) π'_Q is positive for all $\theta \in \text{supp}(F)$ on a given interval (θ_1, θ_2) where $I_F(\theta) > 0$ if $\kappa'(Q(\theta)) < \theta$, as by the envelope theorem,

$$\pi'_Q(\theta) = \theta Q'(\theta) - \kappa'(Q(\theta))Q'(\theta)$$

(recall that an F -ICC Q is differentiable whenever $I_F(\theta) > 0$). Conversely, if π' is negative, then $\kappa'(Q(\theta)) > \theta$. In addition, if p_Q is constant on some interval (θ_1, θ_2) , then one can apply an argument similar to that in Theorem 2's proof to show that Lemma 10-(i) implies that $\pi'_Q(\theta) \leq \pi'_Q(\theta')$ for any $\theta < \theta'$ both in $\text{supp}(F)$. Moreover, if I_F is strictly positive over (θ_1, θ_2) , then a similar argument using Lemma 10-(ii) shows $\pi'_Q(\theta) \geq \pi'_Q(\theta')$, meaning $\pi'_Q(\theta) = \pi'_Q(\theta')$.

We now use the above to explain that $p_Q(\theta) = p_Q(\underline{\theta}_Q)$ for any $\theta \in \text{supp}(F) \setminus \{\bar{\theta}\}$ at which $\kappa'(Q(\theta)) \geq \theta$. Fix any $\tilde{\theta} \in \text{supp}(F)$ such that $\kappa'(Q(\tilde{\theta})) \geq \tilde{\theta}$. Suppose that

$p_Q(\tilde{\theta}) > p_Q(\underline{\theta}_Q)$. By Lemma 9-(iii), p_Q must be constant around $\tilde{\theta}$. Define

$$\theta' = \sup\{\theta \in \Theta : p_Q \text{ strictly increases at } \theta, \theta \leq \tilde{\theta}\}.$$

Observe p_Q cannot be constant around θ' . Therefore, $\theta' < \tilde{\theta}$, and so $\kappa'(Q(\theta')) < \theta'$ by Lemma 9-(iii). But then one can apply the argument in Theorem 2's proof to show that $\kappa'(Q(\tilde{\theta})) < \tilde{\theta}$, a contradiction.

Next we establish that a θ^* exists with three properties. First, every $\theta \in \text{supp}(F)$ has $\kappa'(Q(\theta)) \geq \theta$ only if $\theta \leq \theta^*$, and $\kappa'(Q(\theta)) \leq \theta$ only if $\theta \geq \theta^*$. Second, $I_F(\theta^*) = 0$. And third, $p_{Q-}(\theta^*) = p(\underline{\theta}_Q)$. To do so, pick any $\tilde{\theta} \in \text{supp}(F) \setminus \{\underline{\theta}_Q\}$ such that $\kappa(Q(\tilde{\theta})) \geq \tilde{\theta}$. Let $\hat{\theta} = \sup\{\theta : p(\theta) = p(\underline{\theta}_Q)\}$. Note that $\hat{\theta} > \tilde{\theta}$. As explained earlier, π'_Q is increasing over $(\underline{\theta}_Q, \hat{\theta}) \cap \text{supp}(F)$. Since π'_Q has to be negative (positive) for quality to be above (below) efficient, we get that some $\theta^* \in [\underline{\theta}_Q, \hat{\theta}]$ exists such that for all $\theta \in \text{supp}(F) \cap (\underline{\theta}_Q, \hat{\theta})$, $\kappa'(Q(\theta)) \geq \theta$ only if $\theta \leq \theta^*$, and $\kappa'(Q(\theta)) \leq \theta$ only if $\theta \geq \theta^*$. We now explain that one can pick θ^* to satisfy $I_F(\theta^*) = 0$. First, suppose that $\pi'_Q(\theta)$ is constant at some value x for all $\theta \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$. Notice $x \leq 0$, because $\tilde{\theta} \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$ and $\kappa'(Q(\tilde{\theta})) \geq \tilde{\theta}$. In fact, in this case $\kappa'(Q(\theta)) \geq \theta$ for all $\theta \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$, and so we can take $\theta^* = \hat{\theta}$. To get that $I_F(\theta^*) = 0$ in this case, observe p_Q must be strictly increasing at $\hat{\theta}$ whenever $\hat{\theta} \neq \bar{\theta}$. This concludes the constant π'_Q case. For the second case, suppose we have $\theta, \theta' \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$ with $\theta < \theta'$ such that $\pi'_Q(\theta) < \pi'_Q(\theta')$, and $0 \in [\pi'_Q(\theta), \pi'_Q(\theta')]$. Note $\theta^* \in [\theta, \theta']$. Without loss, we can choose θ and θ' such that the sets $(\theta, \theta^*) \cap \text{supp}(F)$ and $(\theta^*, \theta') \cap \text{supp}(F)$ are both empty. Notice I_F cannot be strictly positive over $[\theta, \theta']$, because then one could apply the argument sketched in the second paragraph of this section to $(\theta - \epsilon, \theta' + \epsilon)$ for sufficiently small $\epsilon > 0$ to get that $\pi'_Q(\theta) = \pi'_Q(\theta')$. Thus, I_F must equal 0 at some $\theta'' \in [\theta, \theta']$, and we can choose $\theta^* = \theta''$.

We now claim that $\underline{\theta}_F = \underline{\theta}_Q$. If not, then since $Q(\bar{\theta}_F) < \bar{q}$, by Lemma 9-(ii),

$$\kappa'(Q(\underline{\theta}_F)) \leq \int \kappa'(Q(\theta)) dF(\theta) = \underline{\theta}_Q < \underline{\theta}_F$$

contradicting the claim that $\underline{\theta}_F \leq \kappa'(Q(\underline{\theta}_F))$.

We now show the monopolist can improve upon (Q, F) by looking at

$$\Theta_{F-} := \{\theta : \kappa'(Q(\theta)) > \theta\}$$

and replacing (Q, F) with $(\hat{Q}, \hat{F})$, where, letting

$$\theta_{F-} = \frac{\int_{\Theta_{F-}} \theta dF(\theta)}{F_-(\theta^*)}$$

we set

$$\hat{F}(\theta) = \begin{cases} 0, & \theta < \theta_{F-} \\ F_-(\theta^*), & \theta \in [\theta_{F-}, \theta^*) \\ F(\theta), & \theta > \theta^* \end{cases}$$

and

$$\hat{Q}(\theta) = \begin{cases} 0, & \theta \leq \theta_{F-} \\ c'(\theta) - c'(\theta_{F-}), & \theta \in (\theta_{F-}, \theta^*) \\ Q(\theta), & \theta \geq \theta^* \end{cases}$$

Notice that $\hat{Q}$ is a $\hat{F}$ -ICC allocation for the marginal price function

$$\hat{p}(\theta) = \begin{cases} -c'(\theta_{F-}), & \theta < \theta^* \\ p_Q(\theta) + c'(\theta_{F-}) - c'(\underline{\theta}_F), & \theta \geq \theta^* \end{cases}$$

The new profit $\Pi_{\hat{Q}} := \int \pi_{\hat{Q}}(\theta) \hat{F}(d\theta)$ is now given by

$$\begin{aligned} \int \pi_{\hat{Q}}(\theta) \hat{F}(d\theta) &= \int_{\theta < \theta^*} \pi_{\hat{Q}}(\theta) \hat{F}(d\theta) + \int_{\theta \geq \theta^*} \pi_{\hat{Q}}(\theta) F(d\theta) \\ &= \int_{\theta < \theta^*} (0) \hat{F}(d\theta) + \int_{\theta \geq \theta^*} [\pi_Q(\theta) + V_Q(\theta_{F-})] F(d\theta) \\ &> \int_{\theta < \theta^*} (0) F(d\theta) + \int_{\theta \geq \theta^*} \pi_Q(\theta) F(d\theta) \\ &\geq \int_{\theta < \theta^*} \pi_Q(\theta) F(d\theta) + \int_{\theta \geq \theta^*} \pi_Q(\theta) F(d\theta) \\ &= \int \pi_Q(\theta) F(d\theta), \end{aligned}$$

where the weak inequality follows from noting that $\pi_Q(\underline{\theta}_Q) = 0$ and $\pi_Q'(\theta) \leq 0$ for all $\theta \in [\underline{\theta}_Q, \theta^*)$. Thus, $(\hat{Q}, \hat{F})$ is an improvement over (Q, F) . A contradiction. $\square$

A.8. Cost Function Characterization

In this section, we show a continuous cost function $C : \mathcal{I} \rightarrow \mathbb{R}$ is affine and strictly increasing in informativeness if and only if a strictly convex continuous function $c : \Theta \rightarrow \mathbb{R}$ exists such that $C(F) = \int c(\theta)F(d\theta)$. To prove this result, note the Riesz representation theorem implies C is continuous and affine if and only if $C(F) = \int \tilde{c}(\theta)F(d\theta)$ for some continuous $\tilde{c} : \Theta \rightarrow \mathbb{R}$. All that remains is to show $\tilde{c}$ must be strictly convex. For this purpose, fix any $x, y, z \in (\underline{\theta}, \bar{\theta})$ such that $y = \beta x + (1 - \beta)z$ for some $\beta \in (0, 1)$. By Lemma 6 in Ravid, Roesler, and Szentes (2020), one can find $F', F'' \in \mathcal{I}$ and $\gamma > 0$ such that $F' \succ F''$, and

$$F' - F'' = \gamma \left(\beta \mathbf{1}_{[x, \bar{\theta}]} + (1 - \beta) \mathbf{1}_{[z, \bar{\theta}]} - \mathbf{1}_{[y, \bar{\theta}]} \right),$$

where for any $w \in \Theta$, $\mathbf{1}_{[w, \bar{\theta}]}$ is the CDF of the distribution that generates w with probability 1. Since C is strictly increasing in $\succeq$, it follows that

$$0 < C(F') - C(F'') = \int c(\theta)(F' - F'')(d\theta) = \gamma (\beta \tilde{c}(x) + (1 - \beta) \tilde{c}(z) - \tilde{c}(y)).$$

The claim follows. □